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Master's Degree in Control Systems Engineering

Network Topology Optimization for Bearing Rigid Multi-Agent Systems via Deep Reinforcement Learning and Graph Neural Networks

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Contents

Abstract **vii**
Sommario . . . ix
List of Figures **xi**
List of Tables **xiii**
Notation **xv**
1 Introduction **1**
1.1 Motivation . . . 1
1.2 Problem statement and approach . . . 2
1.3 Contributions . . . 3
1.4 Scope . . . 4
1.5 Outline . . . 4
2 Background **5**
2.1 Mathematical preliminaries . . . 5
2.1.1 Projections and pseudo-inverses . . . 5
2.1.2 Eigenvalues . . . 6
2.1.3 Rotations . . . 6
2.2 Graphs . . . 7
2.3 Formations and agent domains . . . 8
2.4 Rigidity theory . . . 9
2.4.1 Bearing rigidity . . . 9
2.4.2 Unified framework . . . 10
2.4.3 Minimal bearing rigidity . . . 11
2.4.4 Rigidity eigenvalue . . . 11
2.5 Reinforcement learning . . . 12
2.5.1 Markov decision processes . . . 12
2.5.2 Value-based methods . . . 13
2.5.3 Policy gradient methods . . . 13
2.5.4 Potential-based reward shaping . . . 14
2.5.5 Action masking . . . 14
2.6 Graph neural networks . . . 14
2.7 Combinatorial optimization . . . 15
2.8 Estimation and experiment design . . . 17
2.8.1 Noise on a direction . . . 17
2.8.2 Fisher information . . . 17
2.8.3 Nonlinear least squares . . . 19
2.8.4 Covariance summaries . . . 19

2.8.5 Summary statistics . . . 20

3 Related work **21**

3.1 Bearing rigidity . . . 21
3.2 Topology design for rigid formations . . . 21
3.3 Reinforcement learning for graph construction . . . 22

4 Rigidity analysis for heterogeneous bearing frameworks **23**

4.1 Bearing differential . . . 23
4.2 Heterogeneous rigidity matrix . . . 24
4.2.1 Matrix requirements . . . 24
4.2.2 Node-indexed matrix . . . 26
4.3 Trivial motions . . . 29
4.4 Single measurement values . . . 30
4.4.1 Per-edge block rank . . . 30
4.4.2 Reciprocal pairs . . . 30
4.4.3 Single-edge quantities . . . 31
4.4.4 Flexes . . . 32
4.5 Edge count bound . . . 33
4.6 Estimation robustness . . . 34
4.6.1 Estimation error bound . . . 34
4.6.2 Units and normalization . . . 36
4.6.3 Choice among minimal graphs . . . 36
4.7 Combinatorial structure of the objectives . . . 37
4.7.1 Edge count . . . 37
4.7.2 Rigidity eigenvalue . . . 38
4.8 Summary . . . 40

5 Methodology **43**

5.1 Approach . . . 43
5.2 Environment design . . . 44
5.2.1 Markov decision process . . . 44
5.2.2 Objective and reward . . . 48
5.2.3 Observation design . . . 52
5.3 Policy . . . 58
5.3.1 GNN backbones . . . 59
5.3.2 Action heads . . . 60
5.4 Training . . . 61
5.4.1 Algorithms and hyperparameters . . . 61
5.4.2 Episodes . . . 62
5.4.3 Scoring . . . 63

6 Experiments and results **67**

6.1 Experimental setup . . . 67
6.1.1 Compared methods . . . 67
6.1.2 Objective and measurements . . . 68
6.1.3 Evaluation sets . . . 69
6.2 Reference comparison . . . 70
6.3 Network size . . . 71
6.3.1 Training weights . . . 72

6.4 Domain composition . . . 74
6.5 Conditioning trade-off . . . 79
6.6 Computational cost . . . 82
6.7 Resultant topologies . . . 84
6.8 Formation control . . . 89

7 Conclusions and future work **93**
7.1 Conclusions . . . 93
7.2 Future work . . . 95

A Further analysis **97**
A.1 Trivial motions . . . 97
A.2 Attitude observability . . . 99
A.3 Error validation . . . 99

B Further results **101**
B.1 Training behavior . . . 101
B.2 Training size transfer . . . 101
B.3 Configuration comparisons . . . 101
B.3.1 Algorithm and backbone . . . 101
B.3.2 Action space . . . 101
B.4 Search budget . . . 105
B.5 Selection weight . . . 105
B.6 Stopping policies . . . 105
B.7 Decision quality . . . 106
B.8 Observation channels . . . 108
B.9 Noise validation . . . 108
B.10 Control trajectories . . . 113
B.11 Geometric details . . . 114
B.12 Position trajectories . . . 116

Bibliography **119**

Acknowledgments **125**

Abstract

Multi-agent systems play important roles in various fields, and with the advancements in bearing-only sensors (e.g. cameras), formation control and localization relying on bearing measurements have attracted considerable attention in recent years. Bearing rigidity theory provides the mathematical tools for analysis and development of methods for such systems and networks, and the main tool it provides is infinitesimal bearing rigidity, which allows the recoverability of the formation shape from only the inter-agent bearing measurements. In a non-ideal scenario, which is the case in real-life applications such as robotic swarms or sensor networks, these bearings are subject to measurement noise, and recovering the formation shape (which is useful for various purposes, such as control or localization) requires more bearing measurements to be made for better estimates, though usually these measurements are associated with a sensing cost.

This thesis studies topology optimization of the underlying network of heterogeneous bearing-based multi-agent systems, for achieving rigid formations with consideration to the sensing costs and estimation quality of the formation shape, using graph neural networks trained with deep reinforcement learning. It includes a theoretical analysis based on bearing rigidity theory and combinatorial optimization for heterogeneous bearing-based multi-agent systems, development of the environment, policy architectures, training approach, and a comprehensive set of experiments to analyze the resultant policies and the produced formations, comparing them with classical combinatorial optimization algorithms.

Contributions of this work are as follows: establishment of a node-indexed construction of the extended bearing rigidity matrix for heterogeneous frameworks, an analysis of rigidity objectives from a combinatorial optimization perspective, and development of the methodology for topology optimization with graph neural networks and deep reinforcement learning.

Sommario

I Sistemi Multi-Agente (Multi-Agent Systems) svolgono un ruolo importante in numerosi ambiti e, con i progressi compiuti nei sensori basati esclusivamente sulla misura della direzione (bearing-only), come ad esempio le telecamere, negli ultimi anni hanno ricevuto crescente attenzione il controllo di formazione e la localizzazione basati su misure di bearing. La teoria della bearing rigidity fornisce gli strumenti matematici per l'analisi e lo sviluppo di metodi per tali sistemi e reti, e il principale strumento che mette a disposizione è la Rigidità Infinitesimale di Bearing (Infinitesimal Bearing Rigidity), che consente di determinare la recuperabilità della forma della formazione utilizzando esclusivamente le misure di bearing tra gli agenti. In uno scenario non ideale, come avviene nelle applicazioni reali quali sciami robotici o reti di sensori, tali misure sono soggette a rumore. Il recupero della forma della formazione, utile per diversi scopi quali il controllo o la localizzazione, richiede quindi un numero maggiore di misure di bearing per ottenere stime più accurate, sebbene tali misure siano generalmente associate a un costo di sensing.

Questa tesi studia l'ottimizzazione della topologia della rete sottostante a sistemi multi-agente eterogenei basati su misure di bearing, con l'obiettivo di ottenere formazioni rigide tenendo conto sia dei costi di sensing sia della qualità della stima della forma della formazione, mediante Reti Neurali su Grafi (Graph Neural Networks) addestrate tramite Deep Reinforcement Learning. Il lavoro comprende un'analisi teorica basata sulla teoria della bearing rigidity e sull'ottimizzazione combinatoria per sistemi multi-agente eterogenei basati su bearing, lo sviluppo dell'ambiente, delle architetture delle policy e dell'approccio di addestramento, nonché un insieme completo di esperimenti volti ad analizzare le policy ottenute e le formazioni prodotte, confrontandole con algoritmi classici di ottimizzazione combinatoria.

I contributi di questo lavoro sono i seguenti: l'introduzione di una costruzione indicizzata per nodo della matrice estesa di bearing rigidity per framework eterogenei; un'analisi degli obiettivi di rigidità dal punto di vista dell'ottimizzazione combinatoria; e lo sviluppo della metodologia per l'ottimizzazione della topologia mediante reti neurali su grafi e deep reinforcement learning.

List of Figures

5.1 Action spaces . . . 46

6.1 Paired reference differences . . . 71
6.2 Success across network sizes . . . 72
6.3 Quality across network sizes . . . 73
6.4 Size transfer by training weight . . . 74
6.5 Homogeneous-domain success . . . 75
6.6 Homogeneous-domain quality . . . 76
6.7 Mixed-domain success . . . 77
6.8 Mixed-domain quality . . . 78
6.9 Measurement and conditioning trade-off . . . 80
6.10 Position and attitude conditioning . . . 80
6.11 Paired effects under matched objectives . . . 81
6.12 Search cost across sizes . . . 83
6.13 Results by edit budget . . . 83
6.14 Directed output topologies . . . 84
6.15 Physical formation and topology . . . 85
6.16 Position covariance in the formation . . . 87
6.17 Weakest observable deformation . . . 88
6.18 Control trajectories for one case . . . 90
6.19 Physical control paths . . . 91
6.20 Agent displacement paths . . . 92

B.1 Reference training curves . . . 102
B.2 Exploration schedules . . . 103
B.3 Exploratory training-size transfer . . . 103
B.4 Search budget by training weight . . . 106
B.5 Decision ranks before and after the best graph . . . 107
B.6 Feature perturbations at weight zero . . . 110
B.7 Feature perturbations at high weight . . . 111
B.8 Position-error prediction . . . 112
B.9 Position-error prediction by formation . . . 112
B.10 All control cases . . . 113
B.11 Classical position covariance . . . 114
B.12 Classical weakest modes . . . 115
B.13 Position components over time . . . 116
B.14 Attitude error over time . . . 117

List of Tables

2.1 Agent domains and their projectors . . . 9
2.2 Ranks in homogeneous domains. . . . 11

4.1 Edge-indexed against node-indexed restriction . . . 28
4.2 A rigid three-agent framework the edge-indexed test rejects . . . 28
4.3 Per-domain constants of a homogeneous formation . . . 29
4.4 The out-degree each domain forces on an agent . . . 30
4.5 How the three criteria order the same graphs . . . 35
4.6 What the choice among equally sparse graphs is worth . . . 37
4.7 The quantities derived in this chapter . . . 40

5.1 The action spaces considered . . . 46
5.2 The exchange rate of the unnormalized score . . . 49
5.3 The two forms of the robustness score . . . 51
5.4 The terms of the objective and the reward . . . 53
5.5 The observation channels . . . 54
5.6 Observation invariance under a similarity transform . . . 57
5.7 Information tiers of the observation channels . . . 58
5.8 Parameter counts of the backbones and heads . . . 61
5.9 Training hyperparameters . . . 61
5.10 Initial graph edge count sampling . . . 62
5.11 Quantities used to score an episode . . . 64

6.1 Compared search methods . . . 67
6.2 Principal policy configurations . . . 68
6.3 Size evaluation sets . . . 69
6.4 Reference comparison . . . 70
6.5 Shared subset at $n = 48$ . . . 72
6.6 Stress sample at $n = 96$ . . . 73
6.7 Matched-weight comparison . . . 79
6.8 Reference search cost . . . 82
6.9 Illustrated graph measurements . . . 86
6.10 Aggregate graph structure . . . 86
6.11 Control convergence by case . . . 90

A.1 Measured against predicted position error . . . 100
A.2 Prediction agreement by noise level times error coefficient . . . 100

B.1 Available policy configurations . . . 104
B.2 Algorithm and backbone comparisons . . . 105

B.3 Action-space outcomes 105
B.4 Evaluation-weight sensitivity 108
B.5 Stopping-policy outcomes 109
B.6 Trained observation removal 109

Notation

Graphs, formations, and domains

symbol	meaning
n, m	number of agents, number of directed edges
$\mathcal{G} = (\mathcal{V}, \mathcal{E})$, $\mathcal{K}$	the graph, and the complete digraph on the same vertices
$\mathcal{P} = (\mathcal{V} \times \mathcal{V}) \setminus \Delta$	the $n(n-1)$ ordered pairs, Δ the self-loops
$A \in \{0,1\}^{n \times n}$	adjacency matrix, row i the bearings agent i measures
$\deg_i^{\text{out}}$, $\deg_i^{\text{in}}$	bearings agent i measures, and bearings measured of it
$\mathbf{E}$, $\mathbf{E}_o$, $\bar{\mathbf{E}}$, $\bar{\mathbf{E}}_o$	incidence matrices (2.6) and their lifts by $\otimes I_3$
$\chi = (p, R)$, χ_i	configuration: positions $p_i \in \mathbb{R}^3$, attitudes $R_i \in \mathrm{SO}(3)$; the pose of agent i
$\mathcal{D}_i$, $\mathbb{D}$	agent i's domain, and the set of the five domains (2.10)
$c_i = \dim \mathcal{D}_i$, $c = \sum_i c_i$	degrees of freedom of agent i and of the formation
p_{ij}, $\hat{p}_{ij}$, s_{ij}	$p_j - p_i$, its unit vector, and $\|p_{ij}\|^{-1}$
$b_{ij} = R_i^\top \hat{p}_{ij}$, $b_{\mathcal{G}}$	the bearing i measures of j in its own frame, and the bearing map
α, Q, t	the scale, rotation, and translation of a similarity (2.14)
$L(\chi)$	characteristic length, the formation's RMS radius (4.29)

Linear algebra and rotations

$P(x) = I_d - xx^\top$, $[a]_\times$	orthogonal projector onto $x^\perp$, skew-symmetric matrix
Π	an orthogonal projector, $\Pi = \Pi^\top = \Pi^2$
A^+, s_k	Moore-Penrose pseudo-inverse, singular values in descending order
$\mu_1(A) \le \cdots \le \mu_n(A)$	eigenvalues of a symmetric matrix, ascending
$A \preceq B$	Loewner order, $B - A$ positive semidefinite
ω, $\theta = \|\omega\|$	angular velocity or rotation vector, and rotation angle
$\mathbb{E}$, $\mathbf{1}$	expectation, indicator

Rigidity

S_i, P_i	translational and rotational DOF projectors of agent i, Table 2.1
$\bar{S}$, $\bar{P}$	blkdiag$(S_1, \ldots, S_n)$, blkdiag$(P_1, \ldots, P_n)$
$\mathbf{B}_\mathcal{G} \in \mathbb{R}^{3m \times 6n}$	extended bearing rigidity matrix of $\mathcal{G}$
$\mathbf{B}^\times$, $\mathbf{B}^\bullet$	its edge-indexed (2.15) and node-indexed (4.13) constructions
$\mathbf{U}_{ij}$, $\mathbf{V}_{ij}$	the edge-indexed restriction factors of the unified framework
$\mathbf{D}_p$, $\mathbf{D}_a$	position and attitude blocks (2.16)
$\mathbf{B}_s$	the length-normalized matrix (4.29)
$\mathcal{A}$, A, $J_\mathcal{G}$	admissible subspace (4.5), an orthonormal basis of it, the Jacobian on it
$\delta\chi$	a variation of the configuration
$\rho_\mathcal{K} = \operatorname{rk} \mathbf{B}_\mathcal{K}$	the rank a rigid framework must attain
q_v, q_i	virtual and infinitesimal-flex dimensions (2.17)
q_t, q_r	dimension of the trivial variations, and of their rotational part (4.14)
c_k, $c_{\max}$	rank of edge k's row block (4.15), and its maximum over $\mathcal{K}$
m_{req}, $f(n, d)$	lower bound on the edges a rigid graph needs (4.25), its closed form in $\mathbb{R}^d$ (2.20)
r_i	least out-degree agent i's own domain forces on it, Table 4.4
Z	an orthonormal basis of $\ker \mathbf{B}_\mathcal{G}$
λ, v	rigidity eigenvalue (2.21), and its eigenvector, the softest mode
$f(S)$, $H(d)$	rank of the edge set S (4.31), harmonic number in (2.45)

One edit quantities

$g_{ij} = \operatorname{rk}(b_{ij} Z)$	the rank adding $i \to j$ would contribute (4.18)
η_{ij}	$\|b_{ij} Z\|_F / \|b_{ij}\|_F$, the share of the block outside the row space
$\zeta_{ij} = \|b_{ij} v\|$, ν_i	conditioning an absent edge would add, and slack of agent i (4.20)
H, ℓ_e	block leverage of an existing edge, and the rank its removal loses (4.21)
ξ_e	fraction of the conditioning an existing edge carries (4.22)
F, f_i	flex space (4.23), and freedom of agent i (4.24)

Estimation and conditioning

J_{est}	a measure of the shape estimation error in (1.1)
σ	bearing noise, a per-axis angle in radians
η, η_{ij}	standard normal noise (2.47), and on the bearing of edge $i \to j$ (A.8)
$\mathcal{I}$	Fisher information
A_{opt}, D_{opt}, E_{opt}	the A-, D-, and E-optimality criteria of $\mathcal{I}$ (§2.8.4)
a_{opt}, d_{opt}, e_{opt}	the same criteria of $\mathbf{B}^\top \mathbf{B}$ (4.28)
w_k	the kth nonzero eigenvalue of $\mathcal{I}$, or of $\mathbf{B}^\top \mathbf{B}$ in Chapter 4
$\varepsilon(\mathcal{G}, \chi)$	shape error, the RMS state error per radian of bearing noise (4.30)
ε_p, ε_a	its position and attitude parts (§6.1)

The decision process

$\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{A}(\cdot), f, r, \gamma, \mu_0)$	the process of Definition 5.1
s, a, $\mathcal{A}(s)$	state, action, and the actions admissible at s
f, r, γ, μ_0	transition, reward, discount, initial distribution
T, G_t	final step of an episode, and the return from step t (2.23)
sel	pointer register, the node held between two picks
φ, κ	state score (potential), and the conditioning weight in it (5.8)
κ_t, κ_e	training and evaluation weights
w_{rank}, w_{edge}	the two weights of φ (5.4)
q	robustness score in $(0, 1)$, Table 5.3
g, g_{ref}, ς_g	spectral criterion, its per-episode reference and width, Table 5.3
c_ε, ς_ε	fixed center and width of the shape-error form, Table 5.3
m_0, $T_{\max}$	initial edge count (5.19), horizon (5.20)
s^*, t^*	best visited state, and the step it was first reached (5.21)
$\Delta\varphi$	best-final gap, $\varphi(s^*) - \varphi(s_T)$ (5.22)

Policy and learning

o, X, C, $\mathcal{Z}$	observation (5.11), its node, coordinate, and pair blocks
$h_i^{(l)}$, H	node embedding at layer l, and the backbone width
d_v, d_e	node and pair feature widths
q_a, $\tilde{q}_a$	action value, and its masked form (5.2)
π, V^π, Q^π, A^π	policy, state value, action value, and advantage
θ, θ^-	network parameters, and those of the target network
ε	exploration rate of ε-greedy
ς	generalized-advantage parameter (2.31)
ϱ_t, ϵ_{c}	probability ratio and clipping parameter of PPO (2.32)
τ	Polyak rate of the target network

Formation control

$\chi^\star$, $b^\star_{\mathcal{G}}$	goal configuration, and its bearings over the selected edges
u, k_c	velocity command and controller gain (6.3)
$e_{\mathcal{G}}$	bearing error, $b_{\mathcal{G}}(\chi) - b^\star_{\mathcal{G}}$

1 Introduction

1.1 Motivation

Multi-agent systems, teams of autonomous ground robots and aerial vehicles as examples, often need to keep a formation, and observe their own formation shape, without an external positioning system. It becomes necessary to measure the geometry of the formation in such a scenario. Cameras are widely used for this purpose since they are low-cost, lightweight, and low-power. They are bearing only sensors, meaning they are used to measure *directions* between agents, so the direction to a detected neighbor can be computed directly from its pixel coordinates while its distance needs additional information to be estimated [1]. This is a source of motivation for formation control and localization methods that rely on bearing measurements alone.

Bearing rigidity theory studies how the set of bearing measurements changes as a result of agents' movements (variations), and whether the measurements are enough to determine the shape of a formation. A bearing does not change when two agents move along the line joining them, so a formation whose agents measure too few of each other can deform without any measurement changing, and its shape can then be neither recovered nor controlled. The formation is bearing rigid when its measurements allow the formation shape to be determined, and this allows for e.g. stabilization controllers or localization techniques to be built upon. Whether a formation is rigid depends on which agents measure which, so the sensing graph plays a crucial role in this field.

The cost associated with each measurement, such as processing of the camera images, is another challenge in real life systems, and it is desirable to have as few measurements as possible in a bearing based multi-agent system. A system with too few measurements is not rigid, and with too many measurements comes unnecessary sensing and computation costs. Furthermore, rigid formations with the same number of measurements are not of the same quality, in terms of how well the formation shape can be estimated from noisy bearing measurements. A method that chooses the sensing graph therefore has to account for both the number of measurements and the accuracy they provide.

The agents of a formation are also not necessarily of the same type in many applications. A ground robot modeled as a unicycle moves in a plane and rotates about the vertical axis, a quadrotor is often modeled as moving in three dimensions with a single heading angle, and

a fully actuated aerial vehicle can translate and rotate in every direction. Agents with such different motion capabilities may be part of the same formation. The motions the agents can perform play an important role in bearing rigidity, as each bearing measurement can be thought of as constraining the motion degrees of freedom (DOFs) of agents, therefore allowing the shape of the formation to be ”fixed” if there are enough measurements. Hence, it is desirable to have methods that can work on any type of mixture of agents. In various applications, an agent can leave the formation or a link can fail, and the remaining agents then have to be able to adapt to these types of changes, and the sensing topology needs to be adapted as well. This further supports the need to have methods that work on various network sizes and agent mixtures.

With these motivating factors, this work focuses on the sensing topology of multi-agent systems, with the goal of constructing bearing rigid frameworks, with consideration to the number of measurements, and the quality of the estimation of the formation shape via bearing rigidity theoretic tooling. It uses a learning based approach, in particular reinforcement learning, and tries to solve the combinatorial optimization problem of choosing the edge set of a directed graph representing the multi-agent system. Its purpose is to study optimization of the sensing topology for bearing rigidity, rather than providing a method ready for deployment that solves the problem for every single scenario.

1.2 Problem statement and approach

The problem studied in this work is choosing which directed bearing measurements a formation of heterogeneous agents should have, in order for the formation to be bearing rigid with few measurements, and for its shape to be recoverable accurately from noisy bearings. The problem can be stated as:

> *given n agents at poses χ with domains $\mathcal{D}_1, \dots, \mathcal{D}_n$, choose the directed edge set $\mathcal{E}$ so that $(\mathcal{G}, \chi)$ is infinitesimally bearing rigid, using as few edges as possible, and allowing the shape to be recovered accurately under bearing noise.*

More formally, it can be stated as a multi-objective optimization problem. Let $J_{\text{est}}(\mathcal{G}, \chi)$ denote a measure of the error with which the formation shape can be recovered from noisy bearing measurements. Then the topology optimization problem considered in this work is

$$\min_{\mathcal{E} \subseteq (\mathcal{V}\times\mathcal{V})\setminus\Delta} \left(|\mathcal{E}|, J_{\text{est}}(\mathcal{G}, \chi)\right) \qquad \text{subject to} \qquad \operatorname{rk} \mathbf{B}_{\mathcal{G}}(\chi) = \operatorname{rk} \mathbf{B}_{\mathcal{K}}(\chi), \tag{1.1}$$

where the two objectives represent sensing sparsity and estimation robustness respectively, and these can be viewed as competing objectives. Δ is the set of self-loops, since an agent does not measure a bearing of itself, and $\mathcal{K}$ is the complete graph on the same poses. The constraint $\operatorname{rk} \mathbf{B}_{\mathcal{G}}(\chi) = \operatorname{rk} \mathbf{B}_{\mathcal{K}}(\chi)$ is the rank test for infinitesimal bearing rigidity. Thus, the problem is to select a rigid sensing topology that trades off the number of retained measurements against the accuracy with which the formation shape can be recovered under bearing noise.

In Chapter 5, a scalar objective function is formulated to solve this problem using reinforcement learning. The sensing topology of a given formation is optimized through a sequence of single edge additions and removals, with each edit selected by a graph neural network based policy. A sequential formulation allows each decision to depend on the graph produced by the earlier ones, which is useful when the value of one measurement depends on which measurements are already present.

Reinforcement learning is useful for this type of problem when good graph modifications cannot be identified from their immediate effect alone. A classical local search evaluates candidate edits from the current graph, whereas a learned policy can assign value to an edit through the states that it is expected to lead to after several further decisions. These are also reasons given by [2] for using reinforcement learning on non-canonical graph optimization problems. Another important point is that if optimal solutions can already be constructed through a sequence of locally good decisions, then modeling a longer decision horizon offers little advantage over a classical method in such problems. These points are relevant to the problem at hand, as some parts of the topology optimization objective have enough structure that simple search methods are already strong, while the estimation robustness part depends more strongly on combinations of measurements and can make the value of an edit depend on later edits. Chapter 4 makes this distinction through the combinatorial properties of the rigidity and conditioning objectives.

1.3 Contributions

The bearing rigidity matrix is corrected for the heterogeneous setting, building upon the unified framework of [3]. The rigidity model is formulated so that agents with different translational and rotational degrees of freedom can be treated within the same framework, ensuring that the rank test for infinitesimal bearing rigidity is well-defined, and properties/quantities relevant to the construction of sparse rigid sensing graphs are derived.

The structure of the sparsity and estimation objectives is studied, showing that the two parts of the topology optimization problem have different properties from a combinatorial search perspective.

The main methodological contribution is a learning based formulation of topology optimization as sequential graph editing, for the bearing rigidity problem in particular. A reinforcement learning policy based on graph neural networks is trained to modify the directed bearing topology while balancing rigidity, sparsity, and estimation robustness. The formulation is made so that the trained policy is able to generalize across network sizes and heterogeneous agent compositions.

To the best of the author's knowledge, the literature around bearing rigidity theory focuses more on formation control and localization, and there are limited works regarding the sensing graph of bearing rigid multi-agent systems (see Chapter 3). Existing work on network topology does not cover higher-level agent domains (§2.3), and heterogeneous settings. This thesis considers heterogeneous networks and common domain groups, and presents a first study of a

learning based approach to topology optimization in bearing rigidity.

1.4 Scope

There are some (strong) assumptions in the formulation used throughout this work. These are mainly concerns about centralized access of the trained policy to information. The policy is centralized and observes every pose, the whole edge set, and quantities computed from the rigidity matrix of the whole formation, several of which no single agent could compute. This means that the method developed here is not suitable for a distributed deployment scenario, and that would require a whole new formulation. §5.2.3.4 presents ”tiers” for the information the trained policy has access to, categorizing them based on the assumptions surrounding them.

The formation is assumed to be static, meaning the agents do not move during the optimization process. Agent configurations are assumed to be generic, which excludes a set of measure zero of arrangements, such as collinear formations, where the bearing rigidity theory and the tools it presents fail, for purposes of this work. The bearing measurements between agents are exact, and the uncertainty is only added for evaluation purposes.

During training, the network sizes are kept fixed, with the number of agents up to 20, due to available resources, technical limitations of the implementation, and used libraries. The scope of the networks considered for experimentation is also limited, with network size up to 96 agents.

1.5 Outline

Chapter 2 covers preliminary background knowledge, and Chapter 3 reviews the related literature. Chapter 4 develops the rigidity analysis for heterogeneous formations, covering the node-indexed rigidity matrix, the lower bound on the edge count, the measures of conditioning, and the structure of the two objectives. Chapter 5 poses topology design as a Markov decision process and describes the environment, the policy, and its training. Chapter 6 compares the learned policies with the classical search methods, and Chapter 7 concludes the thesis. The appendices contain derivations and secondary experiments/results.

2 Background

This chapter aims to inform the reader of the necessary background knowledge in various fields, to allow for easier understanding of the explanations in later chapters. This chapter is put together as a concise and dense summary of the required mathematical tools and concepts, without going into specific details or explaining how and why the material provided here is used. The reader is assumed to have a basic understanding of linear algebra, graph theory, machine learning, reinforcement learning, graph neural networks, and statistical estimation, as the information provided here is not comprehensive and covers only the parts utilized in this thesis. Bearing rigidity theory is covered more extensively than others, though the reader is referred to mainly [3] and [4] for further information.

2.1 Mathematical preliminaries

2.1.1 Projections and pseudo-inverses

An orthogonal projector is a matrix Π with $\Pi = \Pi^\top = \Pi^2$, and its eigenvalues are 0 and 1. For a unit vector $x \in \mathbb{R}^d$,

$$P(x) = I_d - xx^\top \tag{2.1}$$

is the orthogonal projector onto the hyperplane orthogonal to x. $P(x)x = 0$ and $\operatorname{rk} P(x) = d-1$.

For $A \in \mathbb{R}^{m\times n}$, the row space and the kernel are orthogonal complements in $\mathbb{R}^n$, and the column space and the kernel of $A^\top$ are orthogonal complements in $\mathbb{R}^m$,

$$\operatorname{row}(A)^\perp = \ker A, \qquad \operatorname{col}(A)^\perp = \ker A^\top. \tag{2.2}$$

Appending rows to A raises its rank if and only if one of the new rows is not orthogonal to $\ker A$. The rank of a matrix stacked from row blocks $A_1, \ldots, A_m$ is at most the sum of the ranks of the blocks,

$$\operatorname{rk}\big[A_1;\ \ldots;\ A_m\big] \ \le\ \sum_{k=1}^{m} \operatorname{rk} A_k. \tag{2.3}$$

The singular value decomposition $A = U\Sigma V^\top$ has orthogonal $U \in \mathbb{R}^{m\times m}$ and $V \in \mathbb{R}^{n\times n}$, and the singular values $s_1 \ge s_2 \ge \cdots \ge 0$ on the diagonal of $\Sigma \in \mathbb{R}^{m\times n}$. The rank of A is the

number of nonzero singular values, the last $n - \operatorname{rk} A$ columns of V span $\ker A$, and the nonzero eigenvalues of $A^\top A$ are the squares s_k^2 of the nonzero singular values.

The Moore-Penrose pseudo-inverse is $A^+ = V\Sigma^+ U^\top$, where $\Sigma^+ \in \mathbb{R}^{n\times m}$ is the transpose of Σ with every nonzero singular value inverted. For any $y \in \mathbb{R}^m$, the vector $x = A^+ y$ is the minimizer of $\|Ax - y\|$ with the smallest norm, and it is orthogonal to $\ker A$. The products A^+A and AA^+ are the orthogonal projectors onto $\operatorname{row}(A)$ and $\operatorname{col}(A)$. For a symmetric positive semidefinite matrix $M = \sum_k w_k v_k v_k^\top$ with orthonormal eigenvectors v_k,

$$M^+ = \sum_{w_k>0} \frac{1}{w_k} v_k v_k^\top, \qquad \operatorname{tr} M^+ = \sum_{w_k>0} \frac{1}{w_k}. \tag{2.4}$$

A symmetric matrix M is positive semidefinite, written $M \succeq 0$, when $x^\top Mx \geq 0$ for every x. For symmetric matrices A and B, the Loewner order $A \preceq B$ means $B - A \succeq 0$.

2.1.2 Eigenvalues

Let the eigenvalues of a symmetric matrix $A \in \mathbb{R}^{n\times n}$ be written in ascending order, $\mu_1(A) \leq \mu_2(A) \leq \cdots \leq \mu_n(A)$.

Lemma 2.1 (Weyl monotonicity)**.** *If $A \preceq B$, then $\mu_k(A) \leq \mu_k(B)$ for every k. In particular, adding a positive semidefinite matrix to A lowers none of its eigenvalues.*

Lemma 2.2 (derivative of a simple eigenvalue)**.** *Let A be symmetric and let $\mu_k(A)$ be a simple eigenvalue with unit eigenvector v. For a symmetric E, the function $t \mapsto \mu_k(A + tE)$ is differentiable at $t = 0$ and*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \mu_k(A + tE) = v^\top Ev. \tag{2.5}$$

2.1.3 Rotations

For $a = [a_1, a_2, a_3]^\top \in \mathbb{R}^3$, let

$$[a]_\times = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}$$

denote the skew-symmetric matrix satisfying $[a]_\times b = a \times b$ for all $b \in \mathbb{R}^3$. The rotation group $\mathrm{SO}(3)$ is the set of matrices $R \in \mathbb{R}^{3\times 3}$ with $R^\top R = I_3$ and $\det R = 1$. A rotation that varies in time satisfies $\dot{R} = [\omega]_\times R$, where ω is the angular velocity expressed in the world frame, and R describes the rotation of a body w.r.t. the world frame.

The exponential map provides a local parametrization of rotations around the identity, from $\mathfrak{so}(3)$ to $\mathrm{SO}(3)$. For a vector $\omega \in \mathbb{R}^3$, the associated skew-symmetric matrix $[\omega]_\times \in \mathfrak{so}(3)$ is mapped to

$$R = \exp([\omega]_\times) \in \mathrm{SO}(3).$$

The vector ω is a rotation vector with its direction defining the rotation axis and its norm $\|\omega\|$ the rotation angle. For $\omega \neq 0$, $\exp([\omega]_\times)$ represents a rotation of angle $\|\omega\|$ about the axis $\omega/\|\omega\|$, while $\omega = 0$ gives the identity rotation.

The logarithmic map provides the inverse local parametrization, mapping rotations in $\mathrm{SO}(3)$ back to the Lie algebra $\mathfrak{so}(3)$. For a rotation matrix R,

$$[\omega]_\times = \log(R),$$

and the corresponding rotation vector is recovered from the skew-symmetric matrix $[\omega]_\times$. For a rotation of angle $\theta \in (0, \pi)$,

$$\theta = \arccos\left(\frac{\mathrm{tr}(R) - 1}{2}\right), \qquad \log(R) = \frac{\theta}{2\sin\theta}(R - R^\top).$$

2.2 Graphs

A directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has a vertex set $\mathcal{V} = \{v_1, \ldots, v_n\}$ and an edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ of ordered pairs, and $(v_i, v_j) \in \mathcal{E}$ does not imply $(v_j, v_i) \in \mathcal{E}$. In this work, an edge (v_i, v_j), also written $i \to j$, means that agent i in a multi-agent system measures the bearing (a unit vector describing the direction) of agent j. The adjacency matrix $A \in \{0,1\}^{n\times n}$ has $A_{ij} = 1$ exactly when $(v_i, v_j) \in \mathcal{E}$, and $m = |\mathcal{E}|$ is the number of edges. The out-degree $\deg_i^{\text{out}} = \sum_j A_{ij}$ counts the bearings agent i measures and the in-degree $\deg_i^{\text{in}} = \sum_j A_{ji}$ the bearings measured of it. The complete digraph $\mathcal{K}$ on $\mathcal{V}$ contains all $n(n-1)$ ordered pairs of distinct vertices, excluding self-loops.

With the edges ordered as $e_1, \ldots, e_m$, the incidence matrices of $\mathcal{G}$ are the $n \times m$ matrices

$$[\mathbf{E}]_{ik} = \begin{cases} -1, & e_k = (v_i, v_j) \text{ for some } j, \\ +1, & e_k = (v_j, v_i) \text{ for some } j, \\ 0, & \text{otherwise,} \end{cases} \quad [\mathbf{E}_o]_{ik} = \begin{cases} -1, & e_k = (v_i, v_j) \text{ for some } j, \\ 0, & \text{otherwise.} \end{cases} \tag{2.6}$$

When every vertex carries a vector in $\mathbb{R}^3$, the matrices are lifted by the Kronecker product,

$$A \otimes I_3 = \begin{bmatrix} a_{11}I_3 & \cdots & a_{1m}I_3 \\ \vdots & \ddots & \vdots \\ a_{n1}I_3 & \cdots & a_{nm}I_3 \end{bmatrix} \in \mathbb{R}^{3n\times 3m},$$

which replaces every entry a_{ik} of $A \in \mathbb{R}^{n\times m}$ by the block $a_{ik}I_3$, giving $\bar{\mathbf{E}} = \mathbf{E} \otimes I_3$ and

$\bar{\mathbf{E}}_o = \mathbf{E}_o \otimes I_3$. For $e_k = (v_i, v_j)$, the transposes of the incidence matrices have the row blocks

$$\left[\bar{\mathbf{E}}^\top\right]_{k,\cdot} = \left[\, \mathbf{0} \cdots \underbrace{-I_3}_{i} \cdots \underbrace{+I_3}_{j} \cdots \mathbf{0} \,\right], \tag{2.7}$$

$$\left[\bar{\mathbf{E}}_o^\top\right]_{k,\cdot} = \left[\, \mathbf{0} \cdots \underbrace{-I_3}_{i} \cdots \underbrace{\mathbf{0}}_{j} \cdots \mathbf{0} \,\right]. \tag{2.8}$$

The reciprocity of an ordered pair (and edge) is $A_{ij}A_{ji}$, which equals one when both edges between the two nodes (agents) are present. The number of two-step paths (common neighbors) from i to j is

$$[A^2]_{ij} = \sum_k A_{ik}A_{kj},$$

which counts the nodes (agents) k such that $i \to k$ and $k \to j$. With $d(i,j)$ the length of the shortest directed path from i to j, and $\mathcal{R}_i$ the set of agents reachable from i, the closeness centrality of agent i is

$$C_i = \frac{|\mathcal{R}_i|}{\sum_{j\in\mathcal{R}_i} d(i,j)} \cdot \frac{|\mathcal{R}_i|}{n-1}, \tag{2.9}$$

which reduces to $(n-1)/\sum_j d(i,j)$ when every agent is reachable. The betweenness centrality of agent i is $\sum_{s\neq i\neq t} \sigma_{st}(i)/\sigma_{st}$, where σ_{st} is the number of shortest directed paths from s to t and $\sigma_{st}(i)$ the number of those passing through i. The eigenvector centrality (or prestige score) x is a normalized dominant eigenvector of $A^\top$, satisfying $A^\top x = \lambda_{\max}(A^\top)x$ (computed with the power iteration $x \leftarrow A^\top x/\|A^\top x\|$ in this work).

2.3 Formations and agent domains

A formation is a set of n agents together with their poses and domains. Agent i has a position $p_i \in \mathbb{R}^3$ and an attitude $R_i \in \mathrm{SO}(3)$, called together its pose $\chi_i = (p_i, R_i)$. The configuration of the formation is denoted by $\chi = (\chi_1, \ldots, \chi_n)$.

The domain of agent i, $\mathcal{D}_i$, specifies the translational and rotational degrees of freedom of its pose. Five domains are considered in this work,

$$\mathcal{D}_i \in \left\{\mathbb{R}^2,\ \mathbb{R}^3,\ \mathbb{R}^2 \times S^1,\ \mathbb{R}^3 \times S^1,\ \mathrm{SE}(3)\right\}, \qquad c_i = \dim \mathcal{D}_i \in \{2,3,3,4,6\}, \tag{2.10}$$

where c_i is the number of degrees of freedom of agent i. An agent in $\mathbb{R}^2$ is a point that moves in a plane, and an agent in $\mathbb{R}^2 \times S^1$ is the same type of point but with a heading direction, e.g. the unicycle model of a ground vehicle. An agent in $\mathbb{R}^3$ is a point that moves freely in 3D space. An agent in $\mathbb{R}^3 \times S^1$ moves freely in space and has a single heading angle about a fixed axis, as in the common model of a quadrotor, and generally the axis of rotation aligns with the vertical axis (z). An agent in $\mathrm{SE}(3)$ translates and rotates in every direction, e.g. a fully actuated aerial vehicle. The agents in $\mathbb{R}^2$ and $\mathbb{R}^3$ have no attitude of their own, and in this work they measure bearings in the world frame (denoted as $R_i = I_3$ for completeness), whereas the others measure bearings in their own frames. The domain of agent i is encoded by two orthogonal projectors

in world coordinates, S_i onto the directions the agent can translate along and P_i onto the axes it can rotate about, listed in Table 2.1.

Table 2.1: Considered agent domains in this work. S_i projects onto the translational directions of agent i and P_i onto its rotation axes, both in world coordinates, and $c_i = \text{rk}\, S_i + \text{rk}\, P_i$. v is the rotation axis of an $\mathbb{R}^3 \times S^1$ agent and e_3 the vertical axis.

$\mathcal{D}_i$	$\mathbb{R}^2$	$\mathbb{R}^3$	$\mathbb{R}^2 \times S^1$	$\mathbb{R}^3 \times S^1$	SE(3)
S_i	$\text{diag}(1,1,0)$	I_3	$\text{diag}(1,1,0)$	I_3	I_3
P_i	0	0	$e_3 e_3^\top$	$vv^\top$	I_3
c_i	2	3	3	4	6

2.4 Rigidity theory

The rigidity theory formulation used in this work mostly follows directly [3], with certain changes discussed in Chapter 4.

2.4.1 Bearing rigidity

The multi-agent system in bearing rigidity theory is modeled as a *framework*. A framework $(\mathcal{G}, \chi)$ as used here consists of a formation configuration χ together with a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, whose vertices represent the agents and edges represent bearing measurements.

In a framework, an edge (v_i, v_j) carries the direction from agent i to agent j, measured in the frame of agent i,

$$b_{ij} = R_i^\top \hat{p}_{ij}, \qquad \hat{p}_{ij} = \frac{p_j - p_i}{\|p_j - p_i\|}, \qquad s_{ij} = \|p_j - p_i\|^{-1}, \tag{2.11}$$

with $R_i = I_3$ for the agents in $\mathbb{R}^2$ and $\mathbb{R}^3$.

The *bearing map* stacks the bearings associated with the $m = |\mathcal{E}|$ edges of $\mathcal{G}$,

$$b_\mathcal{G}(\chi) = \begin{bmatrix} b_1^\top & \cdots & b_m^\top \end{bmatrix}^\top \in (S^2)^m, \qquad b_k = b_{ij} \text{ for } e_k = (v_i, v_j). \tag{2.12}$$

The *bearing rigidity matrix* $\mathbf{B}_\mathcal{G}$ is the Jacobian of the bearing map. For an infinitesimal variation $\delta\chi$ of the configuration,

$$\dot{b}_\mathcal{G}(\chi(t)) = \frac{d}{dt} b_\mathcal{G}(\chi(t)) = \mathbf{B}_\mathcal{G}(\chi(t))\delta\chi, \tag{2.13}$$

where the time dependency of the configuration is written for completeness, although this work mainly deals with static configurations of formations.

The trivial variations are variations of the formation configuration that preserve all bearings of the complete graph. They are the infinitesimal similarity transforms of the formation,

$$p_i \longmapsto \alpha Q p_i + t, \qquad R_i \longmapsto Q R_i, \qquad \alpha > 0,\ Q \in \text{SO}(3),\ t \in \mathbb{R}^3, \tag{2.14}$$

restricted to the motions allowed by the agent domains. They consist of common translation and scaling, together with common rotation when permitted by every agent. The rotational part has dimension zero if any agent is frameless, one when the agents share a single rotation axis, and three when all agents are in $\mathrm{SE}(3)$.

Definition 2.3 (infinitesimal bearing rigidity, adapted from [3])**.** $(\mathcal{G}, \chi)$ is infinitesimally bearing rigid (IBR) when every variation that preserves all bearings of $\mathcal{G}$ preserves all bearings of $\mathcal{K}$, that is when $\ker \mathbf{B}_{\mathcal{G}} = \ker \mathbf{B}_{\mathcal{K}}$, or equivalently when $\operatorname{rk} \mathbf{B}_{\mathcal{G}} = \operatorname{rk} \mathbf{B}_{\mathcal{K}} = \rho_{\mathcal{K}}$.

For every graph on the same configuration $\ker \mathbf{B}_{\mathcal{K}} \subseteq \ker \mathbf{B}_{\mathcal{G}}$, and $\ker \mathbf{B}_{\mathcal{K}}$ is the set of trivial variations.

2.4.2 Unified framework

In [3], every lower-dimensional domain is embedded into $\mathrm{SE}(3)$, giving every agent six coordinates and padding unavailable variations with zeros. The extended bearing rigidity matrix (EBRM) is

$$\mathbf{B}_{\mathcal{G}}^{\times} = \begin{bmatrix} \mathbf{D}_p \, \mathbf{U} \, \bar{\mathbf{E}}^{\top} & \mathbf{D}_a \, \mathbf{V} \, \bar{\mathbf{E}}_o^{\top} \end{bmatrix} \in \mathbb{R}^{3m \times 6n}, \qquad \mathbf{U} = \operatorname{blkdiag}(\mathbf{U}_{ij}), \quad \mathbf{V} = \operatorname{blkdiag}(\mathbf{V}_{ij}), \tag{2.15}$$

where

$$\mathbf{D}_p = \operatorname{blkdiag}\big(s_{ij} R_i^{\top} P(\hat{p}_{ij})\big) , \qquad \mathbf{D}_a = -\operatorname{blkdiag}\big(R_i^{\top} [\hat{p}_{ij}]_{\times}\big) . \tag{2.16}$$

$\mathbf{D}_p$ and $\mathbf{D}_a$ are block diagonal over the edges and hold the derivative of each bearing with respect to the positions and to the attitude of the measuring agent, and $\mathbf{U}_{ij}$ and $\mathbf{V}_{ij}$ define the translational and rotational directions of the bearing b_{ij} in the 3D space with respect to the ith agent frame, listed in Table I of [3] for homogeneous formations.

The rank of the EBRM satisfies

$$\operatorname{rk} \mathbf{B}_{\mathcal{G}} = 6n - q_v - q_i, \tag{2.17}$$

where q_v is the dimension of the virtual variations introduced by the embedding and it equals the number of null columns in the rigidity matrix, and q_i is the dimension of the infinitesimal variations that preserve all bearings of $\mathcal{G}$. For the complete graph,

$$\ker \mathbf{B}_{\mathcal{K}} = \mathcal{S}_t \oplus \mathcal{S}_v, \tag{2.18}$$

where $\mathcal{S}_t$ and $\mathcal{S}_v$ are the trivial and virtual variation spaces.

For a homogeneous formation whose agents have c degrees of freedom each, the trivial variations have dimension $c+1$ with c rigid-body motions allowed by the domain and one uniform scaling. Hence, a framework is infinitesimally bearing rigid iff

$$\operatorname{rk} \mathbf{B}_{\mathcal{G}} = cn - c - 1 \tag{2.19}$$

[3, Theorem 3], and the ranks per domain are summarized in Table 2.2.

Table 2.2: Ranks in homogeneous domains.

domain	rigid rank
$\mathbb{R}^2$	$2n-3$
$\mathbb{R}^3$	$3n-4$
$\mathbb{R}^2 \times S^1$	$3n-4$
$\mathbb{R}^3 \times S^1$	$4n-5$
$\mathrm{SE}(3)$	$6n-7$

2.4.3 Minimal bearing rigidity

A generically bearing rigid graph in $\mathbb{R}^d$ is minimal when it has the fewest edges among generically bearing rigid graphs on the same number of vertices [4]. For a homogeneous framework in $\mathbb{R}^d$ with $n \geq 3$ and $d \geq 2$, the minimum number of edges is

$$f(n,d) = 1 + kd + r + \mathbf{1}[r > 0], \qquad k = \left\lfloor \tfrac{n-2}{d-1} \right\rfloor, \quad r = (n-2) \bmod (d-1). \tag{2.20}$$

A graph is minimally bearing rigid in $\mathbb{R}^d$ if and only if it is generically bearing rigid and has $f(n,d)$ edges. For $3 \leq n \leq d+1$, the minimally bearing rigid graphs are exactly the cycles on n vertices [4]. These results apply to undirected frameworks of points in $\mathbb{R}^d$.

2.4.4 Rigidity eigenvalue

The rigidity eigenvalue λ of the bearing rigidity matrix $\mathbf{B}$ is, with the eigenvalues in ascending order,

$$\lambda \;=\; \mu_{6n-\rho_{\mathcal{K}}+1}\big(\mathbf{B}^\top \mathbf{B}\big) \;=\; s_{\rho_{\mathcal{K}}}^2, \tag{2.21}$$

where s_k are the singular values of $\mathbf{B}$ in descending order. On a rigid framework the kernel has dimension $6n - \rho_{\mathcal{K}}$ and λ is the smallest nonzero eigenvalue of $\mathbf{B}^\top \mathbf{B}$, while on a flexible framework $\lambda = 0$. The eigenvalue bounds the change of the bearings under a variation orthogonal to the kernel from below,

$$\|\mathbf{B}\,\delta\chi\| \;\geq\; \sqrt{\lambda}\,\|\delta\chi\| \qquad \text{for every } \delta\chi \perp \ker \mathbf{B}. \tag{2.22}$$

For frameworks of points, the matrix $\mathbf{B}^\top \mathbf{B}$ computed with the distance factors s_{ij} removed from the rows is called the bearing Laplacian. Two interpretations of the rigidity eigenvalue are as follows.

(i) *Estimation.* To first order, an error in the bearings is amplified by at most $1/\sqrt{\lambda}$ in the recovered configuration, measured orthogonally to the kernel.

(ii) *Convergence.* The convergence rate of the network localization dynamics of [4] is dominated by the corresponding eigenvalue of the bearing Laplacian.

The position columns of **B** carry the factors s_{ij}, which have units of inverse length, and the attitude columns are dimensionless. Scaling a framework of points by $\alpha > 0$ scales λ by $1/\alpha^2$.

2.5 Reinforcement learning

In this section a brief introduction to reinforcement learning (RL) is given, focusing on the relevant concepts for this thesis. The reader is referred to [5], [6], [7] for a comprehensive background in RL and the utilized algorithms in this work.

2.5.1 Markov decision processes

In reinforcement learning an agent interacts with an environment by observing a state s_t, taking an action a_t, and receiving a reward r_t and the next state s_{t+1} [5]. A Markov decision process is $(\mathcal{S}, \mathcal{A}, p, r, \gamma)$, where $\mathcal{S}$ and $\mathcal{A}$ are the state and action sets, $p(s' \mid s, a)$ is the transition probability, $r(s, a)$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. The Markov property means that the distribution of the next state and reward depends on the past only through the current state and action.

An *episode* is a finite sequence of interactions indexed by $t = 0, \dots, T$, where T is the final time step. The *return* following step t is

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k r_{t+k}. \tag{2.23}$$

A *policy* $\pi(a \mid s)$ gives the probability of taking action a in state s. The state value and action value functions are

$$V^\pi(s) = \mathbb{E}_\pi\big[G_t \mid s_t = s\big], \qquad Q^\pi(s, a) = \mathbb{E}_\pi\big[G_t \mid s_t = s,\ a_t = a\big], \tag{2.24}$$

and the advantage is $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$. The optimal action value satisfies

$$Q^*(s, a) = r(s, a) + \gamma \sum_{s'} p(s' \mid s, a) \max_{a'} Q^*(s', a'). \tag{2.25}$$

An on-policy method learns from data generated (via interaction with the environment) by the policy being optimized, while an off-policy method can learn from data generated by a different policy. The ε-greedy rule selects actions according to

$$a = \begin{cases} \text{a random action}, & \text{with probability } \varepsilon, \\ \arg\max\limits_{a' \in \mathcal{A}} Q(s, a'), & \text{with probability } 1 - \varepsilon, \end{cases} \tag{2.26}$$

allowing for exploration of states and actions not selected by a greedy policy which only selects the best action according to the value functions.

2.5.2 Value-based methods

Temporal-difference methods update a value estimate using the reward and the estimated value of the next state. Q-learning applies

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\Big], \tag{2.27}$$

where α is the step size. Q-learning is off-policy [5].

A deep Q-network (DQN) represents the action value by a neural network Q_θ and trains on minibatches sampled from a replay memory [6]. A target network with parameters θ^- is used to compute the target,

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{B}}\Big[\big(y - Q_\theta(s,a)\big)^2\Big], \qquad y = \begin{cases} r, & s' \text{ terminal}, \\ r + \gamma \max_{a'} Q_{\theta^-}(s',a'), & \text{otherwise}, \end{cases} \tag{2.28}$$

[8]. The target parameters can be updated by

$$\theta^- \leftarrow \tau\theta + (1-\tau)\theta^-, \qquad 0 < \tau \ll 1 \tag{2.29}$$

[9]. Exploration is commonly performed with an ε-greedy policy.

2.5.3 Policy gradient methods

Policy gradient methods represent the policy by π_θ. For a discrete action space, a network can output logits l_a and define $\pi_\theta(a \mid s) = \exp(l_a)/\sum_b \exp(l_b)$. A policy-gradient estimator is

$$\hat{g} = \hat{\mathbb{E}}_t\Big[\nabla_\theta \log \pi_\theta(a_t \mid s_t)\, \hat{A}_t\Big], \tag{2.30}$$

where $\hat{A}_t$ estimates the advantage [7].

In an actor-critic method, the policy is the actor and a value network V is the critic. Generalized advantage estimation (GAE) uses the temporal-difference residuals $\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t)$,

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\varsigma)^l \delta_{t+l}, \qquad \varsigma \in [0,1], \tag{2.31}$$

where ς is conventionally denoted by λ, but it is renamed here to distinguish it from the rigidity eigenvalue [10].

Proximal policy optimization (PPO) uses the ratio $\varrho_t(\theta) = \pi_\theta(a_t \mid s_t)/\pi_{\theta_{\text{old}}}(a_t \mid s_t)$ and maximizes the clipped surrogate objective with an entropy bonus,

$$\mathcal{L}(\theta) = \hat{\mathbb{E}}_t\Big[\min\big(\varrho_t(\theta)\hat{A}_t,\ \mathrm{clip}\big(\varrho_t(\theta), 1-\epsilon_{\mathrm{c}}, 1+\epsilon_{\mathrm{c}}\big)\hat{A}_t\big) + c_2\, \mathcal{H}\big[\pi_\theta(\cdot \mid s_t)\big]\Big], \tag{2.32}$$

where ϵ_{c} is the clipping parameter and $\mathcal{H}$ is the policy entropy with coefficient c_2 [7]. PPO is

on-policy and trains on batches collected with the current policy.

2.5.4 Potential-based reward shaping

For a potential function $\Phi : \mathcal{S} \to \mathbb{R}$, a potential-based shaping term is

$$F(s, s') = \gamma\Phi(s') - \Phi(s). \tag{2.33}$$

Its discounted sum over an episode telescopes to

$$\sum_{t=0}^{T-1} \gamma^t \big(\gamma\Phi(s_{t+1}) - \Phi(s_t)\big) = \gamma^T\Phi(s_T) - \Phi(s_0). \tag{2.34}$$

2.5.5 Action masking

Invalid actions can be excluded by masking their policy logits or action values,

$$\tilde{z}_a = \begin{cases} z_a, & a \text{ valid}, \\ -\infty, & a \text{ invalid}, \end{cases} \tag{2.35}$$

where $z_a = l_a$ for a policy-gradient method and $z_a = Q(s, a)$ for a value-based method. Masking before the softmax gives invalid actions zero probability [11], while masking before the maximum excludes them from value-based action selection. Random actions used for ε-greedy exploration are sampled from the valid action set in this work, by applying the same action masking technique.

2.6 Graph neural networks

A graph neural network computes a representation of every vertex of a graph from node features, optionally edge features, and the graph structure. With the node features stacked in $X \in \mathbb{R}^{n \times d}$ and adjacency matrix A, a model F is permutation equivariant if

$$F(\Omega X, \Omega A \Omega^\top) = \Omega F(X, A) \tag{2.36}$$

for every permutation matrix Ω.

Many graph neural networks can be written as message passing networks [12]. In each layer, node i receives messages from its neighbors, aggregates them with a permutation-invariant operation, and updates its representation,

$$h_i^{(l+1)} = \phi^{(l)}\Big(h_i^{(l)}, \bigoplus_{j \in \mathcal{N}(i)} \psi^{(l)}\big(h_i^{(l)}, h_j^{(l)}, e_{ij}\big)\Big), \tag{2.37}$$

where $h_i^{(l)}$ is the representation of node i at layer l, $\mathcal{N}(i)$ is its neighborhood, e_{ij} contains the

features of edge (i, j), $\psi^{(l)}$ is the message function, $\bigoplus$ is a permutation-invariant aggregation operator such as sum or mean, and $\phi^{(l)}$ is the node update function. The functions $\phi^{(l)}$ and $\psi^{(l)}$ are shared by all vertices.

A graph representation can be obtained with a readout such as a sum or mean over the resultant node embeddings [12]. Sum aggregation preserves multiplicity information that mean aggregation can discard [13], while mean aggregation does not grow with the number of aggregated terms.

GINE. The graph isomorphism network (GIN) uses sum aggregation and an MLP update,

$$h_i^{(l+1)} = \mathrm{MLP}^{(l)}\left((1+\epsilon^{(l)})h_i^{(l)} + \sum_{j\in\mathcal{N}(i)} h_j^{(l)}\right), \tag{2.38}$$

with a fixed or learnable scalar $\epsilon^{(l)}$ [13]. GIN can be extended to edge features [14], and the edge-feature variant used in this work, GINE, incorporates the edge features into the neighbor messages,

$$h_i^{(l+1)} = \mathrm{MLP}^{(l)}\left((1+\epsilon^{(l)})h_i^{(l)} + \sum_{j\in\mathcal{N}(i)} \mathrm{ReLU}\big(h_j^{(l)} + W^{(l)}e_{ij}\big)\right), \tag{2.39}$$

where $W^{(l)}$ maps the edge features to the width of $h^{(l)}$.

EGNN. The $E(n)$-equivariant graph neural network (EGNN) of [15] associates each vertex with features h_i and coordinates $x_i \in \mathbb{R}^3$. A layer computes

$$m_{ij} = \phi_e\big(h_i, h_j, \|x_i - x_j\|^2, e_{ij}\big), \tag{2.40}$$

$$m_i = \sum_{j\neq i} m_{ij}, \qquad h_i' = \phi_h(h_i, m_i), \tag{2.41}$$

$$x_i' = x_i + C\sum_{j\neq i}(x_i - x_j)\phi_x(m_{ij}), \tag{2.42}$$

with learned functions ϕ_e, ϕ_h, and ϕ_x, and the constant $C = 1/(n-1)$. The messages and node representations are invariant to translations, rotations, and reflections of the coordinates, given that the inputs are invariant scalars [15].

2.7 Combinatorial optimization

Choosing the edges of a directed graph on n agents amounts to selecting a subset of the $n(n-1)$ possible ordered pairs, giving $2^{n(n-1)}$ possible edge sets. Such problems can be described by a set function

$$f : 2^{\mathcal{E}} \to \mathbb{R},$$

where the ground set $\mathcal{E}$ contains all candidate elements, here the possible directed edges, and $f(S)$ assigns a value to a selected subset $S \subseteq \mathcal{E}$.

The function is *monotone* if adding elements cannot decrease its value,

$$f(S) \leq f(T) \qquad \text{whenever } S \subseteq T,$$

and *submodular* if it satisfies the diminishing-returns property,

$$f(S \cup \{e\}) - f(S) \geq f(T \cup \{e\}) - f(T), \qquad S \subseteq T \subseteq \mathcal{E}, \quad e \in \mathcal{E} \setminus T. \tag{2.43}$$

The *minimum submodular cover* problem tries to find the smallest subset that has the same function value as the full ground set,

$$S^{\star} \in \arg\min_{S \subseteq \mathcal{E}} \{|S| : f(S) = f(\mathcal{E})\}. \tag{2.44}$$

For a monotone, submodular, and integer-valued function, a greedy construction starts from $S = \emptyset$ and repeatedly adds the element with the largest marginal gain,

$$e^{\star} \in \arg\max_{e \in \mathcal{E} \setminus S} \big(f(S \cup \{e\}) - f(S)\big),$$

until $f(S) = f(\mathcal{E})$. The resulting set is not necessarily a minimum cover. However, [16] gives the approximation bound

$$\frac{|S_{\text{greedy}}|}{|S^{\star}|} \leq H(d) = \sum_{k=1}^{d} \frac{1}{k}, \qquad d = \max_{e \in \mathcal{E}} \big(f(\{e\}) - f(\emptyset)\big), \tag{2.45}$$

consequently $H(1) = 1$ and $H(2) = 1.5$. When $d = 1$, as for a matroid rank function, the greedy construction is optimal and returns a minimum cover.

Other example methods search directly over complete candidate solutions by modifying the current set. A local-search method accepts changes that improve the objective and stops when no admissible local modification gives an improvement. Such a point is only a local optimum and not necessarily globally optimal.

Simulated annealing reduces this dependence on local optima by also accepting some worsening changes [17]. For a cost J to be minimized, let

$$\Delta = J(S') - J(S)$$

be the change produced by moving from the current set S to a candidate set S'. The candidate is accepted with probability

$$P_{\text{acc}} = \begin{cases} 1, & \Delta \leq 0, \\ \exp(-\Delta / T_{\text{a}}), & \Delta > 0, \end{cases} \tag{2.46}$$

where $T_{\mathrm{a}} > 0$ is the annealing temperature.

2.8 Estimation and experiment design

2.8.1 Noise on a direction

A unit vector $\mu \in \mathbb{R}^d$ lies on the sphere S^{d-1}. Its tangent space (tangent plane at $d = 3$) at μ is the orthogonal hyperplane

$$T_\mu S^{d-1} = \mu^\perp = \{v \in \mathbb{R}^d : \mu^\top v = 0\}.$$

Directional noise can therefore be modeled by perturbing μ only along directions orthogonal to it and then projecting the result back onto the unit sphere. With $P(\mu)$ the orthogonal projector onto $\mu^\perp$, the noisy direction is

$$x = \frac{\mu + \sigma P(\mu)\eta}{\|\mu + \sigma P(\mu)\eta\|}, \qquad \eta \sim \mathcal{N}(0, I_d). \tag{2.47}$$

The perturbation $\sigma P(\mu)\eta$ is an isotropic Gaussian in the $(d-1)$-dimensional tangent space, while the normalization enforces $\|x\| = 1$.

For small σ, normalization affects the direction only at higher order and the angular error is approximately the norm of the tangent perturbation. Since

$$\mathbb{E}\left[\|P(\mu)\eta\|^2\right] = \operatorname{tr} P(\mu) = d - 1,$$

the root mean square angular error is approximately

$$\sigma\sqrt{d-1},$$

and therefore $\sigma\sqrt{2}$ in three dimensions.

2.8.2 Fisher information

Let an observation y have density $p(y \mid \theta)$ for an unknown parameter $\theta \in \mathbb{R}^k$. The likelihood $p(y \mid \theta)$ describes how compatible a candidate value of θ is with the observed data. Its gradient with respect to θ,

$$\nabla_\theta \log p(y \mid \theta),$$

is called the *score*. The Fisher information is the expected outer product of the score,

$$\mathcal{I}(\theta) = \mathbb{E}\left[\nabla_\theta \log p(y \mid \theta) \nabla_\theta \log p(y \mid \theta)^\top\right]. \tag{2.48}$$

It describes the sensitivity of the measurement distribution to changes in θ, where the directions associated with large information are more strongly constrained by the observations, and

directions in $\ker \mathcal{I}$ cannot be distinguished locally from the data.

Proposition 2.4 (Cramér-Rao bound)**.** *If $\mathcal{I}(\theta)$ is nonsingular, every unbiased estimator $\hat{\theta}$ satisfies, under the usual regularity conditions,*

$$\operatorname{Cov}(\hat{\theta}) \succeq \mathcal{I}(\theta)^{-1}. \tag{2.49}$$

For the linear Gaussian model

$$y = A\theta + \sigma\eta, \qquad \eta \sim \mathcal{N}(0, I),$$

the *maximum-likelihood* estimator chooses the parameter that maximizes the likelihood of the observed measurement,

$$\hat{\theta}_{\mathrm{ML}} = \arg\max_{\theta} p(y \mid \theta),$$

and equivalently, it minimizes the negative log-likelihood. Since

$$-\log p(y \mid \theta) = \frac{1}{2\sigma^2}\|A\theta - y\|^2 + \text{constant},$$

the maximum-likelihood estimator for the linear Gaussian model is

$$\hat{\theta}_{\mathrm{ML}} = \arg\min_{\theta} \|A\theta - y\|^2,$$

which equals the *least-squares* estimator that selects the parameter whose predicted measurement $A\theta$ has the smallest squared residual with respect to y.

For this model, the Fisher information is

$$\mathcal{I} = \frac{A^\top A}{\sigma^2}, \tag{2.50}$$

and hence the information depends on the geometry of the measurement matrix A and decreases with variance σ^2. If A has full column rank, the least-squares solution is unique,

$$\hat{\theta}_{\mathrm{LS}} = (A^\top A)^{-1} A^\top y.$$

When $\mathcal{I}$ is singular, there are parameter directions that cannot be identified from the measurements. A linear function $C\theta$ whose rows lie in range $\mathcal{I}$ can still be estimated, and every unbiased estimator satisfies

$$\operatorname{Cov}\big(\widehat{C\theta}\big) \succeq C\mathcal{I}^{+}C^\top \tag{2.51}$$

[18], where $\mathcal{I}^{+}$ is the Moore-Penrose pseudoinverse.

Taking $C = \Pi$, the orthogonal projector onto $(\ker \mathcal{I})^{\perp}$, (2.51) gives the bound on the identifiable part of the parameter,

$$\operatorname{Cov}\big(\widehat{\Pi\theta}\big) \succeq \Pi\mathcal{I}^{+}\Pi = \mathcal{I}^{+}.$$

Transformations that leave all measurements unchanged generate unobservable directions and

therefore lie in $\ker \mathcal{I}$. These directions are called the *gauge*.

2.8.3 Nonlinear least squares

For the nonlinear measurement model

$$y = h(\theta) + \sigma\eta, \qquad \eta \sim \mathcal{N}(0, I),$$

maximum likelihood gives the nonlinear least-squares problem

$$\min_{\theta} \frac{1}{2}\|h(\theta) - y\|^2.$$

The Gauss-Newton method solves it iteratively by linearizing the residual $r(\theta) = h(\theta) - y$ and computing

$$\delta = \arg\min_{\delta} \|J\delta + r\|,$$

where J is the Jacobian of h. If J is rank deficient, the minimum-norm step is $\delta = -J^{+}r$. When θ contains attitudes, the increments are applied to the rotation matrices through the exponential map of §2.1.3 in this work.

2.8.4 Covariance summaries

The information matrix can be reduced to scalar measures that summarize different aspects of estimation uncertainty [19]. For a nonsingular information matrix $\mathcal{I}$, A-optimality uses

$$A_{\text{opt}} = \operatorname{tr} \mathcal{I}^{-1},$$

which measures the total variance of the estimated parameters. D-optimality uses

$$D_{\text{opt}} = \log\det \mathcal{I}^{-1},$$

which is related to the volume of the confidence ellipsoid. E-optimality uses

$$E_{\text{opt}} = \lambda_{\max}(\mathcal{I}^{-1}) = \frac{1}{\lambda_{\min}(\mathcal{I})},$$

and therefore measures the largest uncertainty along any parameter direction [19]. Smaller values are preferred for all three criteria.

If $\mathcal{I}$ is singular because some directions are unidentifiable, only the identifiable subspace is considered. Let $w_k > 0$ denote the nonzero eigenvalues of $\mathcal{I}$. The corresponding criteria are

$$A_{\text{opt}} = \operatorname{tr} \mathcal{I}^{+} = \sum_k \frac{1}{w_k},$$

$$E_{\text{opt}} = \frac{1}{\min_k w_k},$$

and

$$D_{\text{opt}} = -\sum_k \log w_k.$$

The pseudoinverse in the A-optimality criterion removes the unidentifiable directions (the gauge).

2.8.5 Summary statistics

For a proportion estimated from k successes in N independent trials,

$$\hat{p} = \frac{k}{N},$$

the uncertainty in $\hat{p}$ can be described by a confidence interval. At confidence level $1-\alpha$, the Wilson interval is

$$\frac{\hat{p} + \frac{z^2}{2N}}{1 + \frac{z^2}{N}} \pm \frac{z}{1 + \frac{z^2}{N}} \sqrt{\frac{\hat{p}(1-\hat{p})}{N} + \frac{z^2}{4N^2}}, \tag{2.52}$$

where z is the $1-\alpha/2$ quantile of the standard normal distribution.

When an analytical confidence interval is not available, the sampling uncertainty of a statistic can be estimated by bootstrap resampling. A bootstrap sample is formed by drawing N observations with replacement from the original N observations. The statistic is recomputed on this sample, and the procedure is repeated many times to obtain an empirical approximation of its sampling distribution. A percentile bootstrap confidence interval is then obtained from the corresponding lower and upper percentiles of the bootstrap statistics.

For paired comparisons, each pair is treated as a single observation during resampling. The two values in a pair are therefore always selected together, preserving their dependence and allowing the uncertainty of paired quantities such as differences or ratios to be estimated correctly. These techniques are used for presenting the results in Chapter 6 and Appendix B.

3 Related work

3.1 Bearing rigidity

Bearing rigidity is employed in [20] for the shape control of mobile formations in two-dimensional space, with distributed control laws designed for nonholonomic agents. The directed bearing rigidity matrix and its properties are introduced and studied for $\mathrm{SE}(2)$ in [21]. Bearing rigidity theory is extended for frameworks of points in arbitrary dimensions in [22], and nonlinear distributed formation control laws are proposed. Localization with anchors is studied in [23], and localizability conditions and a distributed localization protocol are given. In [24], an extension of the rigidity theory for directed $\mathbb{R}^3 \times S^1$ frameworks is used in formation control tasks. It is shown in [25] that bearing rigidity is a generic property of the underlying graph, and that Laman graphs are generically bearing rigid in arbitrary dimensions. The observability of bearing-only localization with moving agents is analyzed in [26] with nonlinear observability theory. The survey [1] covers formation control and network localization, and the more recent survey [27] introduces a two-level taxonomy for bearing-based formation control methods. The rigidity formulation in this work mostly follows [3], which gives a general framework for bearing rigidity on any differential manifold in $\mathrm{SE}(3)^n$, with an extended form of the rigidity matrix and conditions for infinitesimal rigidity. Formations of agents that are heterogeneous in terms of actuation capabilities are modeled as generalized frameworks in [28], where a formation controller is designed for them and compared with a leader-follower combination of homogeneous controllers. Formation maneuver control of leader-follower multi-agent systems with time-varying target bearings is studied in [29], where the properties of bearing rigidity are extended to time-varying cases. Signed angle rigidity, which uses signed angle constraints that are independent of the global coordinate frame, is analyzed combinatorially in [30], where it is shown that Laman graphs characterize minimally signed angle rigid graphs. Angle-based localization and rigidity maintenance, which is closely related to bearing rigidity, is studied in [31].

3.2 Topology design for rigid formations

In [32], the construction of networks with desirable algebraic rigidity properties is formulated as a submodular set function optimization problem, and greedy algorithms that achieve global

optimality or a near-optimality guarantee are proposed. The rigidity recovery problem is studied in [33], where the minimum number of additional bearing measurements is analyzed and algorithms to select the edges to add are provided. The design of formations running planar weighted bearing-compass dynamics is studied in [34], where formation topologies are constructed with submodular optimization techniques. Minimal and 1-redundant bearing rigidity are studied in [4], where the number of edges of a minimally bearing rigid graph in $\mathbb{R}^d$ is derived (2.20). In [35], the bearing rigidity graph is used to describe the topology of a group of autonomous underwater vehicles for localization, and a min-weighted bearing rigidity graph generation strategy is developed. In [36], the framework is decomposed into subframeworks so that bearing rigidity, a global property, is expressed as a set of local properties, and a decentralized gradient-based controller preserves bearing rigidity. In [37], bearing rigidity is used for a resilient network design in multirobot coverage control, and a recovery algorithm provides guarantees for the network in the event of robot loss while maintaining a minimally rigid structure. An energy-efficient fault-tolerant control framework for bearing-based fixed-wing UAV formations is developed in [38].

3.3 Reinforcement learning for graph construction

In [39], the construction of a graph is formulated as a decision-making process in which a central agent creates topologies by trial and error and receives rewards proportional to the value of a target objective, and an algorithm based on reinforcement learning and graph neural networks is proposed. SparRL [40] uses deep reinforcement learning for graph sparsification, producing an edge-reduced graph. A critical analysis of incorporating neural network based algorithms into the classical combinatorial optimization framework is presented in [41], studying their performance, transferability, computational cost, and generalization to larger instances. In [42], a deep reinforcement learning algorithm for graph searching is proposed for network topology optimization. The survey [2] reviews reinforcement learning methods for combinatorial optimization problems on graphs, separating the works that optimize the graph structure from the ones that optimize the outcome of a process on a fixed graph.

4 Rigidity analysis for heterogeneous bearing frameworks

This chapter covers the analysis and derivations based on bearing rigidity theory. The theory and results presented here are used within the environment of Chapter 5, for constructing the reward function and informing the policy via the derived variables provided as observations. The main tools established here are a formulation of an extended bearing rigidity matrix for heterogeneous networks, a heuristic algorithm to identify the required number of edges in order for a framework to be IBR, conclusions on the two possible objectives for optimization, and establishing oracles that inform the RL policy.

4.1 Bearing differential

Agent i measures the direction (bearing) to agent j in its own frame. The bearing b_{ij} can be expressed as

$$b_{ij} = R_i^\top \hat{p}_{ij}, \qquad \hat{p}_{ij} = \frac{p_{ij}}{\|p_{ij}\|}, \qquad p_{ij} = p_j - p_i, \qquad s_{ij} = \|p_{ij}\|^{-1} \tag{4.1}$$

where $p_i \in R^3$ is the position of agent i and $R_i \in SO(3)$ is its rotation matrix. Both factors in b_{ij} change under a variation of the configuration and they are differentiated separately. For the direction,

$$\begin{aligned} \frac{d}{dt}\hat{p}_{ij} &= \frac{\dot{p}_{ij}}{\|p_{ij}\|} - \frac{p_{ij}\left(p_{ij}^\top \dot{p}_{ij}\right)}{\|p_{ij}\|^3} = s_{ij}\left(I_3 - \hat{p}_{ij}\hat{p}_{ij}^\top\right)\dot{p}_{ij} \\ &= s_{ij}\, P(\hat{p}_{ij})\,(\dot{p}_j - \dot{p}_i), \qquad P(x) = I_3 - xx^\top. \end{aligned} \tag{4.2}$$

Only the component of relative velocity between agents *perpendicular* to $\hat{p}$ can vary the bearing, while the component along $\hat{p}$ changes the distance between the two agents and leaves the measurement untouched. This results in no change in the bearing in case of a uniform scaling of the whole formation, and along with uniform translation and rotation of the formation, they describe the trivial motions of §4.3. Another observed fact is that a planar edge constrains only one direction, and a spatial edge constrains two, which fix the counting bound of §4.5.

For the attitude, with ω_i the angular velocity in the world frame and $\dot{R}_i = [\omega_i]_\times R_i$, so that

$\dot{R}_i^\top = -R_i^\top [\omega_i]_\times$,

$$\dot{R}_i^\top \hat{p}_{ij} = -R_i^\top [\omega_i]_\times \hat{p}_{ij} = -R_i^\top \left(\omega_i \times \hat{p}_{ij}\right) = +R_i^\top \left(\hat{p}_{ij} \times \omega_i\right) = R_i^\top [\hat{p}_{ij}]_\times \omega_i. \tag{4.3}$$

Rotating an agent's own frame moves its measurements even when nothing translates. Adding (4.2) and (4.3),

$$\dot{b}_{ij} = \underbrace{s_{ij} R_i^\top P(\hat{p}_{ij})}_{\mathbf{D}_p^{(k)}} (\dot{p}_j - \dot{p}_i) \; - \; \underbrace{\left(-R_i^\top [\hat{p}_{ij}]_\times\right)}_{\mathbf{D}_a^{(k)}} \omega_i, \tag{4.4}$$

for the edge $e_k = (v_i, v_j)$.

The *bearing map* $b_{\mathcal{G}}$ stacks b_{ij} over the edges of $\mathcal{G}$ in a fixed order, and a rigidity matrix is the matrix representing its differential. Following sections discuss this matrix and how it is constructed in this work.

4.2 Heterogeneous rigidity matrix

The unified framework of [3] embeds every domain into $\mathrm{SE}(3)^n$, so that a formation mixing agent types is described by a rigidity matrix with $6n$ columns. The same approach is taken in this work. An agent then carries six coordinates whether or not it can vary them, and the coordinates it cannot vary have to be accounted for. The following show that the accounting works if each agent's restriction is applied to that agent's own columns, and construct the rigidity in a node-indexed way to account for individual agents' domains.

4.2.1 Matrix requirements

Let $\bar{S} = \mathrm{blkdiag}(S_1, \ldots, S_n)$ and $\bar{P} = \mathrm{blkdiag}(P_1, \ldots, P_n)$. A variation δ of the configuration is lifted to $\delta^+ \in \mathbb{R}^{6n}$ by padding the coordinates an agent does not have with zeros. The lifted variations fill the *admissible subspace*

$$\mathcal{A} = \mathrm{range}\, \bar{S} \oplus \mathrm{range}\, \bar{P} \subseteq \mathbb{R}^{6n}, \qquad \dim \mathcal{A} = c = \sum_i c_i, \tag{4.5}$$

and with $A \in \mathbb{R}^{6n \times c}$ any matrix with orthonormal columns spanning $\mathcal{A}$ we take $\delta^+ = A\delta$ and write

$$q_v := 6n - c = \dim \mathcal{A}^\perp \tag{4.6}$$

for the number of coordinates the embedding supplies and no agent can use. Let $J_{\mathcal{G}} \in \mathbb{R}^{3m \times c}$ be the Jacobian of the bearing map in the coordinates A induces, so that $J_{\mathcal{G}}\delta = \dot{b}_{\mathcal{G}}$ whenever $\delta^+ = A\delta$.

Two requirements are placed on a candidate for the rigidity matrix, $\mathbf{B} \in \mathbb{R}^{3m \times 6n}$. It must reproduce the differential on motions the agents can perform, and the coordinates they cannot

vary must index identically zero columns. These requirements are expressed as

$$\mathbf{B}A = J_{\mathcal{G}}, \tag{F}$$

$$\mathbf{B}\big(I_{6n} - AA^{\top}\big) = 0. \tag{Z}$$

Condition (F) describes the same bearing rigidity matrix as Definition 13 of [3]. Condition (Z) is the one that decides between the construction of the rigidity matrix of [3] and §4.2. The rank identity (2.17) counts q_v as the number of null columns of the extended matrix, and the rigidity test of Definition 2.3 compares $\operatorname{rk}\mathbf{B}_{\mathcal{G}}$ with $\operatorname{rk}\mathbf{B}_{\mathcal{K}}$. For this comparison, q_v needs to appear on both sides, which requires the null columns to be a property of the agents' domains alone, with no dependence on the network topology.

The edge-indexed construction of (2.15) places a factor $\mathbf{U}_{ij}$ between $\mathbf{D}_p$ and $\bar{\mathbf{E}}^{\top}$. Both $\mathbf{D}_p$ and $\mathbf{U}$ are block diagonal over *edges*, so the product multiplies the whole of row block k of (2.7) by the single matrix $\mathbf{D}_p^{(k)}\mathbf{U}_{ij}$:

$$\big[\mathbf{D}_p\,\mathbf{U}\,\bar{\mathbf{E}}^{\top}\big]_{k,\cdot} = \big[\,\mathbf{0}\;\cdots\;\underbrace{-\mathbf{D}_p^{(k)}\mathbf{U}_{ij}}_{i}\;\cdots\;\underbrace{+\mathbf{D}_p^{(k)}\mathbf{U}_{ij}}_{j}\;\cdots\;\mathbf{0}\,\big]. \tag{4.7}$$

Lemma 4.1 (one factor for two endpoints)**.** *In* $\mathbf{B}_{\mathcal{G}}^{\times}$ (2.15) *the column blocks of* i *and* j *inside row block* k *are* $-\mathbf{D}_p^{(k)}\mathbf{U}_{ij}$ *and* $+\mathbf{D}_p^{(k)}\mathbf{U}_{ij}$*. Any restriction encoded by* $\mathbf{U}_{ij}$ *is therefore imposed on both endpoints of edge* k *at once.*

Acting on a lifted variation, (4.7) returns

$$\big(\mathbf{B}^{\times}\delta^{+}\big)_k = \mathbf{D}_p^{(k)}\mathbf{U}_{ij}\big(\delta_{p,j} - \delta_{p,i}\big) - \mathbf{D}_a^{(k)}\mathbf{V}_{ij}\delta_{a,i}, \tag{4.8}$$

so $\mathbf{U}_{ij}$ acts on the *difference* of the two endpoint variations. Multiplying (2.7) from the *right* by the node-block-diagonal $\bar{S}$ instead scales each column block by its own projector,

$$\big[\mathbf{D}_p\,\bar{\mathbf{E}}^{\top}\bar{S}\big]_{k,\cdot} = \big[\,\mathbf{0}\;\cdots\;-\mathbf{D}_p^{(k)}S_i\;\cdots\;+\mathbf{D}_p^{(k)}S_j\;\cdots\;\mathbf{0}\,\big], \tag{4.9}$$

and acting on a lifted variation returns

$$\big(\mathbf{B}^{\bullet}\delta^{+}\big)_k = \mathbf{D}_p^{(k)}\big(S_j\delta_{p,j} - S_i\delta_{p,i}\big) - \mathbf{D}_a^{(k)}P_i\,\delta_{a,i}. \tag{4.10}$$

The only difference between (4.8) and (4.10) is whether the projector is applied before or after the two endpoint variations are subtracted. When $S_i = S_j = S$ the two agree, since $S\delta_{p,j} - S\delta_{p,i} = S(\delta_{p,j} - \delta_{p,i})$.

Lemma 4.2 (when (F) and (Z) are compatible)**.** *Let* $e_k = (v_i, v_j)$*. If* $\mathbf{B}^{\times}$ *satisfies* (F) *and* (Z) *on the column blocks of* i *and* j*, then*

$$\mathbf{D}_p^{(k)}\big(S_i - S_j\big) = 0, \qquad \textit{equivalently} \qquad \operatorname{range}(S_i - S_j) \subseteq \operatorname{span}\{\hat{p}_{ij}\}. \tag{4.11}$$

In particular, if i is planar and j is spatial then $\hat{p}_{ij} = \pm e_3$, so j lies vertically above or below i.

Proof. Write $\mathbf{U} = \mathbf{U}_{ij}$ and read the two column blocks of (4.7) in turn. By (4.5) the orthogonal projector onto $\mathcal{A}$ acts on agent i's position columns as S_i, so (Z) reads $\mathbf{D}_p^{(k)}\mathbf{U}(I - S_i) = 0$, that is $\mathbf{D}_p^{(k)}\mathbf{U} = \mathbf{D}_p^{(k)}\mathbf{U}S_i$. On the admissible part, (F) requires the block to reproduce (4.10), whose position term at i is $-\mathbf{D}_p^{(k)}S_i$, so $\mathbf{D}_p^{(k)}\mathbf{U}S_i = \mathbf{D}_p^{(k)}S_i$. Combining,

$$\mathbf{D}_p^{(k)}\mathbf{U} = \mathbf{D}_p^{(k)}S_i. \tag{4.12}$$

The same two steps at agent j, whose column block carries $+\mathbf{D}_p^{(k)}\mathbf{U}$ by Lemma 4.1 and must reproduce $+\mathbf{D}_p^{(k)}S_j$, give $\mathbf{D}_p^{(k)}\mathbf{U} = \mathbf{D}_p^{(k)}S_j$. Subtracting yields $\mathbf{D}_p^{(k)}(S_i - S_j) = 0$.

For the second form, $\mathbf{D}_p^{(k)} = s_{ij}R_i^\top P(\hat{p}_{ij})$ with $s_{ij} \neq 0$ and R_i orthogonal, so $\ker \mathbf{D}_p^{(k)} = \ker P(\hat{p}_{ij}) = \operatorname{span}\{\hat{p}_{ij}\}$. With $S_i = \operatorname{diag}(1,1,0)$ and $S_j = I_3$ we have $S_i - S_j = -e_3e_3^\top$, so $\operatorname{range}(S_i - S_j) = \operatorname{span}\{e_3\}$ and $e_3 = \pm\hat{p}_{ij}$, both vectors being of unit norm. □

Equation (4.11) is a nontrivial algebraic condition on (p_i, p_j) whenever $S_i \neq S_j$, and its solution set has measure zero in the configuration space. So no choice of $\mathbf{U}_{ij}$ makes the edge-indexed construction of (2.15) satisfy both requirements on an edge joining two agents with different translational freedom, except on a null set. The obstruction is structural rather than a wrong table entry, and it concerns $\mathbf{U}$ alone, since by (4.8) the factor $\mathbf{V}_{ij}$ meets only agent i, so setting $\mathbf{V}_{ij} = P_i$ restricts exactly the agent whose columns it occupies and causes no difficulty.

Remark 4.3 (why this stays invisible until two graphs are compared)**.** By (F) both constructions have the same rank once restricted to the admissible columns, $\operatorname{rk}(\mathbf{B}^\times A) = \operatorname{rk}(\mathbf{B}^\bullet A) = \operatorname{rk} J_\mathcal{G}$. They differ only in how much rank the inadmissible columns contribute, and by Lemma 4.1 that contribution depends on which edges are present. Definition 2.3 compares $\operatorname{rk}\mathbf{B}_\mathcal{G}$ with $\operatorname{rk}\mathbf{B}_\mathcal{K}$, so a difference in that contribution is detected as a difference in rigidity.

4.2.2 Node-indexed matrix

The bearing rigidity matrix this work uses, constructed by applying the DOF restrictions per agent, is as follows.

$$\boxed{\mathbf{B}_\mathcal{G}^\bullet = \begin{bmatrix} \mathbf{D}_p\,\bar{\mathbf{E}}^\top\bar{S} & \mathbf{D}_a\,\bar{\mathbf{E}}_o^\top\bar{P} \end{bmatrix} \in \mathbb{R}^{3m\times 6n}} \tag{4.13}$$

Throughout the rest of the thesis, this matrix is sometimes referred to as simply $\mathbf{B}$. Every coordinate an agent cannot vary is now an exactly zero column, which is what §4.2.1 asks for.

Proposition 4.4 (uniqueness)**.** *$AA^\top = \operatorname{blkdiag}(\bar{S}, \bar{P})$, and there is exactly one matrix in $\mathbb{R}^{3m\times 6n}$ satisfying both (F) and (Z), namely $J_\mathcal{G}A^\top$, which equals $\mathbf{B}_\mathcal{G}^\bullet$.*

Proof. $\bar{S}$ and $\bar{P}$ are orthogonal projectors and $\mathcal{A}$ splits along the position and attitude coordinate blocks, so the orthogonal projector onto $\mathcal{A}$ is $AA^\top = \operatorname{blkdiag}(\bar{S}, \bar{P})$. Write $\mathbf{M} = \begin{bmatrix} \mathbf{D}_p\bar{\mathbf{E}}^\top & \mathbf{D}_a\bar{\mathbf{E}}_o^\top \end{bmatrix}$ for the differential before any restriction, so that (4.13) reads $\mathbf{B}_\mathcal{G}^\bullet = \mathbf{M}\,AA^\top$. Idempotence of $AA^\top$ and $A^\top A = I_c$ give

$$\mathbf{B}_\mathcal{G}^\bullet\big(I - AA^\top\big) = \mathbf{M}\,AA^\top\big(I - AA^\top\big) = 0, \qquad \mathbf{B}_\mathcal{G}^\bullet A = \mathbf{M}\,AA^\top A = \mathbf{M}A = J_\mathcal{G},$$

the last equality because $\mathbf{M}$ reproduces (4.4) on admissible variations. So $\mathbf{B}^{\bullet}_{\mathcal{G}}$ satisfies both. Conversely, let $\mathbf{B}$ satisfy both. Splitting the identity as $I_{6n} = AA^{\top} + (I_{6n} - AA^{\top})$,

$$\mathbf{B} = \mathbf{B}AA^{\top} + \mathbf{B}\big(I_{6n} - AA^{\top}\big) = \big(\mathbf{B}A\big)A^{\top} = J_{\mathcal{G}}A^{\top},$$

so the matrix is unique and equals $\mathbf{B}^{\bullet}_{\mathcal{G}}$. □

Theorem 4.5 (the rank test is restored)**.** *For every $\mathcal{G}$,* $\operatorname{rk}\mathbf{B}^{\bullet}_{\mathcal{G}} = 6n - q_v - q_i$ *with* $q_v = 6n - c$ *and* $q_i = \dim\ker(\mathbf{B}^{\bullet}_{\mathcal{G}}A)$*. Hence* $\operatorname{rk}\mathbf{B}^{\bullet}_{\mathcal{G}} = \operatorname{rk}\mathbf{B}^{\bullet}_{\mathcal{K}}$ *if and only if $(\mathcal{G}, \chi)$ is infinitesimally bearing rigid.*

Proof. By (Z) we have $\mathbf{B}^{\bullet}_{\mathcal{G}} = \mathbf{B}^{\bullet}_{\mathcal{G}}AA^{\top}$, so the row space is unchanged by restricting to the admissible columns and

$$\operatorname{rk}\mathbf{B}^{\bullet}_{\mathcal{G}} = \operatorname{rk}\big(\mathbf{B}^{\bullet}_{\mathcal{G}}A\big) = \underbrace{c}_{\text{columns of } A} - \underbrace{q_i}_{\dim\ker} = (6n - q_v) - q_i.$$

The same holds on $\mathcal{K}$ with $q_t = \dim\ker(\mathbf{B}^{\bullet}_{\mathcal{K}}A)$ and with the *same* q_v, which by (4.5) depends only on the domains and not on the edge set. Subtracting, $\operatorname{rk}\mathbf{B}^{\bullet}_{\mathcal{G}} - \operatorname{rk}\mathbf{B}^{\bullet}_{\mathcal{K}} = q_t - q_i$, so the two ranks agree if and only if $q_i = q_t$. Since every edge of $\mathcal{G}$ is an edge of $\mathcal{K}$ we have $\ker\mathbf{B}_{\mathcal{K}} \subseteq \ker\mathbf{B}_{\mathcal{G}}$, so equality of these two dimensions is equality of the two kernels, which is Definition 2.3. □

Relation to the heterogeneous formulation of [28]. The node-wise treatment is not new. It is used in [28], where each agent carries its own selection matrices $S_{p,i} \in \mathbb{R}^{3\times c_{p,i}}$ and $S_{o,i} \in \mathbb{R}^{3\times c_{o,i}}$ mapping its command vector to its linear and angular velocity, assembled block-diagonally into $\mathbf{S}_p = \operatorname{diag}(S_{p,i})$ and $\mathbf{S}_o = \operatorname{diag}(S_{o,i})$ in eqs. (3), (4), and (7) there. The bearing rigidity matrix defined there is $\mathbf{B}_{\mathcal{G}}(\mathbf{x}) \in \mathbb{R}^{3m\times c}$, and its kth row block, eq. (9) of [28], carries $S_{p,i}$ in the column block of agent i and $S_{p,j}$ in the column block of agent j. The two endpoints are restricted separately, which is what (4.9) does here. Infinitesimal rigidity is then defined by $\ker\mathbf{B}_{\mathcal{K}} = \ker\mathbf{B}_{\mathcal{G}}$ on those c columns, so neither the lift nor the count q_v ever appears.

What this section adds is that the choice is *forced* rather than convenient. Lemma 4.2 shows the same restriction cannot be expressed in the edge-indexed $\mathbf{B}^{\times}$ off a measure-zero set, and Proposition 4.4 shows $\mathbf{B}^{\bullet}$ is the only matrix in the padded $\mathbb{R}^{3m\times 6n}$ form compatible with the null-column requirement that the rank test relies on.

Proposition 4.6 (agreement with $\mathbf{B}^{\times}$)**.** *Take $\mathbf{U}_{ij} = S_i$ and $\mathbf{V}_{ij} = P_i$. For an edge $e_k = (v_i, v_j)$ with $S_i = S_j$, the kth row blocks of $\mathbf{B}^{\times}_{\mathcal{G}}$ and $\mathbf{B}^{\bullet}_{\mathcal{G}}$ agree. Hence, if $S_i = S_j$ on every edge of $\mathcal{E}$, then $\mathbf{B}^{\times}_{\mathcal{G}} = \mathbf{B}^{\bullet}_{\mathcal{G}}$ at every configuration.*

Proof. Compare (4.8) with (4.10) on row block k. Writing S for the common value of S_i and S_j, linearity of S gives

$$\mathbf{D}^{(k)}_p S\big(\delta_{p,j} - \delta_{p,i}\big) = \mathbf{D}^{(k)}_p\big(S\delta_{p,j} - S\delta_{p,i}\big) = \mathbf{D}^{(k)}_p\big(S_j\delta_{p,j} - S_i\delta_{p,i}\big),$$

and the attitude blocks agree already, since $\bar{\mathbf{E}}_o^{\top}$ meets only agent i and $\mathbf{V}_{ij} = P_i$. Applying this to every row block yields $\mathbf{B}^{\times}_{\mathcal{G}} = \mathbf{B}^{\bullet}_{\mathcal{G}}$.

In the homogeneous case $\bar{S} = I_n \otimes S$ commutes with the incidence matrix, so

$$\mathbf{U}\bar{\mathbf{E}}^\top = \bar{\mathbf{E}}^\top \bar{S}, \qquad \mathbf{V}\bar{\mathbf{E}}_o^\top = \bar{\mathbf{E}}_o^\top \bar{P},$$

with $\mathbf{U}_{ij} = S$ and $\mathbf{V}_{ij} = P$ as in Table I of [3][1]. □

Numerical validation The reference below is the Jacobian of the bearing map along admissible variations, obtained by central differences on the manifold with attitudes updated as $R_i \mapsto \exp([\varepsilon\omega_i]_\times)R_i$.

Over 432 frameworks covering the five manifolds at $n = 3, 4, 6, 9$ and sixteen heterogeneous mixes at four graph densities, both (2.15) and (4.13) reproduce this Jacobian on $\mathcal{A}$ to a worst relative error of 1.1×10^{-9}, as (F) requires of both. The identity $\mathbf{B}^\bullet(I - AA^\top) = 0$ holds to 2.2×10^{-16} for (4.13), and the corresponding quantity for (2.15) is of order one. Over the 1198 non-degenerate frameworks among 1200 random heterogeneous draws with $n \in \{4, 5, 6\}$, Table 4.1 gives the counts.

Table 4.1: The two constructions compared on the 1198 non-degenerate frameworks among 1200 random heterogeneous draws at $n \in \{4, 5, 6\}$. The first row counts frameworks whose nullity accounting fails, the second those where the rank test disagrees with the numerical Jacobian.

	edge-indexed	node-indexed
rk $\mathbf{B} = 6n - q_v - q_i$ fails	504/1198	0/1198
rank test disagrees with the Jacobian	67/1198 (5.6%)	0/1198

For homogeneous formations the two constructions agree bitwise, checked over 25 graphs in each manifold at $n = 3, 4, 6, 8, 12$, and the ranks match the closed forms $2n-3$, $3n-4$, $3n-4$, $4n-5$, $6n-7$.

A three-agent counterexample Take $p_1 = (0, 0, 0)$, $p_2 = (2, 0, 0)$, $p_3 = (1, 1, 1)$ with $\mathcal{D}_1 = \mathcal{D}_2 = \mathbb{R}^2$, $\mathcal{D}_3 = \mathbb{R}^3$, and $\mathcal{E} = \{(v_1, v_3), (v_2, v_3)\}$. Its bearing-preserving motions are two horizontal translations and scaling, on $\mathcal{G}$ and on $\mathcal{K}$ alike, so the framework is rigid.

Table 4.2: Rank of the rigidity matrix on the three-agent rigid framework. The edge-indexed matrix gains a unit of rank on $\mathcal{K}$ from a vertical motion of the two planar agents, so its rank test compares 4 against 5.

	rk $\mathbf{B}_\mathcal{G}$	rk $\mathbf{B}_\mathcal{K}$	verdict
$\mathbf{B}^\times$, (2.15)	4	5	flexible
$\mathbf{B}^\bullet$, (4.13)	4	4	rigid
finite differences	4	4	rigid

The extra rank is not an artifact of the factor chosen on the planar-planar edge, since the z columns of agents 1 and 2 take entries from all three row blocks of $\mathcal{K}$ and $\mathbf{U}_{12}$ governs one.

[1] For $\mathbb{R}^3 \times S^1$ read $\mathbf{V}_{ij}$ in the projector form $P_i = vv^\top$ rather than the $[\,0_{3\times 2}\ v\,]$ form of Table I of [3]. The two differ as matrices but not in rank, nor in q_v. Only the phrase "null column" is sensitive to the choice, which is why (Z) is stated as an identity.

A second mechanism appears at $n = 4$ with three planar agents and one spatial. There the *count* of null columns is itself graph-dependent, 13 on $\mathcal{G}$ against 12 on $\mathcal{K}$, and a minimally rigid framework is again reported flexible, 6 against 8.

4.3 Trivial motions

By Definition 2.3 a framework is rigid when it attains $\rho_{\mathcal{K}} = \operatorname{rk} \mathbf{B}_{\mathcal{K}}$, so that number has to be known before rigidity can be tested. Equivalently, by Theorem 4.5, one has to know q_t, the dimension of the set of variations that preserve every bearing of the complete graph.

Those variations are global translation, rotation, and scaling. A variation is available to the formation only when every agent can perform its part of it, so q_t is the dimension of what survives intersecting these variations with the agents' own freedoms. Writing q_r for the rotational part,

$$q_t = \big(2 \text{ if some agent is planar, else } 3\big) + 1 + q_r, \qquad q_r = \begin{cases} 0, & \text{some agent is frameless,} \\ 3, & \text{every agent is SE}(3), \\ 1, & \text{otherwise,} \end{cases} \tag{4.14}$$

and $\rho_{\mathcal{K}} = \sum_i c_i - q_t$ by Theorem 4.5 applied to $\mathcal{K}^2$. A frameless agent measures in the global frame in the modeling used in this work, so rotating the world changes what it sees and a single such agent anywhere in the formation removes the rotational part entirely. Equation (4.14) assumes the planar agents share a plane and the agents with one rotational freedom share an axis.

Specializing (4.14) to homogeneous formations gives $q_t = c + 1$ in every domain, the c motions of the formation as a single rigid body together with one scaling, which is Theorem 3 of [3] and produces Table 4.3. The last two columns are the subject of §4.4 and §4.5.

Table 4.3: Specializing (4.14) to homogeneous formations. The trivial dimension is the rigid-body motions of the formation together with one scaling, $\rho_{\mathcal{K}}$ the rank a rigid framework must attain, $c_{\max}$ the most rank one edge can carry, and the last column the bound of (4.25) at $n = 16$.

$\mathcal{D}$	DOF per agent	trivial dim	$\rho_{\mathcal{K}}$	$c_{\max}$	m_{req} at n=16
$\mathbb{R}^2$	2	3 (2 transl. + scale)	$2n-3$	1	29
$\mathbb{R}^3$	3	4	$3n-4$	2	22
$\mathbb{R}^2 \times S^1$	3	4 (+1 rotation)	$3n-4$	1	44
$\mathbb{R}^3 \times S^1$	4	5 (+1 rotation)	$4n-5$	2	30
SE(3)	6	7 (+3 rotations)	$6n-7$	2	45

An agent's own attitude enters the matrix only through the bearings that agent takes, so an oriented agent needs to make bearing measurements to "fix" its own attitude. This gives the number of outgoing edges each node needs, shown in Table 4.4 and derived in Appendix A.2.

[2] Appendix A.1 carries the derivation.

Table 4.4: The least number of outgoing bearings r_i an agent needs for its own attitude to be observable, by its domain alone.

$\mathcal{D}_i$	can rotate about	outgoing bearings must be	r_i
$\mathbb{R}^2$, $\mathbb{R}^3$	nothing	-	0
$\mathbb{R}^2 \times S^1$	e_3	one not parallel to e_3	1
$\mathbb{R}^3 \times S^1$	its axis v	one not parallel to v	1
SE(3)	any direction	two that are not parallel	2

4.4 Single measurement values

This section collects quantities useful to infer what a single edge change in the graph does to the rigidity of the framework. These quantities are made available through a decomposition of **B**.

4.4.1 Per-edge block rank

A bearing is one angle in the plane and two in space, so a single measurement can constrain at most two directions, and fewer when the agents it joins cannot move in all of them. That number is the rank of the three rows the edge contributes, and it is what the edge is worth at best.

Substituting $\mathbf{D}_p^{(k)} = s_{ij} R_i^\top P(\hat{p}_{ij})$ and $\mathbf{D}_a^{(k)} = -R_i^\top [\hat{p}_{ij}]_\times$ into (4.10) and dropping the nonzero scalar s_{ij} and the orthogonal $R_i^\top$, neither of which changes a rank,

$$c_k := \operatorname{rk} b_{ij} = \operatorname{rk}\big[\, \underbrace{P(\hat{p}_{ij})S_i}_{i \text{ moves}} \;\big|\; \underbrace{P(\hat{p}_{ij})S_j}_{j \text{ moves}} \;\big|\; \underbrace{[\hat{p}_{ij}]_\times P_i}_{i \text{ turns}} \,\big]. \tag{4.15}$$

Each term lands in $\hat{p}_{ij}^{\perp}$, since motion along the bearing changes no direction, so $c_k \leq 2$ always.

- **Both endpoints planar.** They share the plane, so $\hat{p}_{ij}$ lies in it and $\hat{p}_{ij}^{\perp}$ meets the plane in one direction. Every term reduces to that same direction, the attitude term included, and $c_k = 1$.
- **Either endpoint spatial.** That agent moves in all of $\mathbb{R}^3$, so $P(\hat{p}_{ij})S_j = P(\hat{p}_{ij})$ already has rank 2, and $c_k = 2$.

Writing $c_{\max} = \max_{k \in \mathcal{K}} c_k$ for the most any one edge can carry,

$$c_{\max} = 1 \text{ when every agent is planar}, \qquad c_{\max} = 2 \text{ otherwise.}$$

4.4.2 Reciprocal pairs

What having the reverse of an existing edge changes depends on the domain. With both endpoints frameless, $P_i = P_j = 0$ kills the attitude half of (4.13), and what remains is $P(-\hat{p}) = P(\hat{p})$ and $s_{ji} = s_{ij}$, while $R_i = R_j = I_3$. Reversing the edge therefore only exchanges which

endpoint carries the minus sign, so

$$b_{ji} = -b_{ij}, \qquad \text{hence} \qquad \operatorname{rk}\begin{bmatrix}\mathbf{B}\\ b_{ji}\end{bmatrix} = \operatorname{rk}\mathbf{B} \quad \text{whenever } (v_i, v_j) \in \mathcal{E}. \tag{4.16}$$

The two blocks span the same row space, so a reciprocal pair contributes no more rank than a single one in case both endpoints are frameless.

Once an agent carries a frame the cancellation fails, for a reason visible in (4.10). The attitude term sits in the measuring agent's columns alone, so the block of (v_i, v_j) touches agent i's attitude columns and the block of (v_j, v_i) touches agent j's. Those are different columns, and no sign relation between the two blocks is available.

4.4.3 Single-edge quantities

Z is an orthonormal basis of $\ker \mathbf{B}$ and v the unit eigenvector at the rigidity eigenvalue λ, the direction the measurements constrain least. Both are taken from the length-normalized matrix $\mathbf{B}_s$ of (4.29).

Rank added by an absent edge. Appending a row block raises the rank by however much of it lies outside the current row space. By (2.2) the gain is

$$\operatorname{rk}\begin{bmatrix}\mathbf{B}\\ b_{ij}\end{bmatrix} - \operatorname{rk}\mathbf{B} \;=\; \operatorname{rk}\big(b_{ij}Z\big). \tag{4.17}$$

Two quantities follow from this,

$$g_{ij} = \operatorname{rk}\big(b_{ij}Z\big) \in \{0, 1, \ldots, c_{\max}\}, \qquad \eta_{ij} = \frac{\|b_{ij}Z\|_F}{\|b_{ij}\|_F} \in [0, 1]. \tag{4.18}$$

g_{ij} is the rank gain itself, an integer bounded by $c_{\max}$. The scalar η_{ij} is the fraction of the block's norm lying outside the current row space. It vanishes where g_{ij} does, and it allows ordering the edges that would contribute the same rank to the matrix. Both quantities are zero on an edge that is already present, as its rows lie in the row space of $\mathbf{B}$.

Conditioning added by an absent edge. Using $G_t = \mathbf{B}^\top\mathbf{B} + t\, b_{ij}^\top b_{ij}$, with λ assumed simple, Lemma 2.2 gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mu_{q+1}(G_t) = \big\|b_{ij}v\big\|^2. \tag{4.19}$$

Two quantities that can be used are,

$$\zeta_{ij} = \big\|b_{ij}v\big\|, \qquad \nu_i = \big(\|v_i^{\text{pos}}\|,\ \|v_i^{\text{att}}\|\big). \tag{4.20}$$

These are useful to understand what would happen to the direction (variation) that would change the bearings the least. The first one says how much adding $i \to j$ would stiffen the

direction the framework is weakest in. The second is the norm of the mode at agent i, split into its position and attitude halves, so it says how much of that weakness is from node i. These are only meaningful quantities if the framework is rigid, as the moment the graph becomes rigid its kernel holds only trivial variations, and a trivial variation changes no bearing, so $b_{ij}Z = 0$ for every pair and (4.18) is identically zero after that, and (4.20) take nonzero values.

The derivative is taken at $t = 0$ while adding the edge corresponds to $t = 1$, so ζ_{ij} is not an exact answer. Empirical tests show that against the true $\Delta\lambda$ its log-log correlation is $+0.93$ over an addition and $+0.35$ over a removal. Therefore these are not considered reliable but possibly useful for choosing edges to add.

Rank carried by an existing edge. For an existing edge e with row block b, let E select the three rows of b from $\mathbf{B}$, so that $b = E^\top\mathbf{B}$. The block leverage $H = b(\mathbf{B}^\top\mathbf{B})^+b^\top = E^\top\mathbf{B}\mathbf{B}^+E$ is a principal submatrix of the orthogonal projector $\mathbf{B}\mathbf{B}^+$ onto $\operatorname{col}\mathbf{B}$. For a unit x, $x^\top Hx = \|\mathbf{B}\mathbf{B}^+Ex\|^2 \le 1$, with equality exactly when $Ex \in \operatorname{col}\mathbf{B}$. Hence $0 \preceq H \preceq I_3$, and the eigenspace of H at eigenvalue 1 is $\{x : Ex \in \operatorname{col}\mathbf{B}\}$. A vector $\mathbf{B}\delta$ lies in range E exactly when $\mathbf{B}_{\mathcal{E}\setminus e}\delta = 0$, so this eigenspace has dimension $\dim\ker\mathbf{B}_{\mathcal{E}\setminus e} - \dim\ker\mathbf{B}$, which is the rank the graph would lose,

$$\ell_e := \operatorname{rk}\mathbf{B} - \operatorname{rk}\mathbf{B}_{\mathcal{E}\setminus e} = \#\big\{\text{eigenvalues of } H \text{ equal to } 1\big\}, \tag{4.21}$$

so $\ell_e = 0$ marks an edge that can be deleted without breaking rigidity.

Conditioning carried by an existing edge. Since $\mathbf{B}^\top\mathbf{B} = \sum_k b_k^\top b_k$ over the row blocks, deleting e subtracts $b^\top b$ and nothing else, so

$$\lambda\big(\mathbf{B}_{\mathcal{E}\setminus e}\big) = \mu_{q+1}\big(\mathbf{B}^\top\mathbf{B} - b^\top b\big), \qquad \xi_e = 1 - \frac{\lambda\big(\mathbf{B}_{\mathcal{E}\setminus e}\big)}{\lambda} \in [0,1], \tag{4.22}$$

with ξ_e the fraction of the conditioning the edge carries. It is 1 whenever $\ell_e > 0$, since an edge whose removal breaks rigidity leaves $\lambda = 0$.

4.4.4 Flexes

One further quantity that can be utilized says how much freedom is left at an agent while the framework is still flexible. Working on the length-normalized matrix $\mathbf{B}_s$ of (4.29), let $N = \ker\mathbf{B}_s$, let T be the kernel of the length-normalized matrix of the complete graph, and let

$$F = N \cap T^\perp, \qquad \dim F = \rho_{\mathcal{K}} - \operatorname{rk}\mathbf{B}, \qquad \Pi = \sum_c v_c v_c^\top \tag{4.23}$$

be the *flex space* and its projector, with $\{v_c\}$ any orthonormal basis of F. Its dimension is the rank deficit.

For a single agent, a scalar that can be used is

$$f_i = \sqrt{\operatorname{tr} \Pi[i,i]} \cdot \sqrt{n / \dim F}, \tag{4.24}$$

with $\Pi[i,i]$ the block of Π on the position and attitude coordinates of agent i, describing how free agent i is, zero once the graph is rigid. It is also rotation invariant, since $\Pi[i,i] \mapsto \bar{Q}\Pi[i,i]\bar{Q}^\top$ with $\bar{Q} = \operatorname{blkdiag}(Q, Q)$ under a global rotation leaves a trace unchanged.

4.5 Edge count bound

A question that would be useful to answer is "How few edges can make a given set of poses rigid?". In homogeneous $\mathbb{R}^d$ the answer is (2.20), and the counting argument behind it is that each bearing contributes $d-1$ constraints. Equation (4.15) shows what can be used instead of "$d-1$" in general, and an argument that can be made is that the rank is subadditive over a partition of the rows (2.3), a block's rank does not depend on which other edges are present, and rigidity needs the blocks of the chosen edges to sum to at least $\rho_{\mathcal{K}}$.

Sorting the block ranks of the complete graph in decreasing order as $c_{(1)} \geq c_{(2)} \geq \cdots$ and accumulating until they cover $\rho_{\mathcal{K}}$,

$$\boxed{m_{\text{req}} = \min\left\{ m \;:\; \sum_{k=1}^{m} c_{(k)} \;\geq\; \rho_{\mathcal{K}} \right\}} \tag{4.25}$$

is a lower bound on the number of edges any rigid graph on these poses can have.

In the homogeneous case every block rank equals $c_{\max}$ and (4.25) reduces to $\lceil \rho_{\mathcal{K}} / c_{\max} \rceil$, which is $f(n,d)$ of (2.20) in $\mathbb{R}^d$.

Algorithm 4.1 shows how (4.25) is computed in this work. It depends on the poses and the domain assignment but not on $\mathcal{E}$, so it is a constant of the formation rather than of the graph.

Equation (4.25) does not say the m_{req} largest blocks can be chosen jointly independent, so m_{req} can be strictly smaller than the true minimum number of edges required for rigidity.

In $\mathbb{R}^d$, [4] proves (2.20) attained, and outside $\mathbb{R}^d$ it is an open question. The algorithm here, as some other quantities used within this work, is intended to be used as auxiliary information that can be derived from theory to inform the policy of Chapter 5.

Definition 4.7 (minimal bearing rigidity, as used here)**.** $(\mathcal{G}, \chi)$ is minimally bearing rigid when it is infinitesimally bearing rigid and $m = m_{\text{req}}$. Where m_{req} comes from (4.25) on the complete graph this is a bound.

Where computationally feasible, an exhaustive search is used to check whether a framework is minimally bearing rigid, for the purposes of analysis in Chapter 6.

Algorithm 4.1 m_{req}, the fewest edges that could make these poses rigid. A lower bound by subadditivity, not a guarantee that a rigid graph of this size exists.

Require: positions and attitudes χ, domains $\mathcal{D}_{1:n}$
Ensure: m_{req}
if $n < 2$ **then**
  return 0
end if
if $\mathcal{D}_i = \mathbb{R}^d$ for every i, with $d \in \{2, 3\}$ **then**
  return $f(n, d)$ ▷ *closed form (2.20); the loop below reproduces it*
end if
$\mathbf{B}_{\mathcal{K}} \leftarrow$ rigidity matrix (4.13) of the complete digraph on χ
$\rho_{\mathcal{K}} \leftarrow \operatorname{rk} \mathbf{B}_{\mathcal{K}}$
for all edges k of $\mathcal{K}$ **do**
  $c_k \leftarrow \operatorname{rk}\big(\mathbf{B}_{\mathcal{K}}[3k{:}3k{+}3,\ \cdot\,]\big)$ ▷ *(4.15); one $3 \times 6n$ block, and $\mathbf{B}_{\mathcal{K}}$ is already built*
end for
sort descending, giving $c_{(1)} \geq c_{(2)} \geq \cdots \geq c_{(n(n-1))}$
$s \leftarrow 0, \quad m \leftarrow 0$
for $k = 1$ **to** $n(n-1)$ **do**
  if $c_{(k)} = 0$ **then**
    break ▷ *this block constrains nothing, and the sort makes the rest zero too*
  end if
  $s \leftarrow s + c_{(k)}, \quad m \leftarrow m + 1$
  if $s \geq \rho_{\mathcal{K}}$ **then**
    break
  end if
end for
return $\max(m, 1)$

4.6 Estimation robustness

The rank condition for infinitesimal bearing rigidity is a binary quantity. Among frameworks that are rigid, and even among frameworks that are rigid on the same number of edges, some recover their shape from noisy bearings better than others. This section tries to give quantitative tools to answer ”how good is a given rigid topology”, from the perspective of robustness against bearing measurement noise for recoverability of the formation shape.

4.6.1 Estimation error bound

Since a bearing is a unit vector, its measurement noise can be seen as a small rotation of b_{ij}, which §2.8.1 models as isotropic Gaussian noise of per-axis standard deviation σ. Under it the bearing map linearizes to $\delta z = \mathbf{B}\,\delta\chi + \sigma\eta$ with η standard normal. This is the linear Gaussian model of §2.8.2, so by (2.50) the Fisher information for $\delta\chi$ is

$$\mathcal{I} = \frac{\mathbf{B}^\top \mathbf{B}}{\sigma^2}. \tag{4.26}$$

$\mathcal{I}$ is singular and therefore not invertible, on any graph, and it prevents using Proposition 2.4.

Its kernel is $\ker \mathbf{B}$, with q_v coordinates the embedding supplies and no agent can vary, and the q_t trivial motions of §4.3, which change no bearing of the complete graph. A direction in $\ker \mathcal{I}$ is one along which the likelihood is flat and the data cannot distinguish it, and no unbiased estimator of it exists. This makes $\mathcal{I}^{-1}$ not merely uncomputable but vacuous.

One approach that can be taken to counteract this is fixing a gauge, pinning one agent at the origin and the scale at one, which restores invertibility at the cost of a choice that two topologies have to share before they can be compared. Another way comes from the fact that a linear function $C\delta\chi$ is estimable when the rows of C lie in $\operatorname{range} \mathcal{I}$, and for such a function (2.51) bounds every unbiased estimator by $C\mathcal{I}^{+}C^{\top}$. Taking $C = \Pi$, the orthogonal projector onto $\ker(\mathbf{B})^{\perp}$, covers the whole identifiable part at once, and since $\Pi\mathcal{I}^{+}\Pi = \mathcal{I}^{+}$,

$$\operatorname{Cov}\big(\widehat{\Pi\,\delta\chi}\big) \;\succeq\; \sigma^{2}\big(\mathbf{B}^{\top}\mathbf{B}\big)^{+}. \tag{4.27}$$

The pseudo-inverse inverts on the identifiable subspace and returns zero on the gauge, so no separate quotient step is needed and no agent needs to be fixed artificially.

The spectrum of $\mathbf{B}^{\top}\mathbf{B}$ therefore can be used to assess how well a topology lets the formation be recovered, and the alphabetic optimality criteria can be written to have scalar values for this. With $w_1 \geq \cdots \geq w_{\rho_{\mathcal{K}}}$ the nonzero eigenvalues of $\mathbf{B}^{\top}\mathbf{B}$, so that $w_{\rho_{\mathcal{K}}} = \lambda$,

$$\underbrace{a_{\text{opt}} = \operatorname{tr}\big((\mathbf{B}^{\top}\mathbf{B})^{+}\big) = \sum_{k=1}^{\rho_{\mathcal{K}}} \frac{1}{w_k},}_{\text{A: total MSE}} \qquad \underbrace{e_{\text{opt}} = \frac{1}{\lambda} = \frac{1}{w_{\rho_{\mathcal{K}}}},}_{\text{E: worst mode}} \qquad \underbrace{d_{\text{opt}} = -\sum_{k=1}^{\rho_{\mathcal{K}}} \log w_k\,.}_{\text{D: log-volume}} \tag{4.28}$$

Table 4.5 shows an empirical test of whether these criteria are related, on some of the configurations used in the results.

Table 4.5: Over 288 rigid graphs per configuration. The first two columns correlate E-optimality against A-optimality and against D-optimality, on logarithmic axes. The last gives the median of $a_{\text{opt}}\lambda$, which is 1 when the softest mode carries the whole trace.

configuration	A against E	D against E	median $a_{\text{opt}}\lambda$
n=8, $\mathbb{R}^3$	0.992	0.598	1.77
n=8, SE(3)	0.989	0.797	1.93
n=8, $\mathbb{R}^2 \times S^1$	0.998	0.791	1.21
mixed, n=10	0.992	0.621	1.68

Selecting a topology by λ rather than by the lowest measured error gave an error between 1.01 and 1.08 times that lowest error on the configurations tested. The objective of §5.2.2 used in experimentation in later chapters computes its robustness term from the shape error of (4.30), which is A-optimality on the normalized matrix.

4.6.2 Units and normalization

The criteria of (4.28) are not comparable between different networks. The position columns of $\mathbf{B}$ carry $1/\text{length}$, through the factor s_{ij} in (4.4), and its attitude columns are dimensionless, therefore a spectrum from $\mathbf{B}$ mixes units and depends on the physical size of the formation. The rigidity eigenvalue λ has the same defect, which is why §2.4.4 records that it depends on the scale of the configuration and not only on how well connected the framework is.

Therefore a normalized bearing rigidity matrix to counteract this is used. With

$$L(\chi) = \sqrt{\tfrac{1}{n}\sum_i \|p_i - \bar{p}\|^2}, \qquad \mathbf{B}_s = \mathbf{B} \cdot \operatorname{blkdiag}\big(L\, I_{3n},\ I_{3n}\big), \tag{4.29}$$

the position error is measured in formation radii and the attitude error in radians, both dimensionless. Scaling columns by a positive number leaves the rank and the nullity alone, so the rigidity test is untouched and only the spectrum moves. The quantity used to put a value on the error of the formation shape is

$$\boxed{\varepsilon(\mathcal{G}, \chi) = \sqrt{\frac{a_{\text{opt}}(\mathbf{B}_s)}{n}}\,,} \tag{4.30}$$

the root mean square per-agent *state error* per radian of bearing noise.

Unlike λ, (4.30) is comparable across n and across pose ranges, so two formations of different sizes, or the same formation under two topologies, can be compared directly.

Comparison across *domains* is weaker in two ways. The trace runs over the $\rho_{\mathcal{K}}$ identifiable directions while the divisor is n, so an agent carrying six degrees of freedom contributes more error coordinates than one carrying two, and ε is inflated in the richer domains relative to a per-direction figure. And for an oriented agent the sum adds a position error measured in formation radii to an attitude error measured in radians, at an implicit rate of one radius to one radian, and in $\mathrm{SE}(3)$ the attitude half is the larger of the two. Neither affects a comparison between methods on the same instances, and ε is used throughout, but a number from one domain should not be read against a number from another as though the two measured the same thing. The construction is a linearization, and Appendix A.3 shows how well this approach holds depending on bearing noise.

4.6.3 Choice among minimal graphs

Enumerating every rigid graph at exactly m_{req} edges on one set of poses and scoring each by (4.30) gives Table 4.6. These show that different topologies with the same number of edges differ significantly in terms of how well the formation shape can be estimated, and this is the basis for pursuing robustness against measurement noise as a research direction.

Efforts to improve the error (and similar) are seen in the literature as well, [33] list "the study of optimality criteria for choosing edges, such as algebraic connectivity, or quantities related to the robustness to noise" as future work, and [35] choose among minimum-edge rigid graphs by

Table 4.6: Every rigid graph at exactly m_{req} edges on one set of poses, enumerated and scored by (4.30).

configuration	rigid graphs	worst/best ε
$n{=}4\ \mathbb{R}^3$	48	3.1×
$n{=}4\ \mathbb{R}^2$	192	8.4×
$n{=}4\ \mathrm{SE}(3)$	108	77.6×
$n{=}4\ \mathbb{R}^3 \times S^1$	302	610×
$n{=}5\ \mathbb{R}^3$	4480	9.3×
$n{=}5$ mixed	18648	7079×

minimizing a sum of per-edge weights, each a lower bound on the trace of that edge's inverse Fisher information. A closely related work is [4], which maximizes the smallest nonzero eigenvalue of the bearing Laplacian over graphs with exactly the minimum number of edges. The objective used here does not fix the edge count, and studies instead the trade-off between the number of measurements and the accuracy of the recovered shape.

4.7 Combinatorial structure of the objectives

§4.5 and §4.6 show two candidate objectives that the topology can be optimized for, the number of edges and the conditioning of the recovered shape. This section analyzes their structures from a combinatorial optimization perspective, and presents a motivation to pursue, and when to pursue, a learning based sequential decision making approach to the optimization problem. This builds one of the core arguments of why this thesis work exists, and Chapter 5 is built upon the motivation here.

4.7.1 Edge count

Using $f(S) = \operatorname{rk} \mathbf{B}_S$ for the rank as a function of the edge set (poses and domains fixed), each edge contributes a three-row block, hence a subspace V_e of $\mathbb{R}^{6n}$ spanned by those rows, and the rank of the assembled matrix is the dimension of the sum of those subspaces,

$$f(S) = \operatorname{rk} \mathbf{B}_S = \dim\Big(\sum_{e \in S} V_e\Big). \tag{4.31}$$

Proposition 4.8. *The function f in (4.31) is integer valued, satisfies $f(\emptyset) = 0$, and is monotone and submodular.*

Proof. Integrality and $f(\emptyset) = 0$ are immediate. Write $V_S = \sum_{e \in S} V_e$. If $S \subseteq T$ then $V_S \subseteq V_T$ and hence $f(S) \leq f(T)$, which is monotonicity. For the marginal, the dimension formula for a sum of subspaces gives

$$f(S \cup \{e\}) - f(S) = \dim\big(V_S + V_e\big) - \dim V_S = \dim V_e - \dim\big(V_e \cap V_S\big). \tag{4.32}$$

Now let $S \subseteq T$ and $e \notin T$. From $V_S \subseteq V_T$ we get $V_e \cap V_S \subseteq V_e \cap V_T$, so $\dim(V_e \cap V_S) \leq$

$\dim(V_e \cap V_T)$, and substituting into (4.32) twice gives $f(S\cup\{e\}) - f(S) \geq f(T\cup\{e\}) - f(T)$, which is (2.43). □

The proof does not use bearings and holds for the rank of any matrix assembled by stacking row blocks indexed by a ground set, so the structure is not particular to this problem.

By Definition 2.3 a subset S is rigid exactly when $f(S) = f(\mathcal{E})$, so the edge-count objective of (1.1) is an instance of minimum submodular cover with $d = \max_e f(\{e\}) = c_{\max}$. Wolsey's guarantee (2.45) then applies to the greedy algorithm that repeatedly adds the edge of largest marginal rank gain, at $H(c_{\max})$:

- **Every agent planar.** $c_{\max} = 1$ by (4.15), every marginal of f is 0 or 1, and f is the rank function of a matroid whose bases are the minimum covers. Greedy finds a basis, so it is *exact*.
- **Any agent spatial.** $c_{\max} = 2$ and $H(2) = 1.5$, so greedy is a 1.5-approximation and nothing stronger is guaranteed.

Measured against (4.25), a greedy construction lands between 0 and 5 percent above the bound, far inside what the worst case allows.

The submodularity of the rigidity matrix rank is not specific to the present formulation. For planar frameworks of points, [34] show that the rank of the bearing rigidity matrix, viewed as a function of the selected edge set, is submodular, and use this structure to justify greedy edge selection. Earlier related design for distance rigidity in the plane appears in [32].

4.7.2 Rigidity eigenvalue

The natural next question is whether the rigidity eigenvalue of §2.4.4 has the same structure, since if it did the same guarantee would carry over to it.

Proposition 4.9 (monotonicity)**.** *Let $\lambda(S) = \mu_{q+1}\big(\mathbf{B}_S^\top \mathbf{B}_S\big)$ with $q = 6n - \rho_{\mathcal{K}}$, an index fixed by $\mathcal{K}$ and the domains, so that $\lambda(S) = 0$ exactly when $\mathcal{G}_S$ is flexible. Then $\lambda(S\cup\{e\}) \geq \lambda(S)$ for every S and every $e \notin S$.*

Proof. Adding an edge appends its row block to the matrix, so

$$\mathbf{B}_{S\cup\{e\}}^\top \mathbf{B}_{S\cup\{e\}} = \mathbf{B}_S^\top \mathbf{B}_S + b_e^\top b_e \;\succeq\; \mathbf{B}_S^\top \mathbf{B}_S, \tag{4.33}$$

since $b_e^\top b_e$ is positive semidefinite. Weyl monotonicity, Lemma 2.1, gives $\mu_k(\mathbf{B}_{S\cup\{e\}}^\top \mathbf{B}_{S\cup\{e\}}) \geq \mu_k(\mathbf{B}_S^\top \mathbf{B}_S)$ at every index k, and in particular at $k = q+1$. The index is the same on both sides precisely because q is determined by $\mathcal{K}$ rather than by S. □

The same observation is made for the bearing Laplacian in [4]. Submodularity of λ would require $\lambda(S\cup\{e\}) - \lambda(S) \geq \lambda(T\cup\{e\}) - \lambda(T)$ whenever $S \subseteq T$.

Proposition 4.10 (the rigidity eigenvalue is not submodular)**.** *Let $S \subseteq T \subseteq \mathcal{E}$ and $e \notin T$ be such that (i) $\mathcal{G}_{S\cup\{e\}}$ is flexible, (ii) $\mathcal{G}_T$ is rigid and $\lambda(T)$ is a simple eigenvalue of $\mathbf{B}_T^\top \mathbf{B}_T$ with unit eigenvector v, and (iii) $b_e v \neq 0$. Then*

$$\lambda(S \cup \{e\}) - \lambda(S) \;=\; 0 \;<\; \lambda(T \cup \{e\}) - \lambda(T), \tag{4.34}$$

so λ is not submodular.

Proof. By (i) and $S \subseteq S\cup\{e\}$ both $\mathcal{G}_S$ and $\mathcal{G}_{S\cup\{e\}}$ are flexible, so both sides of the left marginal vanish by the reading of λ in Proposition 4.9.

For the right marginal write $M = \mathbf{B}_T^\top \mathbf{B}_T$ and $M' = M + b_e^\top b_e$. Since $\mathcal{G}_T$ is rigid, $\operatorname{rk} \mathbf{B}_T = \rho_{\mathcal{K}}$, so $\ker \mathbf{B}_T$ has dimension $q = 6n - \rho_{\mathcal{K}}$; it contains $\ker \mathbf{B}_{\mathcal{K}}$, which has the same dimension, so the two coincide. Write $\mathcal{N} = \ker \mathbf{B}_{\mathcal{K}}$, the trivial-motion space of §4.3; then $\ker M = \mathcal{N}$, and the same argument applied to $T \cup \{e\}$ gives $\ker M' = \mathcal{N}$. Both matrices therefore leave $\mathcal{N}^\perp$ invariant and carry exactly q zero eigenvalues, so $\lambda(T)$ and $\lambda(T \cup \{e\})$ are the smallest eigenvalues of M and M' restricted to $\mathcal{N}^\perp$.

For a unit $u \in \mathcal{N}^\perp$,

$$u^\top M' u = u^\top M u + \|b_e u\|^2 \;\geq\; \lambda(T), \tag{4.35}$$

with equality only if u attains the minimum of $u^\top M u$ and $b_e u = 0$. The unit sphere of $\mathcal{N}^\perp$ is compact, so the left side of (4.35) attains its minimum at some $u^\star$. Equality at $u^\star$ would put $u^\star$ in the eigenspace of $M|_{\mathcal{N}^\perp}$ at its smallest eigenvalue, which by (ii) is spanned by v, and would force $b_e v = 0$, contradicting (iii). Hence $\lambda(T \cup \{e\}) > \lambda(T)$. □

Take any $n \geq 4$ agents in $\mathbb{R}^3$, let T be any minimally rigid edge set, let e be any ordered pair outside it, and let S be a single edge of T. Then $\operatorname{rk} \mathbf{B}_{S\cup\{e\}} \leq 2c_{\max} < \rho_{\mathcal{K}}$ for $n \geq 4$, so (i) holds automatically, while (ii) and (iii) are the generic case and fail only on a proper algebraic subset of the pose space.

The mechanism is that λ vanishes below a rigidity threshold, so the marginal of an edge that does not yet confer rigidity is zero however useful it later becomes, and Wolsey's bound needs submodularity on the whole lattice. Empirical tests show a witness at $n = 4$ in $\mathbb{R}^3$ with $\rho_{\mathcal{K}} = 8$ gives a left marginal of exactly 0 against a right marginal of 0.645. Restricted to triples with S and T both rigid, submodularity fails on 887 of 1493 triples across all five domains and a mixture, with worst marginal gap -4.16×10^{-1}.

The conclusion is that the eigenvalues are global and coupled in a way rank is not, so a pair of edges bracing a direction that neither braces alone is invisible to a myopic method that evaluates one edge at a time. This further motivates the use of a learning based method that has longer horizon capabilities (as discussed in [2]) for the conditioning objective. The objective function of §5.2.2 is built as a combination of sparsity and conditioning/robustness.

4.8 Summary

This section acts as a summary of the bearing rigidity theory machinery used in this work via Table 4.7, which collects the quantities derived. The observation of §5.2.3 and the objective of §5.2.2 are built using this list.

Table 4.7: The quantities this chapter defines, with the result each one comes from.

quantity	expression	description
The graph as a whole		
rigidity rank	$\operatorname{rk} \mathbf{B}_{\mathcal{G}}$	Rank of the extended bearing rigidity matrix of the current graph.
rigidity, IBR	$\operatorname{rk} \mathbf{B}_{\mathcal{G}} = \rho_{\mathcal{K}}$	The framework is rigid when its rank reaches that of the complete graph, Theorem 4.5.
unobservable subspace	$\ker \mathbf{B}_{\mathcal{G}}$, basis Z	Variations of the configuration no bearing reports, trivial motions included.
minimum edge count	m_{req}, (4.25)	Fewest edges a rigid graph on these poses could have, (4.25).
rigidity eigenvalue	$\lambda = \min_k w_k$	Stiffness of the weakest identifiable direction, the E-optimality criterion (4.28). It is the value of $\|\mathbf{B}\delta\chi\|^2$ minimized over the unit variations orthogonal to $\ker \mathbf{B}$.
softest mode	$\mathbf{B}^\top\mathbf{B}\, v = \lambda v,\ \|v\| = 1$	The variation attaining that minimum, the direction the measurements determine least.
estimation bound	$\operatorname{Cov}\big(\widehat{\Pi\delta\chi}\big) \succeq \sigma^2(\mathbf{B}^\top\mathbf{B})^+$	Cramér-Rao bound on the covariance of the recovered shape, §4.6.1.
A-optimality	$a_{\text{opt}} = \operatorname{tr}\big((\mathbf{B}^\top\mathbf{B})^+\big)$	Total mean-squared shape error over the identifiable directions, the criterion the shape error is built from.
D-optimality	$d_{\text{opt}} = -\sum_k \log w_k$	Log-volume of the error ellipsoid, with w_k the nonzero eigenvalues of $\mathbf{B}^\top\mathbf{B}$.
shape error	$\varepsilon = \sqrt{a_{\text{opt}}(\mathbf{B}_s)/n}$	Root-mean-square state error per radian of bearing noise (4.30).
What one edge is worth		
block rank	$c_{ij} = \operatorname{rk} b_{ij}$, (4.15)	The rank the pair carries on its own, bounded by $c_{\max}$.
add rank	$g_{ij} = \operatorname{rk}(b_{ij}Z)$	The rank adding $i \to j$ would contribute, exactly, by (4.17).
add independence	$\eta_{ij} = \|b_{ij}Z\|_F / \|b_{ij}\|_F$	Its continuous relaxation, zero precisely on the pairs that would add nothing.
add stiffness	$\zeta_{ij} = \|b_{ij}v\|$	How much adding $i \to j$ would brace the weakest direction, (4.19).

continued on the next page

quantity	expression	description
node slack, freedom	$\nu_i,\ f_i,$ (4.20), (4.24)	How much of the softest mode, and of the remaining freedom, sits at agent i.
block leverage	$H = b_{ij}(\mathbf{B}^\top\mathbf{B})^{+}b_{ij}^\top$	Its unit eigenvalues count the directions an existing edge holds alone.
remove rank	ℓ_{ij}, (4.21)	The rank lost if the existing edge is removed, (4.21).
remove stiffness	$\xi_{ij} = 1 - \lambda(\mathbf{B}_{\mathcal{E}\setminus ij})/\lambda$	The fraction of the conditioning the existing edge carries, (4.22).

5 Methodology

This chapter discusses the design of an environment to train an RL policy, the architecture of the policy, and the approach to training the policy. It is meant to cover the practical work done throughout the duration of this thesis work, explaining the approach taken to the topology optimization problem at hand without going into the implementation details, but providing a practical approach rather than a theoretical one. A Markov decision process is modeled for the topology optimization problem, with consideration to limitations such as normalization of quantities for generalization across networks, using randomized networks for training, and compatibility for graph neural network architectures. The chapter also covers some of the earlier attempts that did not make it to the final implementation such as various action and observation spaces and reward functions.

In the approach taken in this work, the policy is treated as an optimizer over the set of graph topologies, and its behavior is interpreted as a search procedure. Within an episode, the agent poses and domains are fixed, and the search space consists of the $2^{n(n-1)}$ possible configurations excluding self-loops. φ of §5.2.2 defines the objective that is optimized via a scalar value that represents how desirable a state is, and the goal of the policy is to maximize this objective during an episode. The policy will be evaluated by the best state it visits during an episode, measured by the objective function φ.

5.1 Approach

Topology optimization (or design) in this work focuses on a sequential modification of the graph (the framework) by adding/removing edges to/from the edge set at each step of the environment. Other possible approaches include a one-shot method that maps the agent poses and domains directly to an edge set (a multi-arm bandit formulation), constructing the topology starting with an empty graph each episode, or pruning by removing edges starting with a fully connected graph. The decision to adopt a sequential modification approach instead of a one-shot one is motivated by several considerations.

Equation (4.17) shows that the rank gain of the bearing rigidity matrix an edge addition would yield depends on the current topology, and the same is true of what the edge does to the rigidity eigenvalue. More generally, quantities and information derived from bearing rigidity theory

that may be used to inform the policy can depend on the edges that have already been selected. A method that commits to every edge at once has to predict all of those interactions in advance, whereas a method that makes sequential decisions can read the consequences of its earlier choices from the state before making the next one.

A further consideration is the suitability of sequential modification for practical applications in which topology changes must be performed online under time, communication, or modification constraints. For example, a topology may need to be repaired following a perturbation, such as dropped nodes or increased uncertainty in bearing measurements, while making as few modifications as possible. In such cases, existing edges may be retained and only the necessary additions or removals performed. A sequential modification approach naturally supports this setting by treating the current topology as the initial state and searching for a suitable topology through a sequence of local changes.

This choice is also consistent with existing approaches to graph structure optimization using reinforcement learning. In [39], graph construction and improvement are formulated as sequential decision-making problems, in which the graph is incrementally modified through a series of edge additions. More broadly, [2] identify incremental construction and modification of graph structure as a general formulation within Graph Reinforcement Learning, with actions including edge additions, removals, and rewirings.

It is possible to consider the sequential decision-making approach in two categories. *Construction/Pruning* starts from the empty graph and only adds, or starts with a fully connected graph and only prunes edges. *Modification* starts from a graph and both adds and removes. *Modification* allows for reverting previous changes on the topology, or making exploratory decisions such as adding edges that will be removed later on, to expand the covered area in search of the optimal graph topology guided by an objective. In this work, *constructive* and *modification* approaches have been experimented with, and the results presented cover policies that both add and remove edges.

5.2 Environment design

The environment is what the policy interacts with, both during training and when it is used to optimize a topology. It consists of the decision process over edge sets, the objective that scores its states and defines the reward, and the observation passed to the policy at every step.

5.2.1 Markov decision process

Within the frame of bearing rigidity theory, a multi-agent system is modeled as a framework with its corresponding graph. In this work, it is a directed graph with nodes representing agents and the directed edges are the bearing measurements, made in the local frame of the agent who makes the measurement. To be able to interact with this graph in order to optimize its topology by choosing which edges should exist, a Markov decision process is defined as follows.

Let $\mathcal{P} = (\mathcal{V} \times \mathcal{V}) \setminus \Delta$ the $n(n-1)$ ordered pairs (edges), where Δ is the self-loops, and $\mathbb{D} = \{\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^2 \times S^1, \mathbb{R}^3 \times S^1, \mathrm{SE}(3)\}$ the agent domains.

Definition 5.1 (the topology optimization MDP)**.** $\mathcal{M} = \big(\mathcal{S},\ \mathcal{A},\ \mathcal{A}(\cdot),\ f,\ r,\ \gamma,\ \mu_0\big)$, where

$$\mathcal{S} = \Big\{\, s = (\chi,\ \mathcal{D}_{1:n},\ \mathcal{E}) \ :\ \mathcal{D}_{1:n} \in \mathbb{D}^n,\ \chi \in \textstyle\prod_i \mathcal{D}_i,\ \mathcal{E} \subseteq \mathcal{P},\ \Big\}, \tag{5.1}$$

$\mathcal{A}$ is the action set of §5.2.1.1, $\mathcal{A}(s) \subseteq \mathcal{A}$ the admissible actions at s, $f : \{(s,a) : a \in \mathcal{A}(s)\} \to \mathcal{S}$ the transition, r the reward of (5.10), $\gamma \in (0,1)$ the discount, and μ_0 a distribution on $\mathcal{S}$.

The number of agents in the formation, n, is fixed throughout the training procedure, mainly because of implementation related limitations (e.g. action space shape cannot change per episode). This is discussed later on in §6.1.1.

State. The agent poses and domains are fixed in the beginning of an episode via μ_0. The quantities treated as episode constants are functions of $(\chi, \mathcal{D}_{1:n})$ alone.

Transition. f is deterministic and a state change is guaranteed, unless the ”skip” action in §5.2.1.3 is available.

Reward. r is a function of (s,a) alone and is deterministic.

Termination. Unless the ”stop” action in §5.2.1.3 is available, no state is absorbing and $\mathcal{M}$ has no terminal state.

5.2.1.1 Action spaces

Considered action spaces and ways of interacting with the environment are listed in Table 5.1, and Figure 5.1 shows what one action does in each. These correspond to the one-shot, *modification*, *construction* methods as discussed above, a two-step process of toggling a directed edge, and an approval action to toggle an edge proposed by the environment.

The action type that selects nodes sequentially in two steps to toggle a directed edge is also discussed in [2] and [39], used for the same reason that an action set over pairs grows with $O(n^2)$ and this approach grows with $O(n)$. If this action type is used, the state carries an additional variable (named ”stub” in literature) that indicates whether or not a node is selected, and which one if so. The policy network then has to account for whether or not a node should be selected at the current step, or an edge should be toggled whose base is the current selected node. The reduced complexity/size of the action output compared to other action space types (therefore the action head architecture of the policy network) is beneficial during training, allowing for using larger networks. At inference time however, it did not allow for any significant improvements in the experiments conducted in this work, as the architecture passes each node embedding separately through the action head, as discussed in §5.3.2.

Variants of the defined action space types are also implemented (indicated with an $*$ and omitted from the list below), where e.g. the self-loops are kept out of the possible actions rather than masked within the policy network (with the idea of investigating the effects of action masking). Variants with the ”skip/stop” action of §5.2.1.3 in the action space were also implemented. The ”DecideOnEdge” action type is from an earlier abandoned experiment where the advantage of a learned method is compared against a uniformly random policy, and it was utilized by the policy network taking an action to confirm or reject an edge toggle randomly proposed by the environment at each step.

Table 5.1: Each action space type, how many values the policy has to emit to indicate a specific action, and how many actions one edge toggle costs. $*$ marks the types where self-loops and the skip/stop action are included depending on the configuration.

action space	$\lvert\mathcal{A}\rvert$	policy outputs	actions per edge toggle
`AllEdges`$*$	2^{n^2}	$O(n^2)$	-
`AddRemoveEdge`$*$	$2n^2$	$O(n^2)$	1
`AddEdge`$*$	n^2	$O(n^2)$	1
`SelectNodesSequentially`$*$	n	$O(n)$	2
`DecideOnEdge`	2	$O(1)$	1

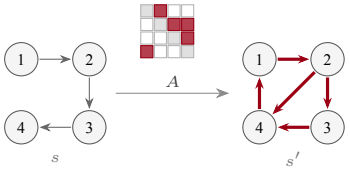
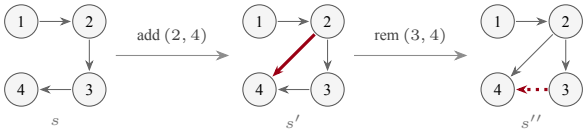
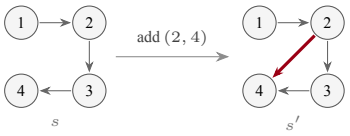
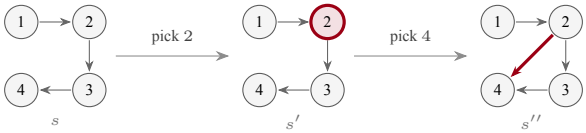
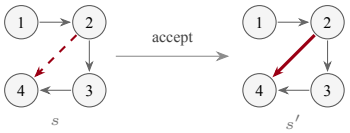
action space	visual representation s
`AllEdges`	
`AddRemoveEdge`	
`AddEdge`	
`SelectNodesSequentially`	
`DecideOnEdge`	

Figure 5.1: Visual representation of the action spaces of §5.2.1.1, each acting on the same state s. Gray arrows are the edges the graph already has, and the red ones mark those the action changed, solid for an edge added and dotted for one removed. In case of ”SelectNodesSequentially”, the selected node is highlighted.

5.2.1.2 Action masking

Invalid actions for a state (such as adding an edge that already exists) are masked out inside the policy network, rather than penalizing via the reward or invalidating within the environment. The state of the environment carries the adjacency information, and this information is passed onto the policy via the observation of §5.2.3. The policy network then masks out the action head's output logits (or q values) with negative numbers large in absolute value, or effectively negative infinity.

$$\tilde{q}_a = \begin{cases} q_a, & a \text{ valid} \\ -\infty, & \text{otherwise,} \end{cases} \qquad \text{valid} = \{\text{add } (i,j) \notin \mathcal{E}\} \cup \{\text{remove } (i,j) \in \mathcal{E}\}. \tag{5.2}$$

In the policy gradient approaches, this masking results in near zero probability of selecting the invalid action after softmax is applied. In value based methods, this ensures that the argmax operator does not select the invalid action. One caveat is in the case of ε-greedy exploratory actions, or e.g. warming up the replay buffer by taking uniformly random actions in the initial stage of training. To accommodate these problems, the same masking mechanism is artificially integrated into the cases where an action may be picked in random.

5.2.1.3 Skip and stop actions

The policy is mainly treated as an optimizer and the experiments are conducted by providing a budget (e.g. a maximum number of steps taken during an episode) and evaluating the output by the best visited state during the episode. Another possible approach is to let the policy itself decide on whether or not the current state is optimized and the procedure should terminate. For this purpose, a "skip/stop" action is implemented as a configurable toggle within various actions space types.

The point of having a "skip" action is to let the policy be able to keep the state at a certain point, which may be desirable in case of a continuous deployment in a dynamic scenario. A problem faced with this is that since the action is a no-op, which means it does not change the state and just increments the step counter within the episode, and since a potential-based reward function is used as described in §5.2.2, the reward seen by the policy during training after taking the "skip" action is zero (if no other reward shaping is in effect, which is the case in most of the training runs). The action is an absorbing cycle of zero reward available at every state, and this easily leads to e.g. an early policy collapse for on-policy methods during training, or a strong dependency to exploration in off-policy methods.

An intuitive explanation for this is that in the initial stages of training, actions taken will not be necessarily "good" actions that increase the objective function's value (on average[1]) and

[1] In fact the policy is not *expected* to always take actions that increase the objective's value, even after the training is completed. This is one of the main motivations of using a learning based approach for the topology optimization problem and what could separate it from a greedy or annealing heuristic. §B.7 shows quantitative results on which actions the policy decides to take, and whether or not these actions result in an increase in φ.

therefore result in a positive reward. The skip action provides a deterministic null reward and this is favored over other actions.

If a ”stop” action is available instead, taking it terminates the episode, so the policy has to decide whether the current graph is optimal. A policy that decides on this by only looking one step ahead stops at the first local optimum of φ and behaves as a hill climber. By §4.7.2 that is something the conditioning objective punishes, since two edges can be worth more together than either is alone and a method choosing one edit at a time cannot see it. The stop action is therefore not a free convenience added to the search framing, and it changes what the policy is able to find.

5.2.2 Objective and reward

To be able to train a policy that optimizes a network topology, there needs to be a definition of what optimal is, and a reward signal that can be used to train the RL policy. For this purpose, a function (φ) that maps a given state to a scalar score, that quantifies how good (or how desired) a state is, is derived. This score acts as the objective that is being optimized, and the reward signal is taken as the change in this objective after taking an action.

Three main components characterize this objective function, in alignment with the optimization problem described in §1.2. These are

- Rigidity component: Should score IBR frameworks higher,
- Sparsity component: Should score sparse graphs higher,
- Robustness component: Should score frameworks more robust against measurement noise higher.

The first one is a necessary condition for the third one, as it is not possible to estimate the poses of agents in a non-rigid framework, using only bearing measurements. The second and third are competing components, a framework whose graph is fully-connected has the lowest estimation error among all configurations and the estimation error (on average) increases as the total edge count decreases. To balance these components, a weighted linear combination of terms each describing individually these components is used, written in (5.8).

5.2.2.1 Score requirements

The defined objective function (and the reward as an extension) needs to have properties that make it invariant to network sizes and heterogeneous compositions. These would allow for training of policies across various networks instead of a single homogeneous network distribution, and provide a stable and well-behaving reward signal.

Three requirements, two by the nature of the problem and one for practicality, can be put on φ. It has to be dimensionless, so that one policy covers different network sizes and domains without the scale of the reward moving under it. It has to be monotone in the two quantities the

problem cares about, rigidity and edge count, increasing in $\operatorname{rk} \mathbf{B}_{\mathcal{G}}$ and decreasing in m. It has to be cheap enough to evaluate at every step.

5.2.2.2 Unnormalized objective

To give an example as to why these requirements are in place, an objective function as in (5.3) is given. This function scores a given state without a consideration for estimation quality of the inverse problem.

$$\varphi_{\text{old}} = w_r \operatorname{rk} \mathbf{B} - w_e\, m, \qquad (w_r, w_e) = (20, 10). \tag{5.3}$$

By §4.4 a rank-adding edge contributes $c_{\max}$, so the score pays what Table 5.2 shows. The gain

Table 5.2: The change in the state's score (gain) after an action in $\mathbb{R}^3$ where $c_{\max} = 2$ and in $\mathbb{R}^2$ where $c_{\max} = 1$. The ratio between the two columns is the preference the score expresses, and it differs between the two domains.

	gain from a rank-adding edge	gain from pruning a redundant edge
$\mathbb{R}^3$ ($c_{\max} = 2$)	$2 \cdot 20 - 10 = +30$	$+10$
$\mathbb{R}^2$ ($c_{\max} = 1$)	$1 \cdot 20 - 10 = +10$	$+10$

from adding an edge is dependent on the network size and domain. An optimal action would result in a gain of 50 at $n{=}4/\mathbb{R}^2$, 270 at $n{=}8/\mathbb{R}^3$, 590 at $n{=}16/\mathbb{R}^3$, which shifts e.g. the critic's target range whenever n or the domain changes.

5.2.2.3 Normalized objective

$$\boxed{\varphi(s) = \frac{w_{\text{rank}} \operatorname{rk} \mathbf{B}_{\mathcal{G}} - w_{\text{edge}}\, m\, c_{\max}}{\rho_{\mathcal{K}}}} \qquad (w_{\text{rank}}, w_{\text{edge}}) = (100, 25). \tag{5.4}$$

$mc_{\max}$ is the maximum rank m edges can carry, so the second term describes the fraction of the required rank spent on edges. Both numerators are ranks and both denominators are $\rho_{\mathcal{K}}$, so φ is dimensionless.

For an edge contributing Δr,

$$\begin{aligned} \Delta\varphi &= \frac{w_{\text{rank}} \Delta r - w_{\text{edge}} c_{\max}}{\rho_{\mathcal{K}}}, \\ \text{so at } \Delta r = c_{\max}: \quad \Delta\varphi &= \frac{(w_{\text{rank}} - w_{\text{edge}}) c_{\max}}{\rho_{\mathcal{K}}} > 0 \iff w_{\text{rank}} > w_{\text{edge}}, \end{aligned} \tag{5.5}$$

for any geometry, domain mix and n, because the same factor $c_{\max}/\rho_{\mathcal{K}}$ appears on both sides. Pruning a redundant edge ($\Delta r = 0$) gains $w_{\text{edge}} c_{\max}/\rho_{\mathcal{K}}$, so

$$\frac{\text{value of a rank-adding edge}}{\text{value of pruning a redundant one}} = \frac{w_{\text{rank}} - w_{\text{edge}}}{w_{\text{edge}}} = 3, \tag{5.6}$$

reproducing $\mathbb{R}^3$'s 3:1 preference identically in every domain. The optimum is

$$\varphi^\star = w_{\text{rank}} - w_{\text{edge}}\frac{m_{\min} c_{\max}}{\rho_{\mathcal{K}}} \;\le\; w_{\text{rank}} - w_{\text{edge}} = 75, \tag{5.7}$$

with equality iff the poses admit a perfectly packed rigid graph. The edge term is normalized by $\rho_{\mathcal{K}}$ and not by m_{req}. By (4.25) the latter is a lower bound rather than the true minimum, so it can only understate how many edges are needed, and an understated denominator over-penalizes every edge. It would also make (5.5) contingent on that bound being tight, which is not something the formulation can assume.

5.2.2.4 Robustness term

Equation (5.4) rewards rigidity and sparsity but says nothing about how accurately the shape of the formation can be recovered from noisy bearings. Two rigid graphs with the same number of edges score the same under it, although by §4.6.3 their accuracy can differ by orders of magnitude, and that accuracy is governed by the conditioning of $\mathbf{B}$. Under Gaussian bearing noise $\mathbf{B}^\top\mathbf{B}$ is, up to the noise variance, the Fisher information of the formation's shape, so its scalar summaries are the classical criteria of optimal experimental design. The rigidity eigenvalue λ is the E-optimality criterion and bounds the error along the worst direction, $a_{\text{opt}} = \operatorname{tr}\big((\mathbf{B}^\top\mathbf{B})^+\big)$ is the A-optimality criterion and measures the total error, and the log-determinant d_{opt} is the D-optimality criterion and measures the volume of the error ellipsoid.

With a positive weight, these criteria are maximized by the complete graph, which is in line with the expectation that the most robust graph is the fully-connected one. However, the values of these criteria have no fixed scale. For instance, λ carries the units of $\mathbf{B}$, so it decays with n and scales with the size of the formation, and using it directly in (5.8) would therefore not meet the requirements.

The robustness term therefore can/should enter through a bounded, dimensionless score. With $q(s) \in (0,1)$,

$$\boxed{\varphi(s) = \underbrace{\frac{w_{\text{rank}}\,\operatorname{rk}\mathbf{B}}{\rho_{\mathcal{K}}}}_{\text{rigidity}} - \underbrace{\frac{w_{\text{edge}}\,m\,c_{\max}}{\rho_{\mathcal{K}}}}_{\text{sparsity}} + \underbrace{\kappa\,\frac{w_{\text{edge}} c_{\max}}{\rho_{\mathcal{K}}}\,\mathbf{1}[\text{IBR}]\;q(s)}_{\text{robustness}}} \tag{5.8}$$

In a flexible (non-rigid) framework, the shape cannot be recovered and no criterion is defined, and the $\mathbf{1}[\text{IBR}]$ term is the binary toggle that turns on the component only if the framework is rigid. Since $q \ge 0$, becoming rigid can only raise the score. The factor $w_{\text{edge}} c_{\max}/\rho_{\mathcal{K}}$ is exactly what one edge costs in (5.4), so multiplying by it expresses the term in edges. Its value lies between 0 and κ edges for any n and any mixture of domains, which is what lets one value of κ mean the same thing across configurations.

The criteria vary by factors rather than by amounts, therefore it makes sense to have q as a logistic function of the logarithm of a criterion. The logistic sigmoid is used because it maps that logarithm onto $(0,1)$, which keeps the term bounded, while still responding to every im-

provement near its middle. Two constants play important roles with this approach. The center is the value of the criterion at which $q = \frac{1}{2}$, so it decides what counts as an average graph, and the width, measured in decades, sets how quickly q changes around the center. If the width is too narrow, or the center is far from the values the policy actually visits, most graphs get a score close to 0 or 1, and if the width is too wide most graphs get a score close to $\frac{1}{2}$. In both cases the objective stops telling good graphs from bad ones.

Table 5.3: The two forms of the robustness score q in (5.8). w_k are the nonzero eigenvalues of $\mathbf{B}^\top\mathbf{B}$ and $\mathbf{B}_s$ is the length-normalized matrix of (4.29). Both forms are defined only on rigid frameworks and cost one matrix decomposition per step.

criterion	expression	description
Spectral form, $q = \mathrm{sigmoid}\big((g - g_{\mathrm{ref}})/\varsigma_g\big)$. The center g_{ref} is the median of g over k graphs built with the constructive method at every reset.		
E-optimality	$g = \log_{10}\lambda,\ \lambda = \min_k w_k$	Stiffness of the weakest identifiable direction.
A-optimality	$g = -\log_{10} a_{\mathrm{opt}},\ a_{\mathrm{opt}} = \sum_k 1/w_k$	Total mean-squared shape error.
D-optimality	$g = -d_{\mathrm{opt}},\ d_{\mathrm{opt}} = -\sum_k \log w_k$	Log-volume of the error ellipsoid.
Shape-error form, $q = \mathrm{sigmoid}\big(-(\log_{10}\varepsilon - c_\varepsilon)/\varsigma_\varepsilon\big)$. The center $c_\varepsilon = 1.35$ and width $\varsigma_\varepsilon = 0.70$ are fixed in advance, so nothing is computed at reset.		
Shape error	$\varepsilon = \sqrt{a_{\mathrm{opt}}(\mathbf{B}_s)/n}$	Root-mean-square shape error per radian of bearing noise.

Two forms of q were implemented, summarized in Table 5.3. The spectral form uses the criteria directly, but since none of them has a fixed scale, the center is computed at the beginning of each episode as the median criterion value of k rigid graphs built on the same poses with the constructive method of §6.1.1, so $q = \frac{1}{2}$ means as well conditioned as a typical constructed graph.

Reasons to use a constructive greedy approach to create reference graphs are that randomly generated graphs do not have the guarantee of being rigid, have high edge counts on average (though it is possible to overcome this as in §5.4.2.1), and that the topologies generated with the constructive greedy will be similar enough for the average case of the learned policy, meaning they will have minimal (or close to minimal) number of edges for rigidity, and have decent eigenvalues to further incentivize the policy to learn to do better during training. The constructions require many rank computations at every reset, which grow quickly with n and are paid again whenever the policy is used (once per episode), and the objective depends on the output of another heuristic method rather than on the graph alone. This motivates to drop the dependency to a costly operation to have reference values for the chosen criterion.

The shape-error form avoids the dependence on the heuristic algorithm to get reference values for the criteria. The shape error ε of (4.30) is the A-optimality criterion computed on the length-normalized matrix $\mathbf{B}_s$ and divided by n, and it is read directly from the singular values of $\mathbf{B}_s$, without solving the estimation problem (e.g. with the Gauss-Newton method of §2.8.3). Since

the normalizations remove the dependency on the size of the formation and on the number of agents, its center and width can be fixed once in advance empirically, from the shape errors of greedy near-minimal graphs over a range of network sizes and domains. The trained policies in Chapter 6 use this form.

Since q lies in $(0, 1)$, the robustness term is always worth less than κ edges. For two rigid graphs on the same poses, where the second one has k more edges and the scores are q_1 and q_2,

$$\varphi(s_2) - \varphi(s_1) = \frac{w_{\text{edge}} c_{\max}}{\rho \kappa} \big(\kappa\,(q_2 - q_1) - k\big), \tag{5.9}$$

so the graph with more edges can score higher only if $\kappa > k$. For $\kappa = 0$ the objective reduces to (5.4), for $\kappa \leq 1$ a rigid graph never scores higher than a rigid graph with fewer edges, so among rigid graphs the term only decides between those with the same number of edges, and for $\kappa > 1$ it allows trading edges for better conditioning, with κ as an upper bound on that trade. There is no principled value for κ, and it would be dependent on the use case, so it is treated as an experimental parameter in §6.5. All constants in (5.8) are fixed within an episode, so φ is a function of the state alone.

5.2.2.5 Reward function

Though the maximum of the objective function cannot be known, it allows to compare states quantitatively. This information is a useful signal that can be used as a reward during training. However, using the function's value directly as the reward led to reward hacking during training, as the value of the objective does not change much near its maxima and the return of an episode becomes the sum of the value of the objective function at each step's state, hence the policy does not have a strong motivation to pursue the best possible state rather than a good enough one.

Therefore a potential-based reward shaping approach (§2.5.4) is taken, and the step reward is the improvement in the objective:

$$r_t = \gamma\varphi(s_{t+1}) - \varphi(s_t) \quad \big[-\text{time penalty} + \text{action reward} + \text{terminal bonus, when enabled} \big]. \tag{5.10}$$

Many works in the literature, as pointed out in [2], use a single reward signal at the final step of an episode, which may cause problems because of a sparse reward signal, or credit assignment issues. In this work, since evaluating the objective function is not a costly operation, using a potential-based shaping approach allows to provide a dense reward signal without a significant slowdown in the training loop.

5.2.3 Observation design

The state of the environment consists of the agents with their poses and domains, and the adjacency of the formation graph. However this state information is not directly passable to the policy, and there is a need to construct better formatted observations that meet the practical requirements such as being in a tensor form, and it is also desirable to have these observations

Table 5.4: The terms of (5.8) and (5.10). The three terms of φ are the three components the objective has to carry, and each of them is a fraction of $\rho_{\mathcal{K}}$, so the whole score is dimensionless.

term	expression	description
reward	$r_t = \gamma\varphi(s_{t+1}) - \varphi(s_t)$	The improvement of the objective, so the shaping is potential based.
objective	$\varphi(s)$, (5.8)	The score of a state, the sum of the three terms below.
rigidity	$w_{\text{rank}} \operatorname{rk} \mathbf{B}_{\mathcal{G}} / \rho_{\mathcal{K}}$	The fraction of the required rank the graph attains.
sparsity	$w_{\text{edge}}\, m\, c_{\max} / \rho_{\mathcal{K}}$	The fraction of the required rank spent on edges, since $mc_{\max}$ is the most m edges can carry.
robustness	$\kappa\,(w_{\text{edge}} c_{\max} / \rho_{\mathcal{K}})\, \mathbf{1}[\text{IBR}]\, q(s)$	How well conditioned the graph is, worth at most κ edges and only on a rigid framework.
cost of one edge	$w_{\text{edge}} c_{\max} / \rho_{\mathcal{K}}$	What one edge costs in φ, which is what denominates the robustness term in edges.
robustness score	$q(s) \in (0, 1)$, Table 5.3	A criterion of Table 4.7, squashed onto a bounded, dimensionless scale.

contain information that is derived by bearing rigidity theory and classical analysis to further inform the policy. The design of the observation discussed in this section aims to allow for training informed policies that can generalize across networks and be able to learn to exploit the underlying bearing rigidity mechanism of the task. Because of the considered policy architectures (graph neural networks), the observation channels correspond to node/edge features of the graph and the adjacency matrix. There are certain considerations when constructing the observation vector, separated by ”channels” for easier analysis and discussion, that come from architectural constraints and policy network capabilities, requirements from the optimization task and generalizability goals, and computational cost.

The explicit information contained within the state, the agents and the graph, is limited by itself, but it allows for further derivations. In this work, the aim is to use/exploit existing theory and algebra to inform the policy to allow for easier learning and better performance. It is possible in theory to simply pass the state information to a policy and expect it to learn the underlying bearing rigidity machinery and to learn to solve the optimization task, however there are limitations with such an approach, especially for generalization capability.

Infinitesimal bearing rigidity is invariant to similarity transforms of the whole formation, global translation, rotation, and scaling. For a policy to construct formation topologies that result in IBR frameworks, its output also needs to be invariant to these transformations. Therefore using absolute positions and orientations in the observation would require that the policy can generalize with these inputs. It is possible to augment the networks the policy sees during training by applying these transformations, however there is still the risk that result will not be invariant. The problem is that the positions (and orientations) are continuous vectors that can

take any value, and invariance to the similarity transformations should be embedded in either the observation provided, or the architecture (§5.3.1.2).

The central pieces of information the task relies on are the bearing measurements. Bearing measurements suffer the same problem as positions or orientations, they are continuous vectors that can take an infinite number of values. However, by nature, bearing measurements (in the local frames of the agents) are invariant to the global transformations. This allows for easier use of them within the observation vector. One other issue is that the policy architecture used to take a graph as input contains a graph neural network, which relies on node (and edge) features as input. It is not possible to use the bearing measurements made by an agent as features for the corresponding node, as the number of these measurements changes and the GNN architecture expects a fixed-size node feature vector, and using the bearing measurements in the node feature vector would make it dependent on the node's degree. Therefore the bearing measurements are given to the policy network as *edge* features, and this necessitates use of GNN architectures that allow edge features as inputs.

5.2.3.1 Channels

The observation is a collection of node and pair (edge) blocks,

$$o = \big(\underbrace{X \in \mathbb{R}^{n\times d_v}}_{\text{node}}, \underbrace{C \in \mathbb{R}^{n\times 3}}_{\text{coords}}, \underbrace{\mathcal{Z} \in \mathbb{R}^{n\times n\times d_e}}_{\text{pair}}, \underbrace{A \in \{0,1\}^{n\times n}}_{\text{adjacency}}, \underbrace{\text{sel} \in \{0,1\}^{n}}_{\text{pointer state}}\big), \tag{5.11}$$

where X holds the channels of each agent, C the normalized positions, $\mathcal{Z}$ the channels of each ordered pair, A the adjacency matrix, and sel the pointer register of the pointer action space, which is zero under the other action spaces. Table 5.5 lists every channel with its expression, its width, and the feature set it belongs to. The base set is always present, and every other set can be left out as a whole, which is how the observation arms of §B.8 are built. Several of the channels are quantities that no distributed agent could compute, and §5.2.3.4 sorts the channels by what each one requires an agent to know.

Most of the rigidity channels are the one-step quantities of §4.4.3, which describe what a single addition or removal would do to the graph, and Table 4.7 collects the quantities of the rigidity matrix they are read from.

Table 5.5: Every channel of (5.11), its expression, its width and the feature set it belongs to. The global channels are single numbers tiled over the agents, so they sit in X like the other node channels. Z is a basis of $\ker \mathbf{B}_G$, v the softest mode, Π the flex projector of (4.23), and b_{ij} the row block edge $i \to j$ contributes. The rigidity channels are evaluated on the length-normalized matrix $\mathbf{B}_s$ of (4.29), and the scale of the channels marked with a dagger is removed as in (5.13).

channel	expression	width	set	description
Node channels X				

continued on the next page

channel	expression	width	set	description
domain	$e_{\mathcal{D}_i} \in \{0,1\}^5$	5	base	Which of the five domains agent i lives in.
degrees	$\widetilde{\deg}_i^{\text{in}}$, $\widetilde{\deg}_i^{\text{out}}$	2	base	Bearings agent i takes, and bearings taken of it.
closeness	$C_i = \frac{n-1}{\sum_j d(i,j)}$	1	centralities	How near agent i is to the others along the graph.
eigenvector	Perron vector of A	1	centralities	Agent i is central when the agents measuring it are.
node betweenness	$B_i = \sum_{s\neq i\neq t} \sigma_{st}(i)/\sigma_{st}$	1	centralities	Share of the shortest paths that pass through agent i.
node freedom†	$f_i = \sqrt{\operatorname{tr} \Pi[i,i]} \cdot \sqrt{n/\dim F}$	1	null space	How free agent i still is (4.24).
node slack†	$\nu_i = \left(\lVert v_i^{\text{pos}} \rVert,\ \lVert v_i^{\text{att}} \rVert\right)$	2	softest mode	How much of the softest mode sits at agent i, position and attitude apart.
Global channels, tiled over the agents				
rank deficit	$(\rho_{\mathcal{K}} - \operatorname{rk} \mathbf{B}_{\mathcal{G}})/\rho_{\mathcal{K}}$	1	rigidity state	How far the graph is from being rigid.
edge budget	m/m_{req}	1	rigidity state	Edges spent against the bound (4.25).
rigidity	$\mathbf{1}[\text{IBR}]$	1	rigidity state	Whether the shape is identifiable at all.
state quality	$\mathbf{1}[\text{IBR}] \cdot q(s)$, (5.12)	1	state quality	How well conditioned the current graph is, on the scale the objective uses.
Coordinates C				
positions	$\tilde{C}_i$, (5.13)	3	base	Normalized positions, read by the EGNN backbone through squared distances only.
Pair channels $\mathcal{Z}$, over every ordered pair				
bearing	$b_{ij} = R_i^\top \hat{p}_{ij}$	3	base	The bearing agent i measures, or would measure, of agent j.
edge exists	A_{ij}	1	base	Whether the pair carries an edge in the current graph.
reciprocity	A_{ji}	1	base	Whether the reverse edge is present.
common neighbors	$\lvert\mathcal{N}_i \cap \mathcal{N}_j\rvert$, normalized	1	base	How much of their neighborhood the two agents share.
edge betweenness	B_{ij}, as B_i over edges	1	centralities	Share of the shortest paths that use the edge.

continued on the next page

channel	expression	width	set	description
add independence	η_{ij}, (4.18)	1	null space	How independent of the current measurements the pair would be.
block rank	$c_{ij}/c_{\max}$, (4.15)	1	edge rank	The rank the pair carries on its own.
add rank	$g_{ij}/c_{\max}$, (4.18)	1	edge rank	The rank adding $i \to j$ would contribute, exactly, by (4.17).
add stiffness†	$\zeta_{ij} = \|b_{ij}v\|$	1	softest mode	How much adding $i \to j$ would brace the weakest direction (4.19).
remove rank	$\ell_{ij}/c_{\max}$, (4.21)	1	removal	The rank lost if the existing edge is removed.
remove stiffness	$\xi_{ij} = 1 - \lambda(\mathbf{B}_{\mathcal{E}\setminus ij})/\lambda$	1	removal	The fraction of the conditioning the existing edge carries.
Remaining blocks of (5.11)				
adjacency	$A \in \{0,1\}^{n\times n}$	n	base	Which actions are admissible, used by the mask (5.2).
pointer register	sel $\in \{0,1\}^n$	1	base	The agent held between the two picks of the pointer action space, zero otherwise.

The state quality channel is the only one that describes the current graph itself,

$$\text{quality}(s) = \mathbf{1}[\text{IBR}] \cdot q(s) \ \in [0,1), \tag{5.12}$$

where q is the robustness score of Table 5.3, the same one the objective uses.

Candidate bearings. The bearings are for every possible edge and not the ones present in the current state of the formation. This is a modeling choice, because an agent cannot know a bearing it has not measured yet, and §5.2.3.4 discusses the availability of information in the observation design.

Graph-theoretic features. These are the standard centralities used in graph representation learning. They are the most expensive part of the observation, since closeness and betweenness both require shortest paths between all pairs at every step. None of them is a rigidity quantity, and the out-degree already carries the per-agent information that Table 4.4 shows to be relevant for rigidity.

5.2.3.2 Normalization

$$\tilde{C}_i = \frac{p_i - \bar{p}}{L(\chi)}, \qquad \widetilde{\deg}_i = \frac{\deg_i}{m_{\text{req}}/n}, \qquad \tilde{\zeta}_{ij} = \frac{\zeta_{ij}}{\bar{\zeta}}. \tag{5.13}$$

Positions are centered and scaled to unit root-mean-square radius before they are given to the policy. Bearings are already invariant to this transform, but the EGNN backbone of §5.3.1.2 reads coordinates through squared distances $\|x_i - x_j\|^2$, which are not. The rigidity quantities are computed from the true poses, and only the copy given to the network is normalized.

The degree and common-neighbor counts are divided by m_{req}/n, the mean degree of a minimally rigid graph, rather than by $n - 1$. Targeted rigid graphs are sparse, so dividing by the mean degree of the complete graph would push these channels toward zero as n grows, and the same structure would look different at two network sizes.

The rigidity channels are computed on the length-normalized matrix $\mathbf{B}_s$ of (4.29), so they do not change when the formation is rescaled. The rank channels are divided by $c_{\max}$, so they lie between zero and one in every domain, and the stiffness lost by removing an edge is a fraction of λ and lies between zero and one as well. The channels built from the softest mode, the stiffness gain of a pair and the slack of an agent, are divided by their own mean over the graph, and the freedom of an agent is scaled to unit root-mean-square. These channels then state which pair or agent is soft relative to the others rather than how soft it is, which is the part that carries over between formations. The independence channel of (4.18) is instead normalized per pair by $\|b_{ij}\|$. On a rigid framework its raw values are all at machine zero, and dividing them by their spread over the graph would turn rounding error into a feature of order one.

5.2.3.3 Invariance

To verify that the designed observation allows for inputs to the policy that are invariant to the similarity transformations bearing rigidity is naturally invariant to, networks are transformed and every channel is compared before and after, in each domain, which gives Table 5.6.

Table 5.6: The largest change in any observation channel when the whole formation is transformed, measured by transforming the network and differencing every channel. Translation and uniform scaling leave the observation unchanged in every domain. Rotation does not, in exactly the two domains whose agents carry no frame.

domain	translate	uniform scale	rotate
$\mathbb{R}^2$, $\mathbb{R}^3$	10^{-14}	10^{-14}	~ 0.8 **(bearings)**
$\mathbb{R}^2 \times S^1$, $\mathbb{R}^3 \times S^1$, SE(3)	10^{-14}	10^{-13}	10^{-13}

Translation and uniform scaling leave every channel unchanged in every domain, so the trained policy with weights relying on these channels is invariant to those. Rotation does not, and the reason is (4.1). An agent that carries a frame reports $R_i^\top \hat{p}_{ij}$, which rotates together with the world and is therefore invariant, while an agent in $\mathbb{R}^2$ or $\mathbb{R}^3$ has no frame and reports the direction in the world frame, which is not. The task is invariant to rotations but the observation is not, in exactly the two frameless domains, so a policy trained there has to learn the invariance from data. To reduce the effect, a random global rotation of the formation is applied at the start of every episode, which leaves the task unchanged and shows the policy the same kind of formation in many orientations.

5.2.3.4 Information tiers

The choice of the information given to the policy via the observation, independent of the implementation, not only determines what the policy requires during training, but also inference time. Depending on the application and use case, certain information may not be available. For example if the network consists of a high number of agents, a decentralized optimization protocol would be desirable, and the complete observation with every channel of Table 5.5 allows only a centralized formulation with strong assumptions about availability of information, such as ”candidate” bearing measurements not currently measured. Therefore it makes sense to consider what assumptions are being made in order to inform a policy, and the designed observation channels are sorted in three tiers in Table 5.7.

Table 5.7: What each channel would require an agent to know if the policy were distributed. Tier 1 is always local, tier 2 depends on a modeling choice about sensing range that is not made here, and tier 3 is global algebra no agent could compute on its own.

tier	availability	channels
1	always	own pose and domain, in/out degree, the bearings currently measured, one-hop (or k-hop) messages
2	under a sensing-radius assumption	bearings to agents *not* currently measured, i.e. all-pairs bearings
3	not available	rk $\mathbf{B}$, rank deficit, c_k, IBR, every null-space channel, the centralities

An agent does not know $\hat{p}_{ij}$ before measuring it, so whether the all-pairs bearings are a legitimate local observation depends on a modeling choice that is not made here. If detection such as omnidirectional vision within some radius is available, and only a maintained link has a considerable cost, then the all-pairs bearings may be available to some extent. This is not only a question in a distributed/decentralized case, but also a centralized one, as it is not about an individual agent being able to access some information but rather being able to compute it.

Most of the *derived* channels, those that are not immediately available but calculated using theory, are tier 3. This means that in this work, the bearing rigidity informed policies are centralized and expect to have calculated quantities provided as input. Whether policies that have only tier 1 inputs reconstruct the classical bearing rigidity tooling within their neural networks is unknown and not investigated in this work. §B.8 shows experimental results comparing when the tiered observation channels are available or removed.

5.3 Policy

The policy model here maps an observation to an output for each possible action, logits (which define action probabilities) in policy-gradient methods, and Q-values in value-based methods. The policy is a neural network model, and consists of two parts. A graph neural network backbone turns the observation into an embedding vector per agent, and an action head turns these

embeddings into scores for the actions of the configured action space. Invalid actions are masked inside the head, as described in §5.2.1.2. Both backbones produce embeddings of the same width H, so the same heads are used with either of them.

5.3.1 GNN backbones

Both backbones perform message passing over all ordered pairs of agents rather than restricting message passing to the edges in the current topology, which is a deliberate design choice. Since the topology itself is the object being optimized, restricting message passing to the current adjacency would constrain the policy to the existing connections when evaluating alternative topologies. Instead, a fully connected computational graph is used so that the policy can access information from every potential agent-agent connection and learn which connections are relevant for constructing a topology that satisfies the objective. The adjacency used for message passing is artificially fully-connected, and the actual edge set reaches the model through the edge_exists channel in the observation. Both backbones use three layers with embeddings of width $H = 128$.

Both backbones aggregate with a mean rather than a sum. A sum leaves the width of an embedding unchanged, but over a fully connected computational graph it adds $n - 1$ messages, so the magnitude of every activation grows with the size of the formation and a policy trained at one size meets activations of a different scale at another. A mean keeps that magnitude roughly fixed as n changes.

5.3.1.1 GINE

The first backbone is the edge-feature variant of the graph isomorphism network [13], [14]. Starting from $h_i^{(0)} = X_i$, a layer updates the embedding of agent i as

$$h_i^{(l+1)} = \mathrm{MLP}^{(l)}\Big(h_i^{(l)} + \frac{1}{n-1}\sum_{j\neq i}\mathrm{ReLU}\big(h_j^{(l)} + W^{(l)}\mathcal{Z}_{ij}\big)\Big), \tag{5.14}$$

where $W^{(l)}$ maps the pair channels to the width of the node features at that layer and $\mathrm{MLP}^{(l)}$ has one hidden layer of width 128 with a LeakyReLU. The messages are averaged instead of summed, to have consistent values with different network sizes, and ϵ is fixed at zero. Separately, the pair channels are collected along outgoing pairs, so agent i receives the channels of the pairs (i, j), which are the bearings it measures or could measure, rather than those of (j, i).

5.3.1.2 EGNN

The second backbone is the $E(n)$-equivariant graph neural network [15]. An EGNN layer keeps the width of its input, so the node channels are first embedded to width H,

$$h_i^{(0)} = W_2\,\mathrm{LeakyReLU}\big(W_1 X_i\big), \qquad W_1 \in \mathbb{R}^{H\times d_v},\ W_2 \in \mathbb{R}^{H\times H}, \tag{5.15}$$

and each layer then computes

$$m_{ij} = \phi_e\big(h_i,\ h_j,\ \|x_i - x_j\|^2,\ \mathcal{Z}_{ij}\big), \qquad h_i \leftarrow h_i + \phi_h\Big(h_i,\ \frac{1}{n}\sum_{j=1}^{n} m_{ij}\Big), \tag{5.16}$$

where x_i are the normalized positions $\tilde{C}_i$ of (5.13), and ϕ_e and ϕ_h are MLPs with one hidden layer and SiLU activations, ϕ_h applied to the concatenation of h_i with the aggregated message and added back to h_i. The aggregation of (2.41) is a sum over $j \neq i$. Here it is a mean, for the same reason as in the previous section, and it runs over every j including i itself, since messages are formed for all n^2 pairs and the self pair is not excluded. The positions enter only through squared distances, so the embeddings do not change under a translation or rotation of C.

5.3.2 Action heads

The graph neural network backbones in the policy network provide an embedding vector for each *node* after a forward pass. In order to provide a single value as the policy network output for each possible *edge*, edge embeddings are constructed within the network via ordered (since edges are directed) concatenation of the embeddings of the end points of an edge. These edge embeddings are then passed through an MLP together with their initial feature channels to get the corresponding output value:

$$\big[q_{ij}^{\text{add}},\ q_{ij}^{\text{rem}}\big] = \text{MLP}\big([\,h_i,\ h_j,\ \mathcal{Z}_{ij}\,]\big), \qquad q^{\text{skip}} = \text{MLP}\Big(\tfrac{1}{n}\textstyle\sum_i h_i\Big), \tag{5.17}$$

where both MLPs have one hidden layer of width 256 with a LeakyReLU. The pair channels in the observation are given to the head directly, in addition to reaching it through the backbone. Otherwise the channels of a single pair would reach its score only after being averaged with the other pairs of the same agent in every GNN layer, and the one-step rigidity channels are the ones that describe a single pair most precisely. The scores of adding an existing edge, removing an absent one, and skipping when the skip action is disabled are then masked as in (5.2).

For the pointer action space the head scores agents instead of pairs,

$$q_j^{\text{sel}} = \text{MLP}\big([\,h_j,\ h_{\text{sel}}\,]\big), \qquad h_{\text{sel}} = \sum_k \text{sel}_k\, h_k, \tag{5.18}$$

where h_{sel} is the embedding of the agent held in the register and zero when no agent is selected, and the skip score is computed as in (5.17).

For PPO the value function uses a separate network with a backbone of the same type, followed by a mean over the agent embeddings and an MLP with one hidden layer that outputs the value of the state. For DQN the head above gives the Q-values directly.

5.3.2.1 Parameter counts

The two backbones are designed to produce embeddings of the *same width*, but not with the same number of parameters.

Table 5.8: Number of trainable parameters at the widths of the reference configuration, with $d_v = 14$ node channels, $d_e = 12$ pair channels, embedding width $H = 128$, three backbone layers, and head width 256.

component	parameters
GINE backbone	87,990
EGNN backbone, embedder included	957,252
pair head (5.17)	69,378
skip head (5.17)	33,281

5.4 Training

5.4.1 Algorithms and hyperparameters

Two algorithms are used, DQN [6], a value-based off-policy method, and PPO [7], a policy-gradient on-policy method. Table 5.9 lists the settings used for the reference training runs. Each run collects experience from a single environment.

Table 5.9: Hyperparameters of the reference runs.

setting	value
Both algorithms	
training steps	4×10^5
discount γ	0.99
learning rate	3×10^{-4}
backbone layers, embedding width H	3, 128
head hidden width	256
DQN	
replay memory	10^4 transitions
uniformly random steps before learning	10^4
batch size	256
steps between updates	4
updates between target updates, Polyak rate τ	200, 0.005
exploration ε	linear from 0.8 to 0.02 over 2×10^5 steps
PPO	
rollout length, memory size	256, 256
learning epochs, mini-batches	2, 8
GAE parameter	0.95
ratio clip, value clip	0.2, 0.2
entropy coefficient, value loss scale	0.01, 2.5
gradient norm clip, KL threshold	0.5, 0.015
value normalization	running standardization

The discount factor is below one for both algorithms. The horizon $T_{\max}$ is as given in (5.20). For DQN, exploration is ε-greedy, and the random actions are drawn among the admissible actions of the state, as described in §5.2.1.2. For PPO, every update uses the last rollout and nothing older, and the advantages are computed over that rollout.

5.4.2 Episodes

5.4.2.1 Initial formations

To train a policy using the established reward, while allowing it to generalize across heterogeneous configurations (the generalization capability for network sizes is offloaded onto the policy architecture via a GNN), it makes sense to use various networks in the episodic training scheme. A simple approach is to use random graphs, agent poses, and domains to construct the initial formation. All random quantities are from uniform distributions, the agent positions are drawn from $[-1, 1]$ for all coordinates, the orientations are drawn from $[0, 2\pi]$ as Euler angles, and the domains are from the set shown in (2.10). The number of directed edges in the random graph is drawn uniformly from $\{0, \dots, n(n-1)\}$.

One caveat with this approach is that the number of edges in the graph has expected value $E[m] = n(n-1)/2$. The number of edges for a minimally bearing rigid framework grows linearly with n by the closed forms of §2.4.3 (and it is also dependent on the domains) whereas the expected value for the number of edges grows quadratically with n. This discrepancy is illustrated by several example network sizes and domains in Table 5.10. This means that on average, the graphs generated will need pruning of edges more than they will need addition in order to be optimized. This leads to an imbalance in the policy, as it learns to only remove edges. To counteract this problem, random graphs at every episode are generated using the number of edges drawn from a distribution centered around the minimum number of edges required for rigidity (this number is acquired via Algorithm 4.1).

Table 5.10: Expected number of edges in the uniformly sampled initial graph compared with the minimum number of edges required for bearing rigidity. The expected edge count is $\mathbb{E}[m] = n(n-1)/2$, while the minimum requirement is obtained from Algorithm 4.1.

n	Domain	$\mathbb{E}[m]$	m_{req}	$\mathbb{E}[m]/\text{MBR}$
4	$\mathbb{R}^2$	6.0	5	$1.2\times$
8	$\mathbb{R}^2$	28.0	13	$2.2\times$
8	$\mathbb{R}^3$	28.0	10	$2.8\times$
12	$\mathbb{R}^3$	66.0	16	$4.1\times$

$$m_0 \sim \mathcal{N}\big(m_{\text{req}},\ (0.5\, m_{\text{req}})^2\big) \text{ clipped to } [0, n^2 - n]. \tag{5.19}$$

m_0/m_{req} is centered on 1 at every n and domain, as there is no dependence on n in the distribution. The spread on the distribution is proportional to m_{req} rather than to the gap up to the complete graph, and this is because the latter grows with n^2, which makes the ratio m_0/m_{req} a different distribution at each size.

5.4.2.2 Episode length

In case action space types without a termination action are used[2], there needs to be a mechanism to terminate the episodes. This is done via a maximum step count allowed, and it acts as the ”search budget” given to the policy during an episode. This step count limit, i.e. horizon, is set as

$$T_{\max} = 4\, m_{\text{req}} + 10. \tag{5.20}$$

The horizon is set from m_{req} rather than from n, and the argument is about data efficiency during training. A training run of a fixed number of steps sees total steps$/T_{\max}$ distinct instances, and a memory element used with the training algorithms of fixed size holds memory$/T_{\max}$ of them. A horizon that grows with n, such as n^2 if inspired by the possible directed edge counts, therefore limits the number of distinct formations a policy ever sees more strongly than a horizon that grows with m_{req}, as per the same argument made in §5.4.2.1 and Table 5.10.

The horizon also sets the budget given in inference time, and has to leave room for the edits, addition or removal, a solution actually takes. Without a terminal action every remaining step must apply an edit, so a horizon much longer than the number of edits needed produces a long tail of edits made after the answer has been found. How much of (5.20) turned out to be needed is reported in §B.4.

5.4.3 Scoring

An episode is truncated when the step count reaches $T_{\max}$ of (5.20), and if the “stop” action of §5.2.1.3 is part of the action space, it is terminated if the policy takes that action.

Since the policy is treated as a search procedure over edge sets, and every step has to apply an edit when there is no stop action, the last state of an episode is not a good representative of what the policy has found. A policy can reach a good topology early in the episode and move away from it later, only because it is forced to take actions. For this reason an episode is scored by the best state it visited,

$$s^* = s_{t^*}, \qquad t^* = \min\Big\{\, t \in \{0, \dots, T\} \;:\; \varphi(s_t) = \max_{0 \le \tau \le T} \varphi(s_\tau) \Big\}, \tag{5.21}$$

where T is the length of the episode and s_0 is the initial state.

Scoring by s^* alone would hide whether the policy stays on a good graph once it finds one, so two more quantities are reported with it. The best-final gap

$$\Delta\varphi = \varphi(s^*) - \varphi(s_T) \;\ge 0 \tag{5.22}$$

is zero exactly when the episode ends on its best state, and positive when the policy found a better graph and moved away from it. It separates two questions that the final state mixes, whether the

[2] A limit on the step count for truncation is also present in case there is a termination action.

policy can find a good topology and whether it can stay on one. The best step t^* is the number of steps taken to reach s^*. It distinguishes a policy that finds its answer quickly from one that reaches the same graph late in the episode, and the budget analysis of §B.4 is built on it.

Besides the score, each state is described by quantities that have a direct meaning for the problem, listed in Table 5.11. They are recorded for the best state s^*, the final state s_T, and the average over the steps of the episode.

Two of them measure minimality and differ only in which count they compare against,

$$\text{at bound} = \mathbf{1}\big[\text{IBR} \ \wedge \ m = m_{\text{req}}\big], \tag{5.23}$$

$$\text{minimal} = \mathbf{1}\big[\text{IBR} \ \wedge \ m = m'_{\text{req}}(\mathcal{E})\big], \tag{5.24}$$

where m_{req} is the bound (4.25) computed on the complete graph and m'_{req} is the same accumulation recomputed from the current edge set.

Table 5.11: The quantities used to score an episode. The per-state quantities are recorded at the best state, at the final state, and averaged over the steps of the episode.

quantity	expression	description
Episode		
best state	s^*, (5.21)	The graph the episode returns.
best step	t^*, (5.21)	Steps taken to reach the best state.
best-final gap	$\varphi(s^*) - \varphi(s_T)$	Zero when the episode ends on its best state.
State		
score	$\varphi(s)$, (5.8)	The objective the policy is trained on.
edge count	$m = \lvert\mathcal{E}\rvert$	Number of bearing measurements kept.
rigid	$\mathbf{1}[\operatorname{rk}\mathbf{B}_{\mathcal{G}} = \rho_{\mathcal{K}}]$	Whether the framework is infinitesimally bearing rigid.
at bound	(5.23)	Rigid on exactly m_{req} edges, the proven lower bound.
minimal	(5.24)	Rigid and minimal by the heuristic count on the current edges.
shape error	ε, (4.30)	Predicted error of the recovered shape per radian of bearing noise.

5.4.3.1 Decision quality

Further metrics that can be used to determine if the policy is taking meaningful actions are,

$$\text{useful} = \frac{\#\{t : \varphi(s_{t+1}) > \varphi(s_t)\}}{T}, \qquad \text{wasted} = \frac{\#\{\text{no-op}\} + \#\{\text{skip}\}}{T}, \tag{5.25}$$

$$\text{overshoot} = \frac{\max(0,\ m_T - m_{\text{req}})}{m_{\text{req}}}, \qquad \text{converge} = \frac{t^*}{T}, \tag{5.26}$$

$$\text{edit efficiency} = \frac{|m_T - m_0|}{\#\text{edits}} \in [0, 1]. \tag{5.27}$$

The useful fraction is the share of steps that improved the score, though the policies are not expected to take useful actions all the time. The wasted fraction counts the steps that did not change the graph. The overshoot is the number of edges the final graph carries above m_{req}, relative to m_{req}, and the convergence fraction says how early in the episode the best state was reached. The edit efficiency is one when every edit moved the edge count in the same direction and zero when the edits only went back and forth, which is what a policy without a stop action does once it has found its answer.

6 Experiments and results

6.1 Experimental setup

The policy acting on the environment, either trained or as a classical algorithm, selects directed bearing measurements between agents whose poses remain fixed during an episode, searching for the best topology it can find. A useful output must recover the observable degrees of freedom of the agents while balancing the number of measurements against sensitivity to bearing noise. Rigidity, measurement count, and estimation error are therefore reported separately rather than represented only by the value of the optimization objective, which would also cause an unfair comparison since the learned policies may not be necessarily trained on the same objective.

6.1.1 Compared methods

Table 6.1 lists the methods that are compared using the same agent poses and domain assignments. The editing methods receive the same initial graph, while the constructive search starts from the empty graph.

Table 6.1: Baseline methods used to evaluate the effectiveness of the trained policies via comparison. All methods have the same role, which is a search procedure to find the best topology according to the evaluation objective.

Method	Search procedure
Greedy	Evaluate every admissible single-edge toggle and take the largest strict objective improvement. Stop when none improves the objective or the edit limit is reached.
Spectral	Rank candidate edits using rank and spectral information (from the observation), then evaluate the objective on a shortlist of five candidates. Stop when the shortlist offers no strict improvement or the edit limit is reached.
Degree	Add edges using endpoint degrees until rigidity is obtained, then remove edges while preserving rigidity.
Constructive (greedy)	Traverse candidates in randomized order, retaining additions that increase rank until rigidity is obtained. Return the best of five constructions under the evaluation objective.
Annealing	Accept some objective-decreasing edits and retain the best visited graph. The number of objective evaluations is matched to greedy on each instance.
Learned	Select edits from the trained policy and return the best graph visited within the prescribed horizon. Policies with a stop action instead return the final graph.

The main policies use DQN with a GINE encoder and direct edge actions, with the training configurations summarized in Table 6.2. Each training episode has independently sampled agent domains from $\{\mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^2 \times S^1, \mathbb{R}^3 \times S^1, \mathrm{SE}(3)\}$, admissible random poses, and a random initial edge set. Initial edge counts are sampled around the heuristic lower bound as explained in §5.4.2.1, with a standard deviation proportional to that bound. The policies are therefore trained on varying compositions and initial graphs, though with a constant number of agents n, since the action size depends on n and cannot change during training because of the implementation.

Table 6.2: Training configurations used in the main/principal comparisons. All rows use $n = 10$, randomized domain assignments, direct edge actions, and the normalized shape-error objective. The transition count is the configured training budget.

Policy	Training weight κ_t	Transitions	Training repetitions
DQN/GINE	0	400,000	One
DQN/GINE	10	400,000	Seeds 0, 1, and 2
DQN/GINE	50	400,000	One
DQN/GINE	200	400,000	One

During evaluation, DQN chooses the action that has the largest predicted value without exploration, and PPO samples its categorical policy. A budgeted search may revisit a graph and is credited with the best graph it has found, since its final mandatory edit need not improve that graph, and this differs from a stopping policy which chooses its termination and is evaluated on the graph where it stops. The stopping and sequential-selection variants, along with some other experimentation are examined in Appendix B.

The main ”opponent” to the trained policies was considered the greedy method, as it performed better on average than the others throughout the experiments historically. However, it has a higher computational cost than some of the others, and it is not definitive which method performed best overall, hence the experiments and comparisons include all the methods.

6.1.2 Objective and measurements

The training weight κ_t of the shape error term identifies the objective used to learn a policy, while the evaluation weight κ_e identifies the objective used in the environment, which means the baseline methods use it. For the principal experiments, the state objective has the form, as established in §5.2.2

$$\varphi_{\kappa_e}(s) = \frac{w_{\text{rank}}\,\mathrm{rk}\,\mathbf{B}}{\rho_{\mathcal{K}}} - \frac{w_{\text{edge}}\,m\,c_{\max}}{\rho_{\mathcal{K}}} + \kappa_e \frac{w_{\text{edge}} c_{\max}}{\rho_{\mathcal{K}}}\,\mathbf{1}[\mathrm{IBR}]\; q(s), \qquad (w_{\text{rank}}, w_{\text{edge}}) = (100, 25). \tag{6.1}$$

A larger weight κ increases the value assigned to conditioning relative to an additional measurement, and policies trained with different values for this weight are experimented with.

Important note regarding learned policy outputs. For a learned policy evaluated by its best visited state, the returned graph is the visited graph with the largest value of φ_{κ_e}, using the

evaluation weight κ_e rather than the policy's training weight κ_t. Changing κ_e can therefore change which graph is reported as the policy's output, even when the policy visits exactly the same sequence of graphs.

For a rigid output, the error coefficient is

$$\varepsilon(\mathcal{G}, \chi) = \sqrt{\frac{1}{n} \operatorname{tr}\left[(\mathbf{B}_s^\top \mathbf{B}_s)^+\right]}, \tag{6.2}$$

where the position columns of $\mathbf{B}_s$ are scaled by the formation's root-mean-square (RMS) radius. In the local small-noise approximation, independent tangent-plane bearing noise with standard deviation σ radians gives a predicted per-agent state RMS error of $\sigma\varepsilon$. Position errors are expressed relative to formation radius and attitude errors in radians, with unobservable variations removed. This coefficient describes the combined position and attitude error. The empirical noise experiment in Appendix B.9 measures the position component separately.

The position and attitude contributions are also reported separately where the distinction is relevant. With $C = (\mathbf{B}_s^\top \mathbf{B}_s)^+$ partitioned into position and attitude blocks, their coefficients are $\varepsilon_p = \sqrt{\operatorname{tr}(C_{pp})/n}$ and $\varepsilon_a = \sqrt{\operatorname{tr}(C_{aa})/n}$. Both use the total number of agents, so $\varepsilon^2 = \varepsilon_p^2 + \varepsilon_a^2$ for each graph. The corresponding small-noise predictions are $\sigma\varepsilon_p$ in formation radii and $\sigma\varepsilon_a$ in radians. Domains without attitude degrees of freedom have $\varepsilon_a = 0$. These are predicted error coefficients rather than measured reconstruction errors.

The error coefficient is summarized geometrically because its values span several orders of magnitude. Method vs. method error ratios use only instances on which both outputs are rigid, therefore their sample count can be smaller than the number of evaluated formations, as outputs may turn out to be flexible.

6.1.3 Evaluation sets

The size experiments use frozen poses, domain assignments, and initial graphs, with the sample sizes and policy horizons in Table 6.3. Each method is evaluated on the same instances within a comparison. The domain experiments use separate frozen sets at the training size, and the matched-weight experiments reuse a common subset of the reference benchmark. The smaller diagnostic benchmark is a different instance set and is not pooled with the reference benchmark.

Table 6.3: Instances and policy action horizons for the randomized-domain size experiments.

Agents n	8	10	16	24	32	48	96
Instances N	200	200	100	50	25	25	8
Policy horizon	94	118	190	286	382	574	1150

The main comparisons use evaluation weight 10 unless stated otherwise. Rigidity, edge count, and the error coefficient do not depend on this weight for a fixed graph. The weight does affect which graph is returned, since the best visited graph is selected by its score, and it also guides the objective-based classical searches. The comparisons below concern the physical properties

of those returned graphs. Training weights identify the preferences the learned policies were trained to follow, rather than different definitions of rigidity or error.

Proportion intervals are 95% Wilson intervals [43], and other displayed intervals use 4,000 bootstrap resamples of instances with shared instances resampled together for paired comparisons. Error ratios are computed on a logarithmic scale.

6.2 Reference comparison

The reference comparison in Table 6.4 gives the outcomes on the same randomized-domain formations under the matched objective. Rigidity alone does not separate them on this set. Their differences lie in how many measurements they retain and how well those measurements constrain the observable state.

Table 6.4: Outputs on 200 shared $n = 10$ formations with $\kappa_e = 10$ ($\kappa_t = 10$ for the DQN policies). Rigid and at-bound entries are counts over all instances, $\bar{m}$ is the arithmetic mean edge count among rigid outputs, and $\varepsilon_{\text{geom}}$, $\varepsilon_{p,\text{geom}}$, and $\varepsilon_{a,\text{geom}}$ are geometric means of the combined, position, and attitude error coefficients over rigid outputs.

Method	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$	$\varepsilon_{p,\text{geom}}$	$\varepsilon_{a,\text{geom}}$
Greedy	200/200	84/200	17.62	7.04	5.67	3.79
Spectral	200/200	6/200	21.05	5.45	4.34	2.98
Degree	200/200	58/200	18.07	54.06	40.29	28.25
Constructive	200/200	11/200	18.83	8.13	6.46	4.48
Annealing	200/200	97/200	17.56	6.78	5.55	3.70
DQN, seed 0	200/200	100/200	17.48	6.51	5.27	3.60
DQN, seed 1	200/200	117/200	17.36	6.29	5.06	3.55
DQN, seed 2	200/200	113/200	17.34	6.84	5.53	3.73

Degree-based search obtains sparse graphs but has a substantially larger error coefficient than the conditioning-aware methods. Spectral search obtains a smaller error coefficient than greedy while retaining more edges, so its better conditioning cannot be interpreted independently of its measurement cost. Randomized construction also produces rigid graphs, but its output does not match greedy's combination of sparsity and conditioning. Annealing is closer to greedy on both measures, showing that improvement over greedy is not confined to the learned approach.

All three DQN seeds use fewer edges on average than greedy and attain the lower bound more often. Figure 6.1 gives paired edge-count differences and error ratios, so the comparison accounts for the same formation being easy or difficult for both methods. The edge-count differences are consistently below zero, though the differences are not substantial. The combined-error reduction is supported for seeds 0 and 1, while the interval for seed 2 includes equal error. The position component follows the same pattern, while the attitude error intervals include equal error for all three seeds. The evidence therefore supports only a modest improvement in sparsity, with a less uniform improvement in conditioning across training seeds.

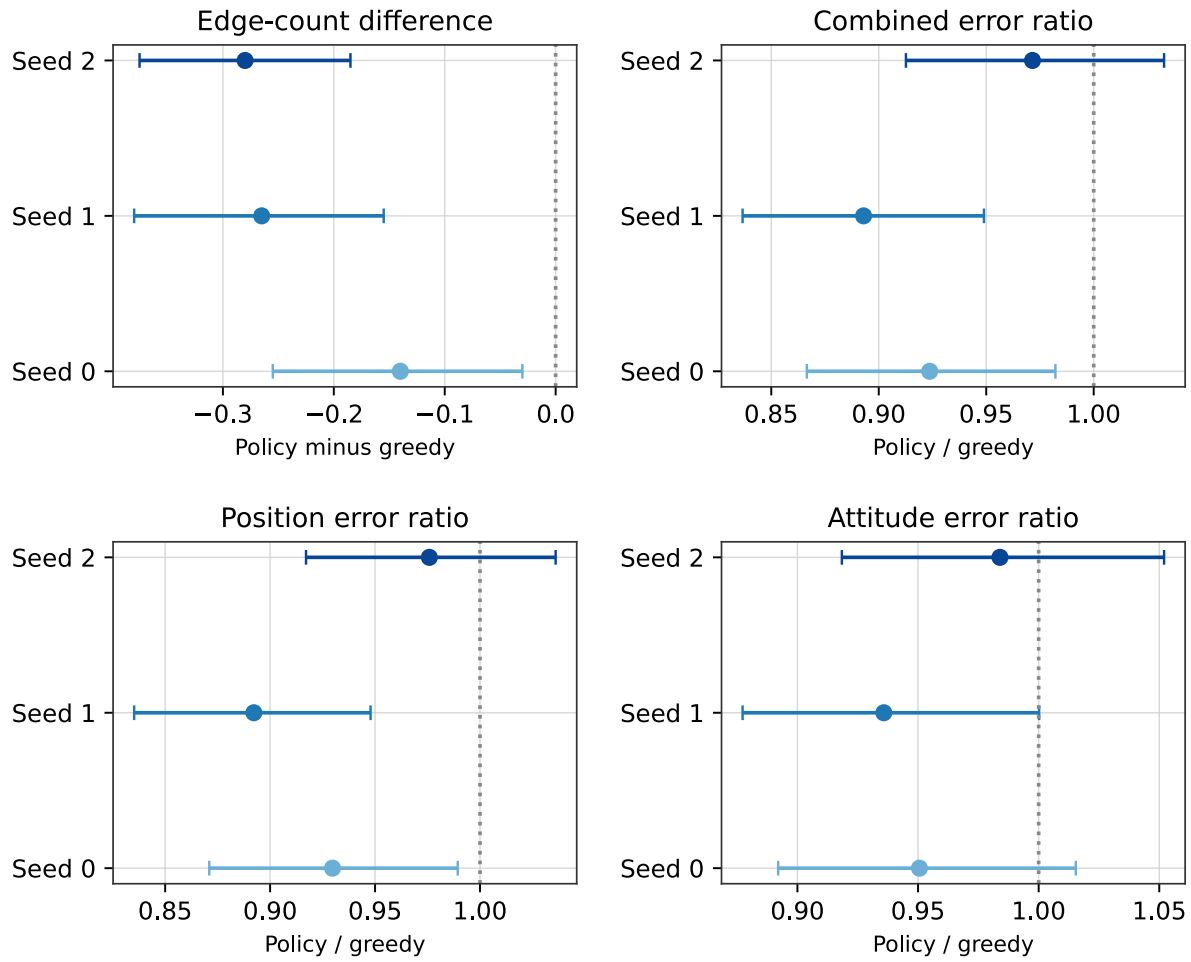


Figure 6.1: Mean edge-count differences and geometric mean combined, position, and attitude error ratios for each reference seed relative to greedy on the same 200 formations. Intervals use paired bootstrap resampling. The vertical dashed lines mark equal edge count and equal error.

6.3 Network size

Figure 6.2 reports rigidity and minimum edge count attainment for the reference policy and the classical baselines, while Figure 6.3 reports the sparsity and robustness of their rigid outputs across network sizes.

The learned policy remains reliable in transfer across smaller network sizes, but its rigidity failures increase on the larger formations. Degree-based and constructive search preserve rigidity on the tested sizes, while the spectral shortlist also produces some flexible outputs at larger sizes. An observation is that the increasing graph size exposed a difference between a method that explicitly continues construction toward the required rank and a learned policy whose actions are not necessarily working towards the required rank. At the largest tested size, the learned policy did not produce rigid frameworks most of the time, despite rigidity being rewarded during training.

The learned policy's smaller shape error does not give reliable insight at larger sizes since the edge ratio and the error coefficient are reported only for the instances with rigid outputs. At $n = 96$, the learned policy's outputs are flexible in over 80% of cases, and hence the metrics

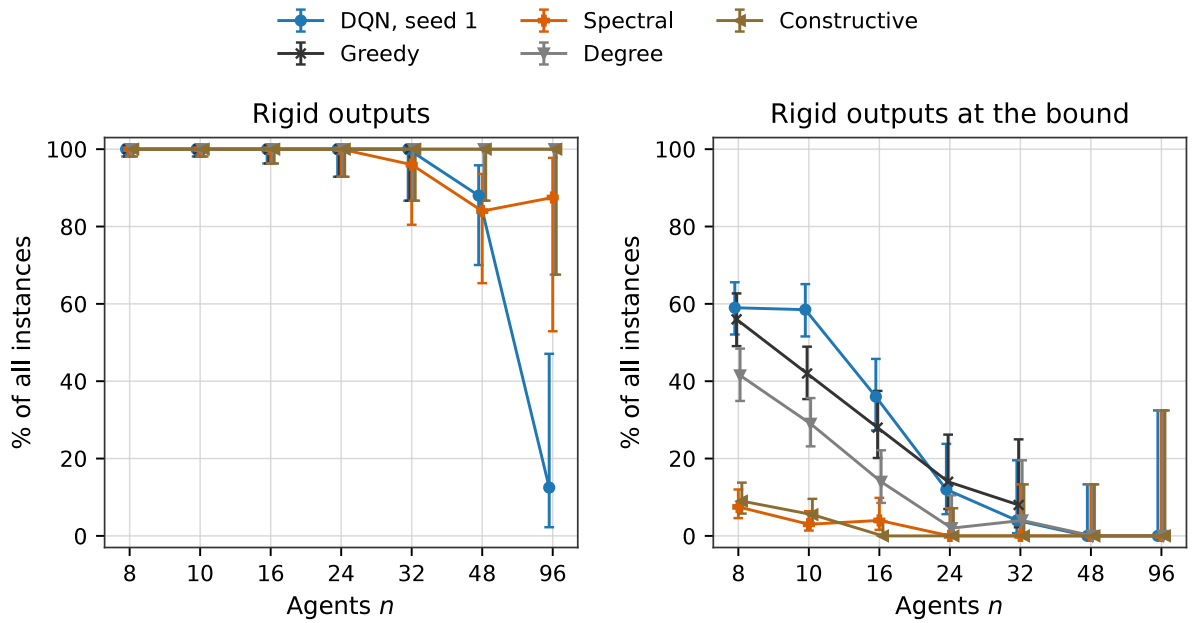


Figure 6.2: Rigidity and lower-bound attainment over all instances in Table 6.3 (different instance counts). The DQN policy is seed 1, with $\kappa_t = \kappa_e = 10$. The intervals are Wilson intervals.

reported only on the rigid outcomes do not reflect the behavior of the policy.

The two tables Table 6.5 and Table 6.6 refer to different network sizes, although each contains eight formations. Table 6.5 uses the first eight formations from the frozen $n = 48$ set, the cases for which greedy was evaluated. The other methods were evaluated on the full set of 25 formations, but their rows in this table are restricted to the same eight cases so that the comparison is fair and uses common inputs. Table 6.6 instead uses the complete available set at $n = 96$, where greedy was not evaluated due to computational cost.

Table 6.5: Outcomes from the shared eight-formation subset of the $n = 48$ benchmark. The final three columns give geometric mean combined, position, and attitude error coefficients among each method's rigid outputs.

Method	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$	$\varepsilon_{p,\text{geom}}$	$\varepsilon_{a,\text{geom}}$
DQN, seed 1	7/8	0/8	109.86	8.89	7.38	4.85
Greedy	8/8	0/8	89.38	30.78	24.00	17.73
Spectral	7/8	0/8	99.43	20.22	16.18	11.87
Degree	8/8	0/8	92.00	3085.19	1843.83	2398.24
Constructive	8/8	0/8	96.50	35.86	25.14	23.18

6.3.1 Training weights

Figure 6.4 compares size transfer for the policies trained with different conditioning weights. All policies use the same formations and best-visited selection at weight $\kappa_e = 10$, so the curves show how the trained policies differ in rigidity, sparsity, and conditioning as the network grows.

The higher-weight policies retain lower error on many of the larger formations, but this does not consistently come with better rigidity success or fewer edges. The policy trained at weight

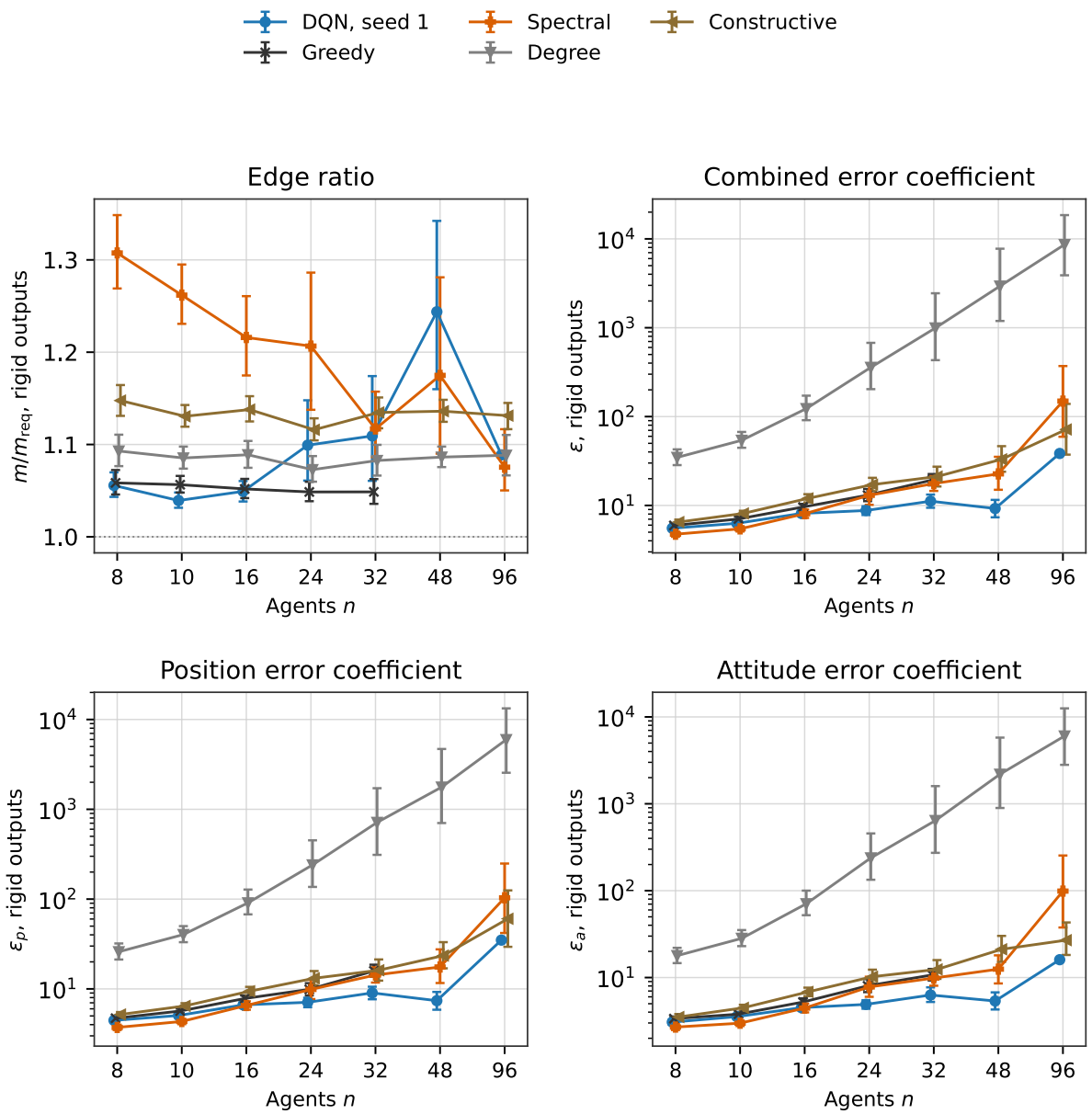


Figure 6.3: Mean edge ratio and geometric mean combined, position, and attitude error coefficients *on rigid outputs*. Intervals use instance bootstrap resampling, with no interval estimated from a single rigid output. The sample sizes and methods are those of Figure 6.2.

Table 6.6: Outcomes from the complete $n = 96$ set with eight formations. The final three columns give geometric mean combined, position, and attitude error coefficients among rigid outputs. Each DQN error entry is a single observation, as the policy did not achieve rigidity in the other cases.

Method	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$	$\varepsilon_{p,\text{geom}}$	$\varepsilon_{a,\text{geom}}$
DQN, seed 1	1/8	0/8	184.00	38.45	34.94	16.05
Spectral	7/8	0/8	184.57	149.33	103.39	98.90
Degree	8/8	0/8	186.88	8633.50	5965.79	6027.42
Constructive	8/8	0/8	194.25	71.64	60.40	26.93

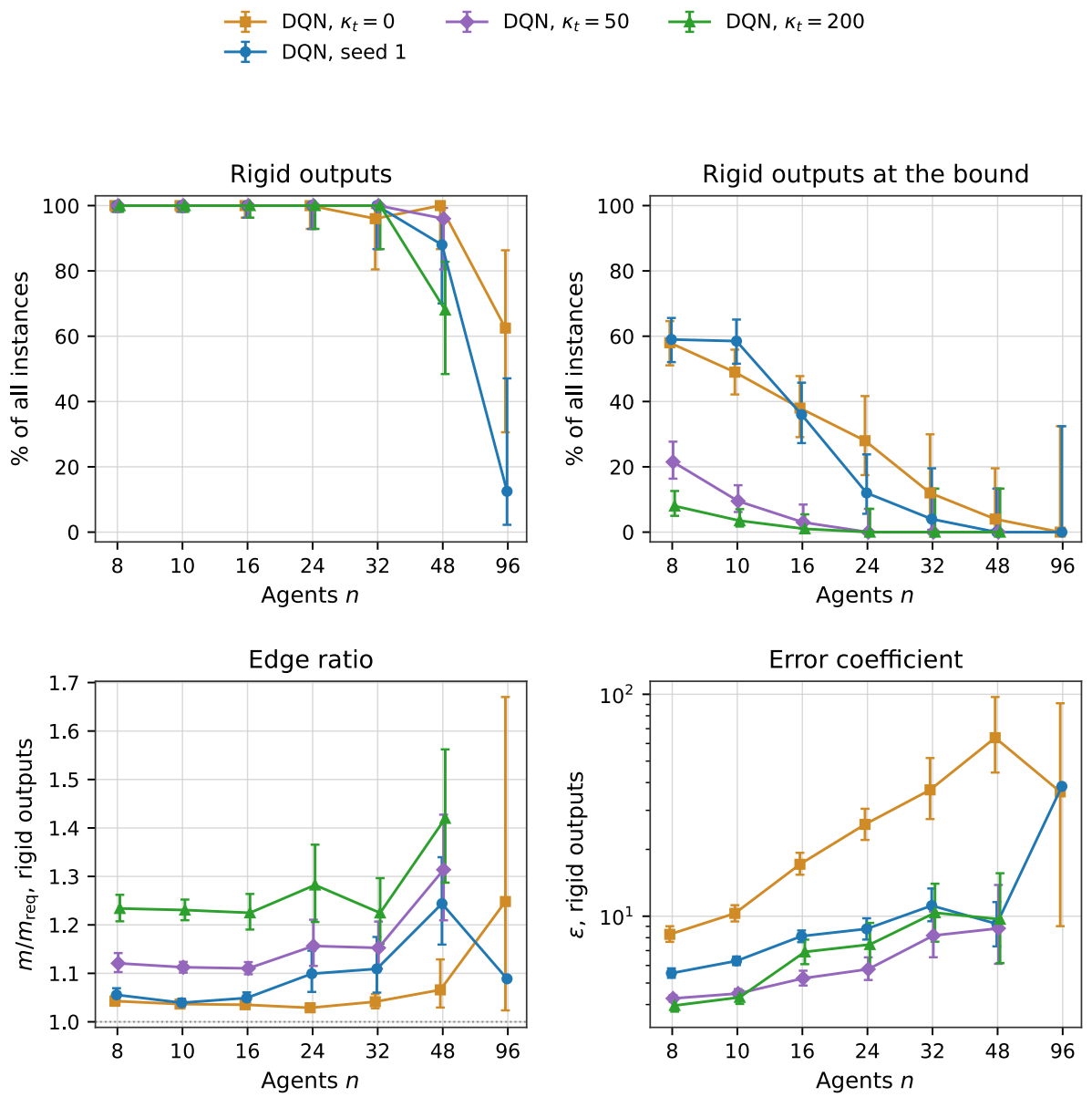


Figure 6.4: Comparison of the outputs of the learned policies with different training objectives across network sizes. The instance counts are those of Table 6.3.

$\kappa_t = 50$ preserves rigidity more often than the $\kappa_t = 200$ policy on larger networks, which suggests that the $\kappa_t = 200$ policy cannot zero-shot transfer across network sizes, at least not as well as the $\kappa_t = 50$ one. The policy with no error term in its training objective produces flexible outputs at the largest size, though it has overall better success at finding rigid topologies. Increasing the training weight therefore does not produce a simple ordering in size generalization, and the error curves must be read together with the rigidity and edge count success rates.

6.4 Domain composition

These measurements test whether policies trained on networks with randomized agent domains (statistically heterogeneous almost always) can perform well in homogeneous networks, with

the same number of agents as the training networks.

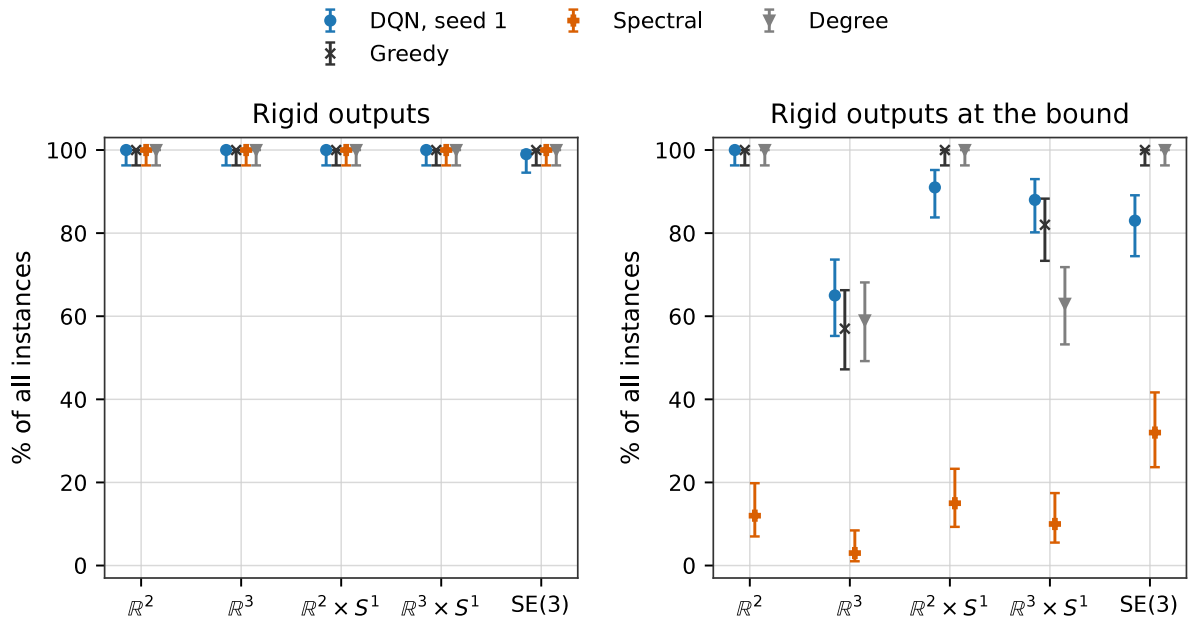


Figure 6.5: Rigidity and edge count lower bound attainment on 100 formations per homogeneous domain at $n = 10$. The Wilson intervals use all evaluated formations.

The reference policy obtains rigid graphs across almost all homogeneous cases, but its sparsity behavior is not the same on all domains. In $\mathbb{R}^2$, both it and greedy attain the lower bound throughout the sample, leaving the distinguishing factor as the shape error metric. In $\mathbb{R}^2 \times S^1$ and $\mathrm{SE}(3)$, greedy attains the bound more consistently. A minimum edge count output is therefore not an automatic consequence in homogeneous networks, with a mixed training distribution.

Conditioning also changes the interpretation of the sparsity results. In the planar domain with no orientations, the policy's smaller error is obtained at the same minimum edge count as greedy, hence proving a desired outperforming behavior. In the spatial domain with no orientations, their edge counts are similar and greedy has a slightly smaller geometric mean error. In some domains with attitude freedom, a reduction in error is accompanied by additional measurements, and though they do not provide a direct comparison, they show that the policy manages a trade-off between sparsity and lower error, as its objective dictates with $\kappa_t = 10$.

The heterogeneous tests of Figures 6.7 and 6.8 use the pairs $\mathbb{R}^2$ with $\mathrm{SE}(3)$ and $\mathbb{R}^3$ with $\mathbb{R}^2 \times S^1$. Both are evaluated with independently assigned random domains and with a balanced composition containing five agents of each type in randomized node order. The figures compare the three reference seeds with greedy on these scenarios.

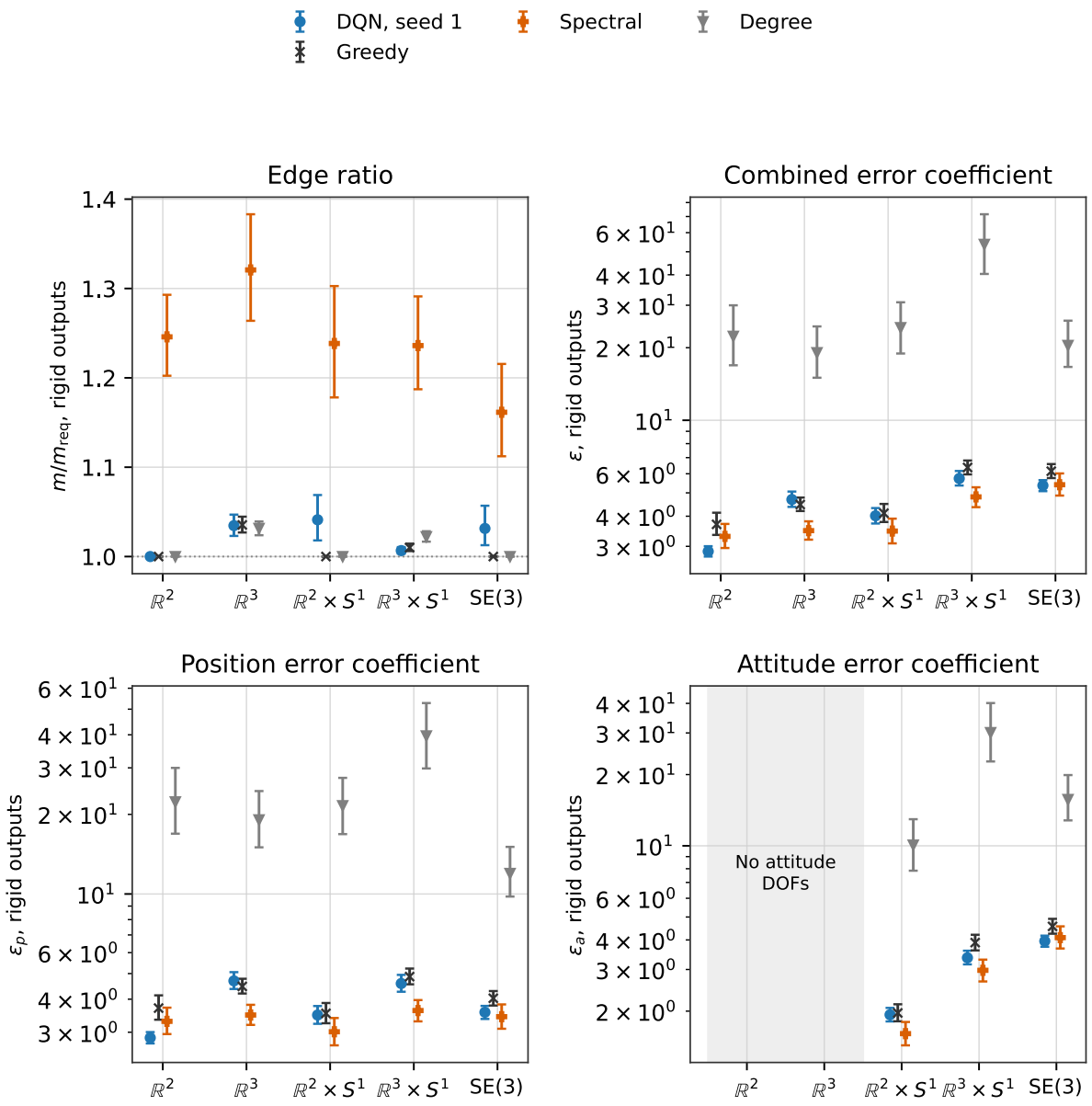


Figure 6.6: Mean edge ratio and geometric means of the combined, position, and attitude error coefficients among rigid outputs on the homogeneous networks. Intervals use instance bootstrap resampling.

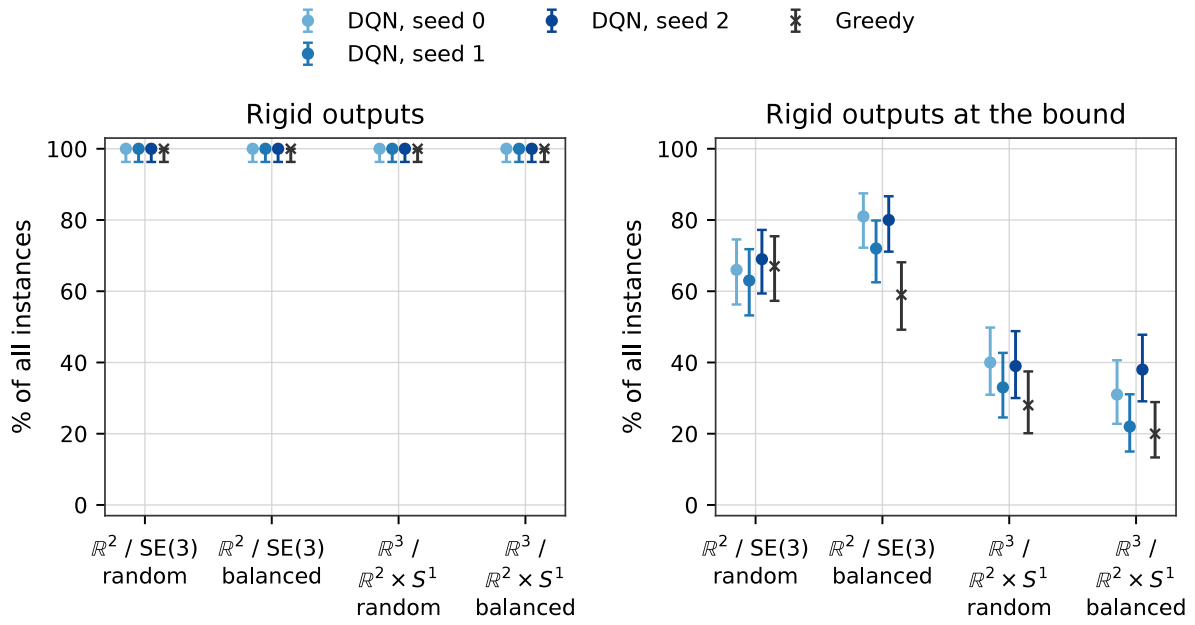


Figure 6.7: Rigidity and lower-bound attainment on 100 formations on the dual domain assignment schemes at $n = 10$. Random assignments draw each node's domain independently, while balanced assignments use five agents of each type. All reference seeds have $\kappa_t = \kappa_e = 10$.

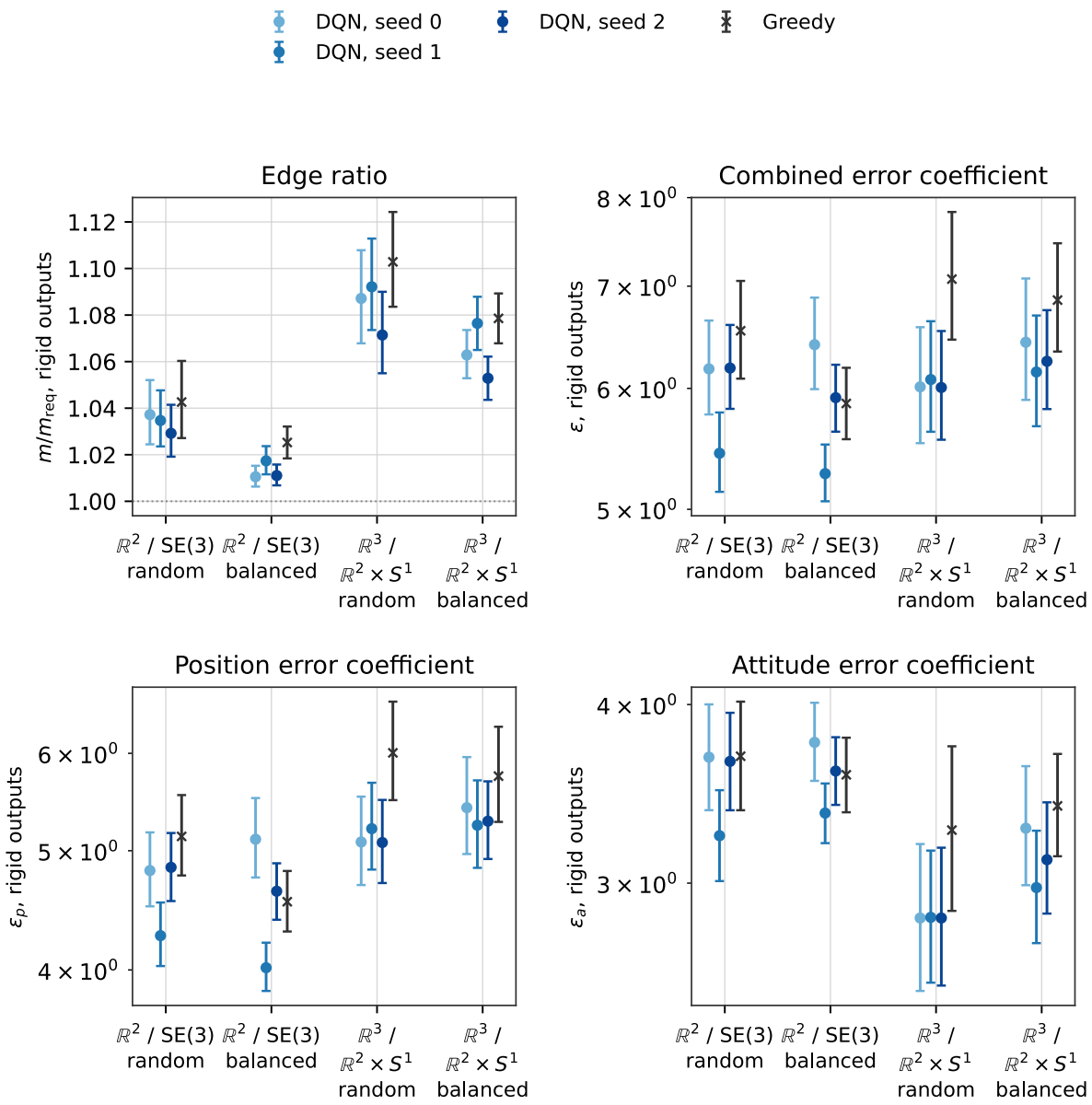


Figure 6.8: Mean edge ratio and geometric means of the combined, position, and attitude error coefficients on the same mixture sets as Figure 6.7.

6.5 Conditioning trade-off

The experiment results presented here compare each policy with greedy under the objective used to train that policy. Table 6.7 reports results on the same benchmark subset. Increasing the weight changes which graphs are desirable, acting as a knob between sparsity and lower estimation error, so the relevant questions are whether the learned policy obtains that desired trade-off effectively and how it compares to greedy.

Table 6.7: Greedy and the trained policy at each weight, evaluated on 100 shared random domain formations with $n = 10$ and $\kappa_t = \kappa_e$. The weight 10 policy refers to DQN seed 1.

$\kappa_t = \kappa_e$	Method	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$
0	Greedy	100/100	45/100	17.86	46.73
0	DQN	100/100	83/100	17.27	51.97
10	Greedy	100/100	47/100	17.80	7.16
10	DQN	100/100	61/100	17.63	6.16
50	Greedy	100/100	6/100	18.78	3.53
50	DQN	100/100	5/100	19.20	4.01
200	Greedy	100/100	0/100	22.49	2.03
200	DQN	100/100	0/100	23.31	2.70

Figure 6.9 shows the edge counts and error coefficients on the same plot. Lower error and lower edge count is the desired behavior, and the measurements show that the policy only outperforms greedy if the objective it is trained on decides on the trade-off between error and edge count with $\kappa_t = 10$, with all evaluations being done on the same objective. This is a strong finding given the extent of the experiments, the model architecture, and the training scheme. It is hard to generalize this conclusion to training and evaluation on different networks, though it shows that the trained policies do not always perform better in-distribution and evaluated on the same objective they were trained on.

Figure 6.10 separates the position and attitude contributions for the same returned topologies. Increasing the conditioning weight reduces both coefficients in these samples, rather than improving one component at the expense of the other. This shows that the combined error is also reduced, and it shows that the objective function provides the expected outcome that the κ value leads to less estimation error, at the expense of more measurements. The reference policy ($\kappa = 10$) has lower geometric means than greedy in both components, while greedy has lower means in both at the larger weights. The figure's results are in line with those of Figure 6.9.

Without the conditioning term, the policy attains the edge lower bound more frequently than greedy but does not show a corresponding error advantage. This is consistent with an objective that rewards rigidity and removing edges without distinguishing equally sparse graphs by their sensitivity to noise. Adding a moderate conditioning weight reduces error for both methods while their edge counts remain close to the lower bound.

Note. Policies except for the one with $\kappa = 10$ have only one trained checkpoint, so the comparison describes those policies rather than an established effect across training repetitions.

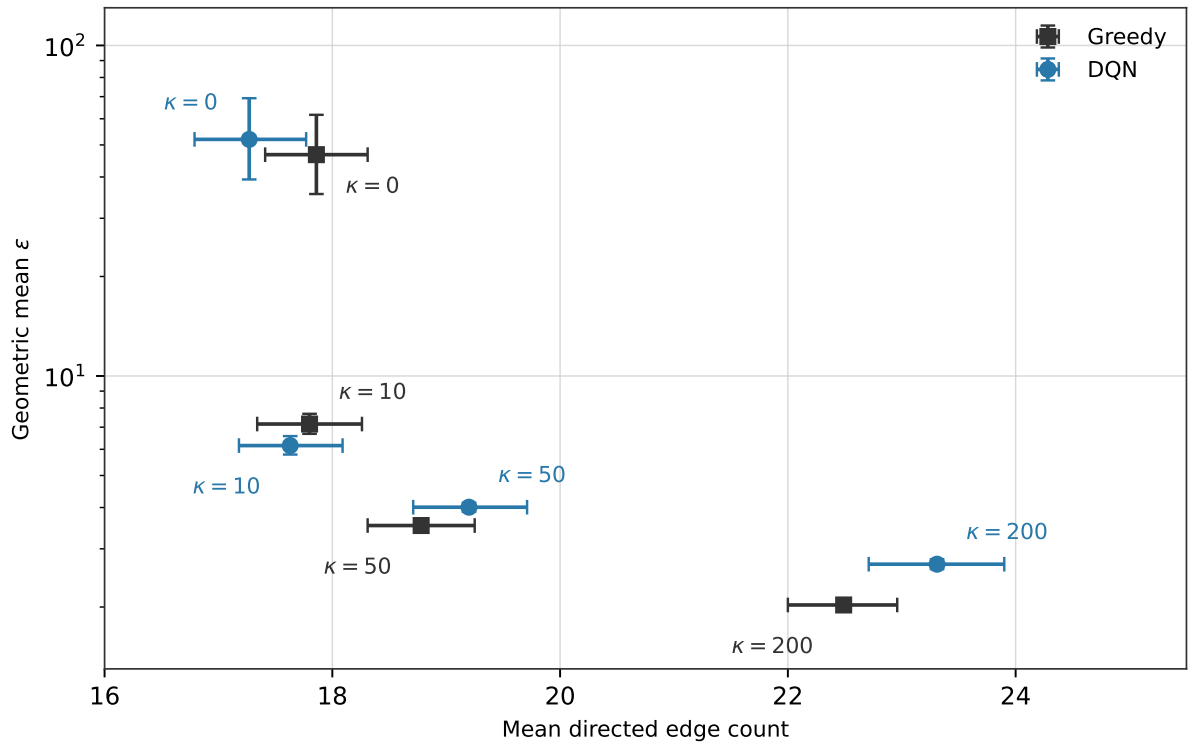


Figure 6.9: Arithmetic mean edge count and geometric mean error coefficient on the same 100 formations for each weight. Error bars are instance bootstrap intervals.

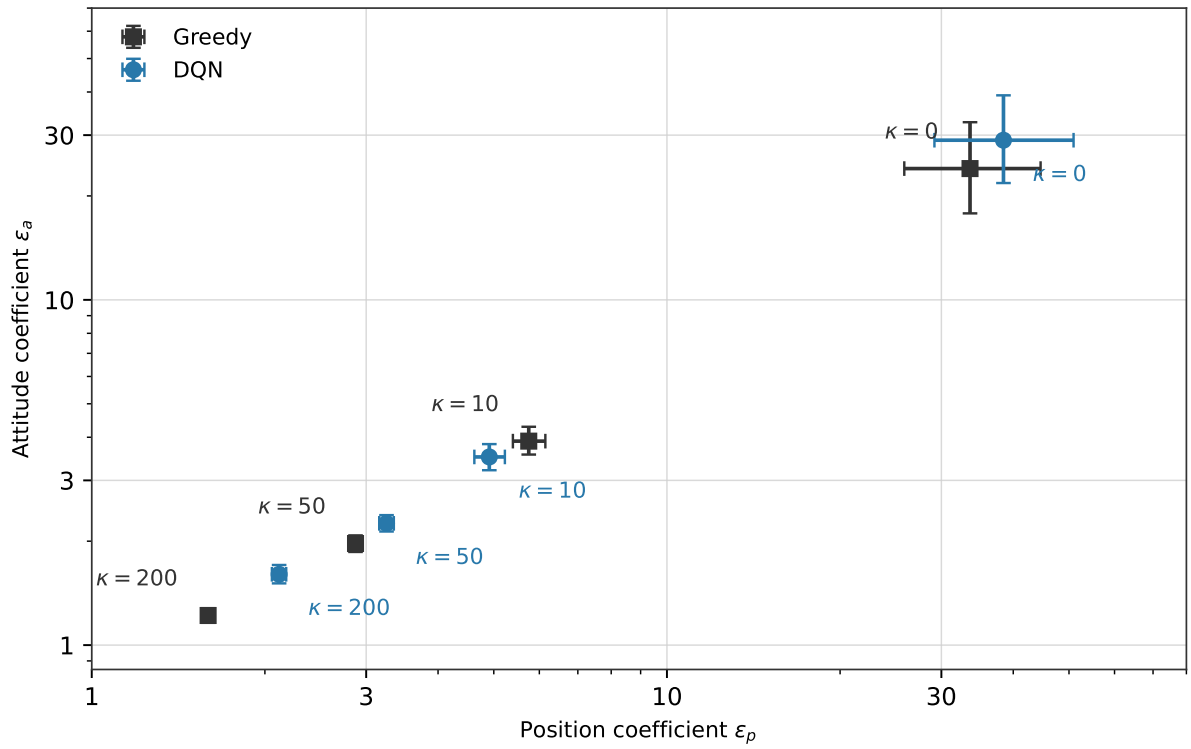


Figure 6.10: Geometric mean position and attitude error coefficients on the same 100 formations as Figure 6.9. The components and the combined error coefficient of Figure 6.9 are averaged separately.

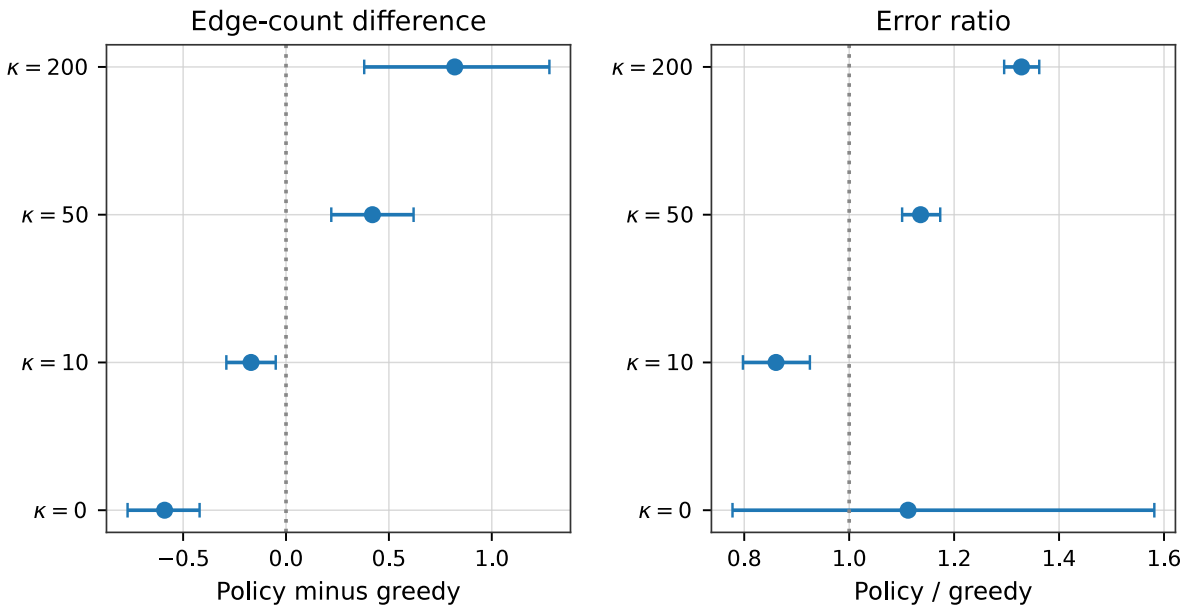


Figure 6.11: Edge count differences and error ratios at each matched training and evaluation weight.

The effect of changing the evaluation weight κ_e for a fixed policy, on larger formations, is examined separately in Appendix B.5.

6.6 Computational cost

Table 6.8 presents the recorded counts to certain primitive function calls, and the wall time, where a primitive call is one invocation of a named calculation. The total is therefore not a count of interchangeable arithmetic operations, but shown for accounting purposes.

Table 6.8: Arithmetic mean cost per formation on the 200-instance reference benchmark at evaluation weight 10. The counts are individual calls to the primitive functions. Matrix builds, rigidity matrix decompositions, and policy forward passes are shown separately. Forward passes are excluded from the primitive total but included in recorded wall time.

Method	Calls	B builds	SVDs	Forwards	ms
Greedy	2327.9	941.5	1386.4	0.0	545.1
Spectral	128.9	43.4	54.2	0.0	33.1
Degree	68.5	34.2	0.0	0.0	18.8
Constructive	465.0	455.0	10.0	0.0	237.2
Annealing	2673.8	942.5	1731.3	0.0	638.5
DQN, seed 1	1065.8	129.2	236.0	118.0	638.4

The policy uses fewer counted numerical calls than greedy at this network size, but it does not have a shorter mean recorded runtime, though neural network inference and the different mix of matrix operations prevent a direct (approximate) conversion from call counts to elapsed time. Degree and spectral search are less expensive on both measures, with the output quality differences already shown in Table 6.4.

Training is excluded from the assessment of the computational cost required in the learned method. The recorded evaluations use CPU linear algebra and GPU policy inference, on a consumer system, and the runtime numbers are specific to the setup.

Figure 6.12 shows how the cost changes with network size. Greedy's complete single step search becomes substantially more expensive as the number of candidate edges grows. The policy avoids rescoring every candidate with the full objective, but still computes derived observation channels and processes a larger candidate edge graph. The sampled sizes and the unequal costs do not allow for establishing an asymptotic complexity exponent, however it is possible to consider the individual complexities of the operations to have a decision on which methods are more suitable for a given practical task, but this type of an analysis is not pursued in this work.

Figure 6.13 reports the best graph available at different episode budgets given to the policies, on a smaller diagnostic benchmark. The edit axis should be interpreted as a search depth measure and not an equal computation budget. These show what an imposed smaller budget would have resulted, but the time (the step index) at which the best graph appears is only known retrospectively.

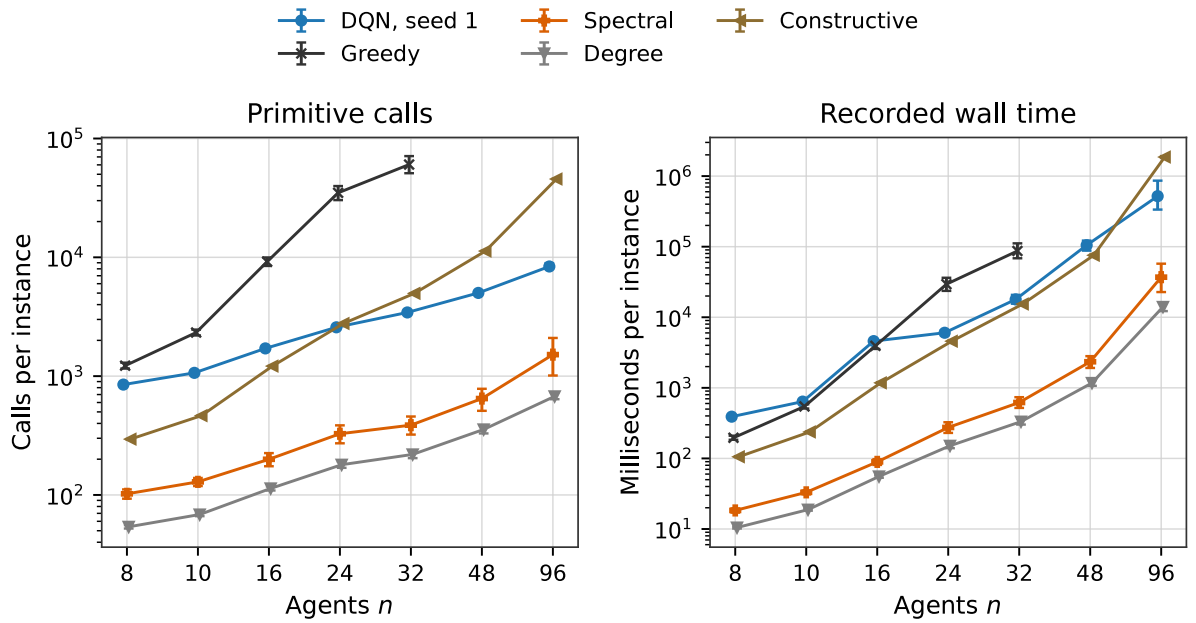


Figure 6.12: Arithmetic mean call count and recorded wall time per formation on different sized networks. Intervals are bootstrap intervals over instances. Costs cover the full recorded search, including searches that returned flexible topologies.

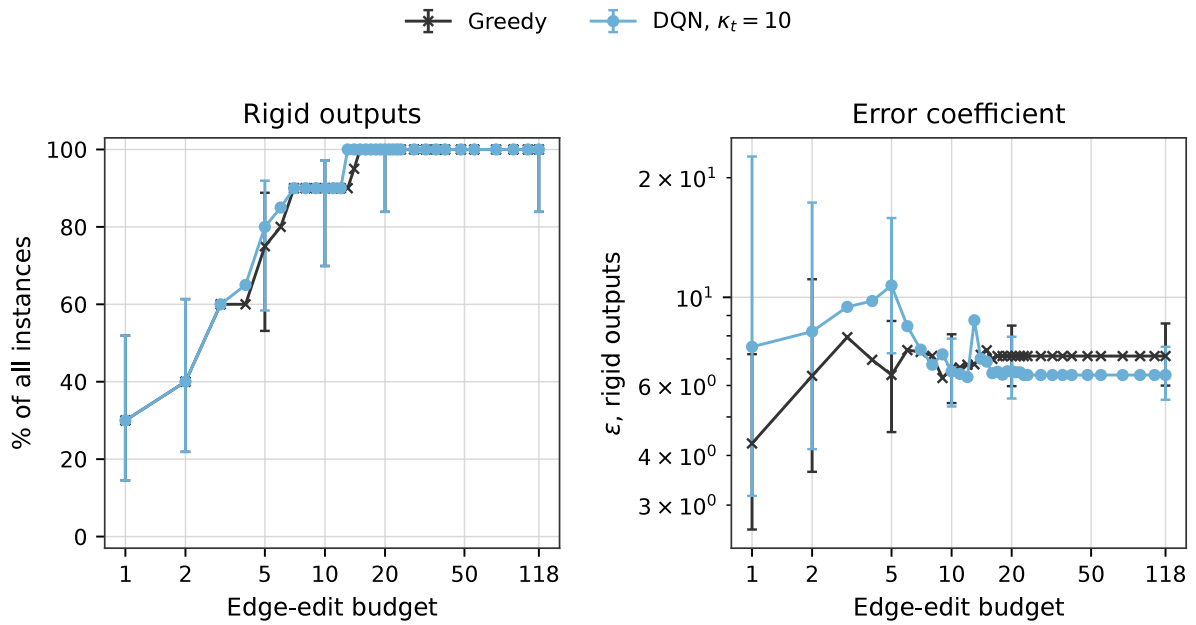


Figure 6.13: Rigidity and error coefficient of the best graph found within each edit limit on 20 shared diagnostic formations. The DQN policy is seed 0 and both methods use weight 10. Error is measured only on rigid outputs.

6.7 Resultant topologies

Figure 6.14 shows the graph connections as a result of topology optimization by different methods on the same formation instance. The associated measurements in Table 6.9 connect the illustrated graphs to their performance results. Figure 6.15 places the same directed graphs in three dimensional space, presenting the actual geometry of the formation.

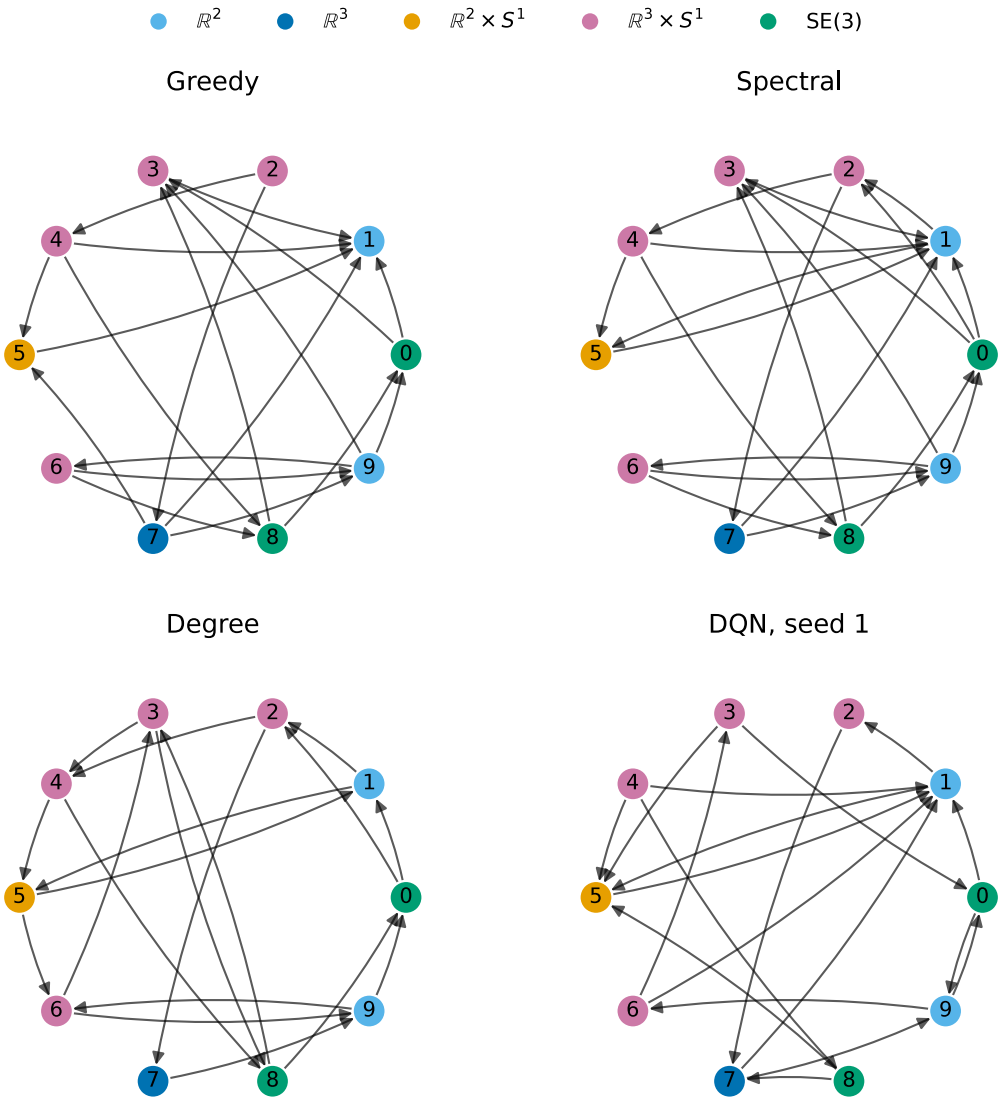


Figure 6.14: Graphs returned by four methods on the same formation. Node colors identify domains and arrows point from the measuring agent to its target.

A sparse graph can satisfy the rank condition while allocating its measurements poorly, and the degree method demonstrates this. It has the same number of edges as greedy and the learned policy, however it has a substantially higher error. Table 6.10 shows further analysis with some quantities from the resultant topologies.

Most selected edge blocks contribute a rank of 2 to the rigidity matrix (in isolation), and the out-degree requirements are satisfied throughout these outputs. These are ”local” properties and they do not establish that all selected edges contribute when considered jointly. The usefulness

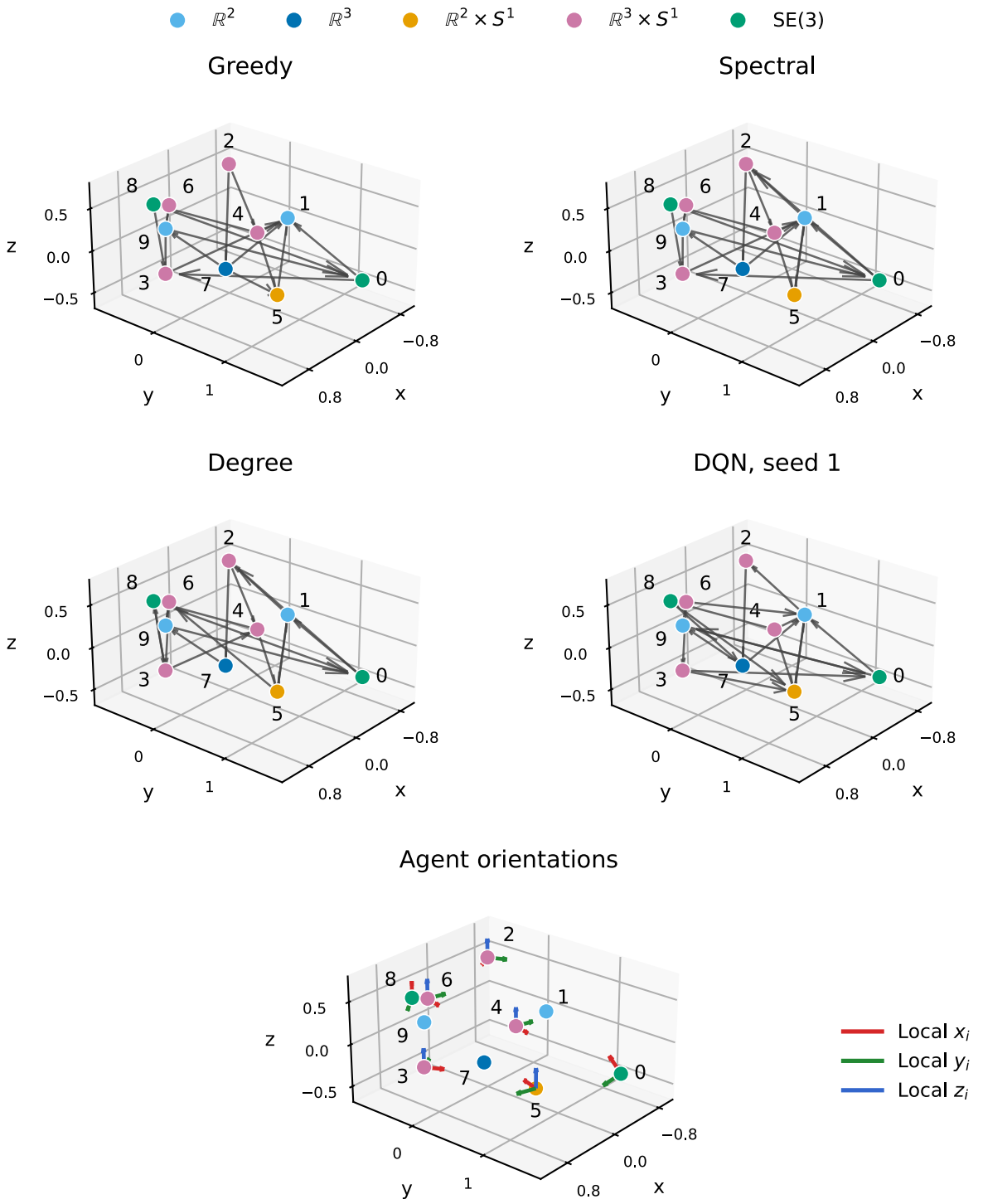


Figure 6.15: The same instance and edge sets as Figure 6.14, shown in 3D. Positions are centered and divided by the formation radius for display, with a common viewing angle and scale in all panels. Colors identify domains, labels identify agents, and arrows run from measuring agents to their targets. The bottom-center panel shows the agent orientations with red, green, and blue arrows denoting local x_i, y_i, and z_i for agents with orientation freedom.

Table 6.9: Measurements for the formation in Figure 6.14.

Method	Rigid	At bound	m	ε
Greedy	1/1	0/1	19.00	5.10
Spectral	1/1	0/1	21.00	6.08
Degree	1/1	0/1	19.00	36.20
DQN, seed 1	1/1	0/1	19.00	5.27

Table 6.10: Instance means over 200 reference outputs per method. Reciprocity is the percentage of directed edges whose reverse is present. Uniform is the expected reciprocity percentage for a uniformly sampled directed graph of the same size as the one the method produced. Rank two is the percentage of edge blocks in the rigidity matrix with rank two. Degree floor is the percentage of nodes satisfying their domain dependent necessary out-degree condition (Table 4.4). Length is the mean edge length divided by formation radius.

Method	Recip.	Uniform	Rank two	Degree floor	Length
Greedy	15.6	18.7	92.6	100.0	1.40
Spectral	21.9	22.5	88.5	100.0	1.40
Degree	15.8	19.2	85.9	100.0	1.40
DQN, seed 0	13.1	18.5	94.7	100.0	1.40
DQN, seed 1	12.2	18.4	94.6	100.0	1.41
DQN, seed 2	13.9	18.4	95.6	100.0	1.41

of reciprocal edges depends on whether the endpoint agents carry orientational freedom. These measurements do not by themselves explain the conditioning differences.

Figure 6.16 shows the uncertainty (covariance) per agent position in the inverse problem of estimating the agent positions (and orientations) from the set of bearing measurements. An elongated ellipsoid indicates that the position error grows in the respective axis more under bearing noise. Figure 6.17 shows the positional part of the softest mode of the normalized rigidity matrix, which is the set of variations the formation is least resistant to. The spectral and degree formations are shown in Appendix B.11. The attitude variations are omitted in the figures.

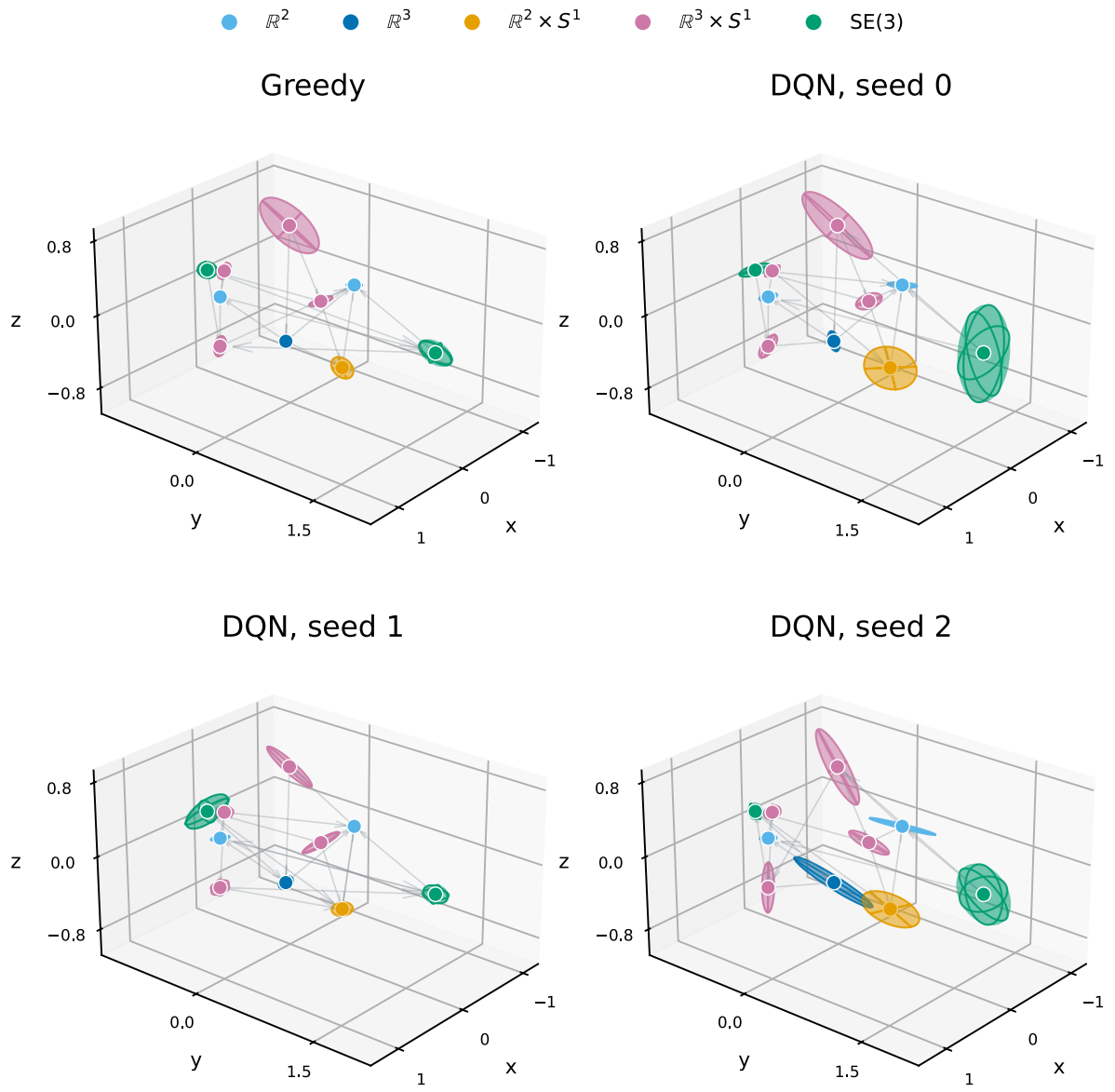


Figure 6.16: Position covariance blocks of $(B_s^\top B_s)^+$ for greedy and all three reference DQN seeds on the formation in Figure 6.15. Ellipsoid semiaxes are proportional to the square roots of the block eigenvalues, using one display factor for all agents in these four panels. Their sizes are comparable across panels, but the surfaces are illustrative and do not denote a stated probability region.

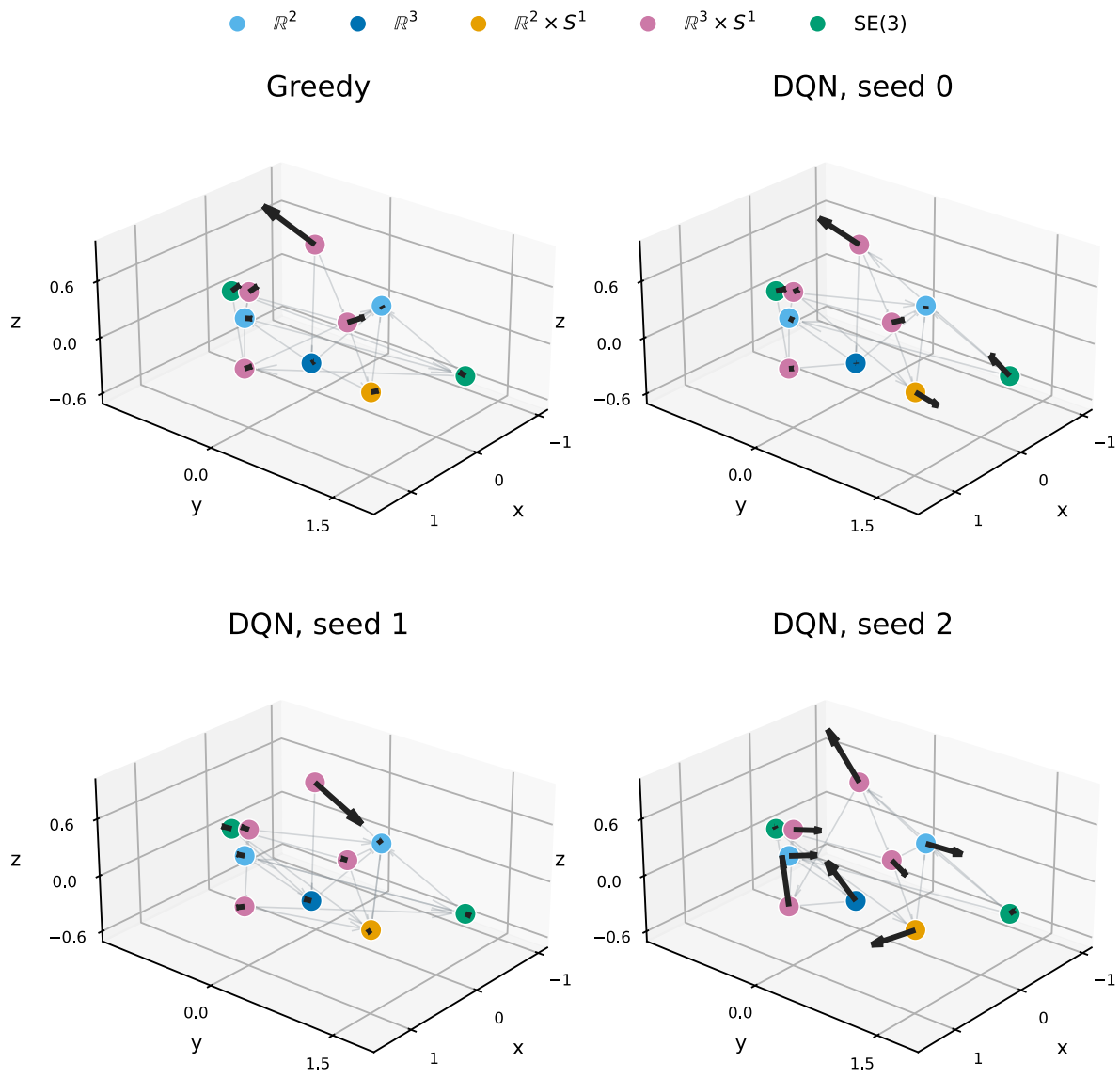


Figure 6.17: Position components of a unit eigenvector for the smallest positive eigenvalue of $B_s^\top B_s$ for greedy and all three reference DQN seeds on the formation in Figure 6.15. The longest arrow is scaled to the same display length in each panel, but the arrow lengths do not compare conditioning between methods. Eigenvector sign is arbitrary and attitude components are not shown.

6.8 Formation control

The experiment whose results are presented here fixes a topology selected on a goal formation and uses that topology throughout a bearing based gradient control simulation.

The controller is the stabilization law of [28]. With $b_{\mathcal{G}}^{\star} = b_{\mathcal{G}}(\chi^{\star})$ the bearings of the goal configuration $\chi^{\star}$ over the selected edge set, and $\mathbf{B}_{\mathcal{G}}(\chi)$ (4.13) evaluated at the current configuration χ, every agent is commanded by

$$u = \begin{bmatrix} \dot{p} \\ \omega \end{bmatrix} = k_c \, \mathbf{B}_{\mathcal{G}}(\chi)^{\top} b_{\mathcal{G}}^{\star}, \tag{6.3}$$

where $\dot{p} \in \mathbb{R}^{3n}$ stacks the linear velocities, $\omega \in \mathbb{R}^{3n}$ stacks the angular velocities in the world frame, and $k_c > 0$ is the gain. The command descends $b_{\mathcal{G}}(\chi) - b_{\mathcal{G}}^{\star}$ (derivation omitted for brevity), and the bearing error evolves as

$$\dot{e}_{\mathcal{G}} = -k_c \, \mathbf{B}_{\mathcal{G}}(\chi) \mathbf{B}_{\mathcal{G}}(\chi)^{\top} e_{\mathcal{G}}. \tag{6.4}$$

The configuration is advanced by an explicit Euler step, $p_i \leftarrow p_i + \Delta t \, \dot{p}_i$ and $R_i \leftarrow \exp\big(\Delta t \, [\omega_i]_{\times}\big) R_i$, with the attitude quaternion renormalized after each step.

Each method receives the same goal bearings and perturbed start within a case. The initial position displacement is Gaussian with standard deviation 0.15 in the sampled position units, restricted to the agents' admissible motions, initial attitudes equal the goal attitudes, and the goal bearings and the current bearings use the same directed edges which are determined via optimizing the topology with the respective method/policy. Changing the topology consequently changes the measurements available to the same controller rather than changing the control law.

The simulations use equal translational and rotational gains and a time step of 10^{-3}. Convergence requires the RMS bearing residual per edge to fall below 10^{-3} within $200{,}000$ steps. Dividing by the number of edges makes the residual criterion comparable between a sparse graph and the complete graph. Position error is measured relative to formation radius after projecting out the unobservable variations, and its interpretation away from the goal is limited by that linear projection.

Table 6.11 reports every shared case, and Figure 6.18 shows the bearing and position trajectories for the first case. The full set of bearing trajectories is shown in Appendix B.10. Topologies for this experiment are selected with $\kappa_e = 10$.

The attitude error trajectory starts near zero because the initial attitudes equal the goal attitudes, and rises while the controller corrects the displaced positions. The recorded attitude quantity is a formation level RMS after gauge projection. Figure 6.19 shows the physical paths for the same first control case.

The displacement paths in Figure 6.20 remove each agent's initial position from its trajectory, making the corrections visible at a common scale, and it is applied separately to each agent. The component traces are shown in Appendix B.12.

Table 6.11: Steps to the per-edge bearing tolerance on five cases. A dash denotes failure to reach the tolerance within the simulation limit.

Method	Reached	Case 1	Case 2	Case 3	Case 4	Case 5
Greedy	4/5	35200	–	54800	182800	85800
Complete graph	5/5	1400	1800	8000	2000	1400
DQN, $\kappa_t = 0$	4/5	76400	94000	85400	55600	–
DQN, $\kappa_t = 10$, seed 0	5/5	76800	128800	59200	53600	121800
DQN, $\kappa_t = 50$	5/5	48400	33600	126000	46600	52200
DQN, $\kappa_t = 200$	4/5	52800	–	36800	23000	44000

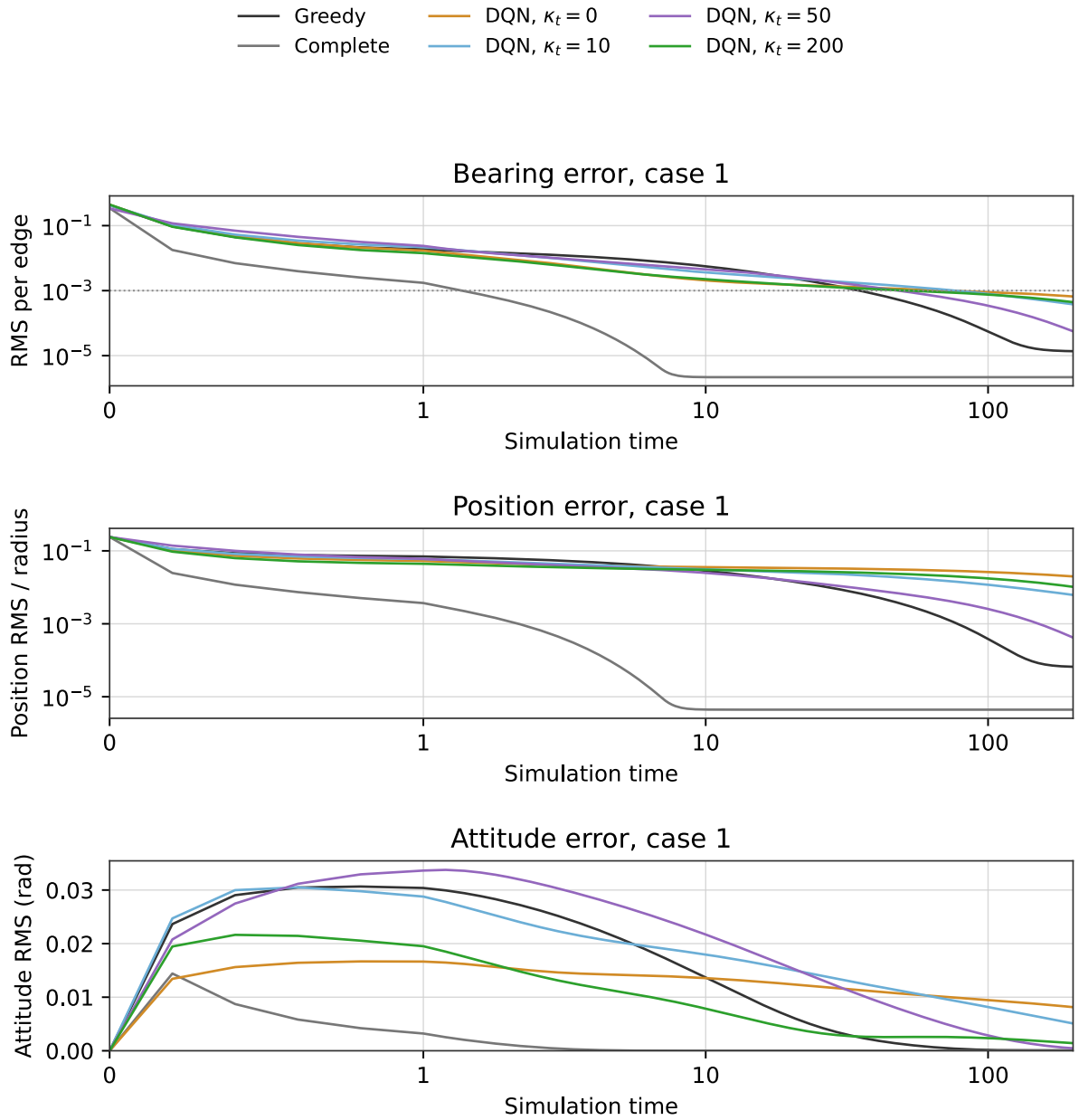


Figure 6.18: Bearing RMS per edge and gauge-projected position and attitude RMS for the first case. The time axis is linear up to $t = 1$ and logarithmic thereafter.

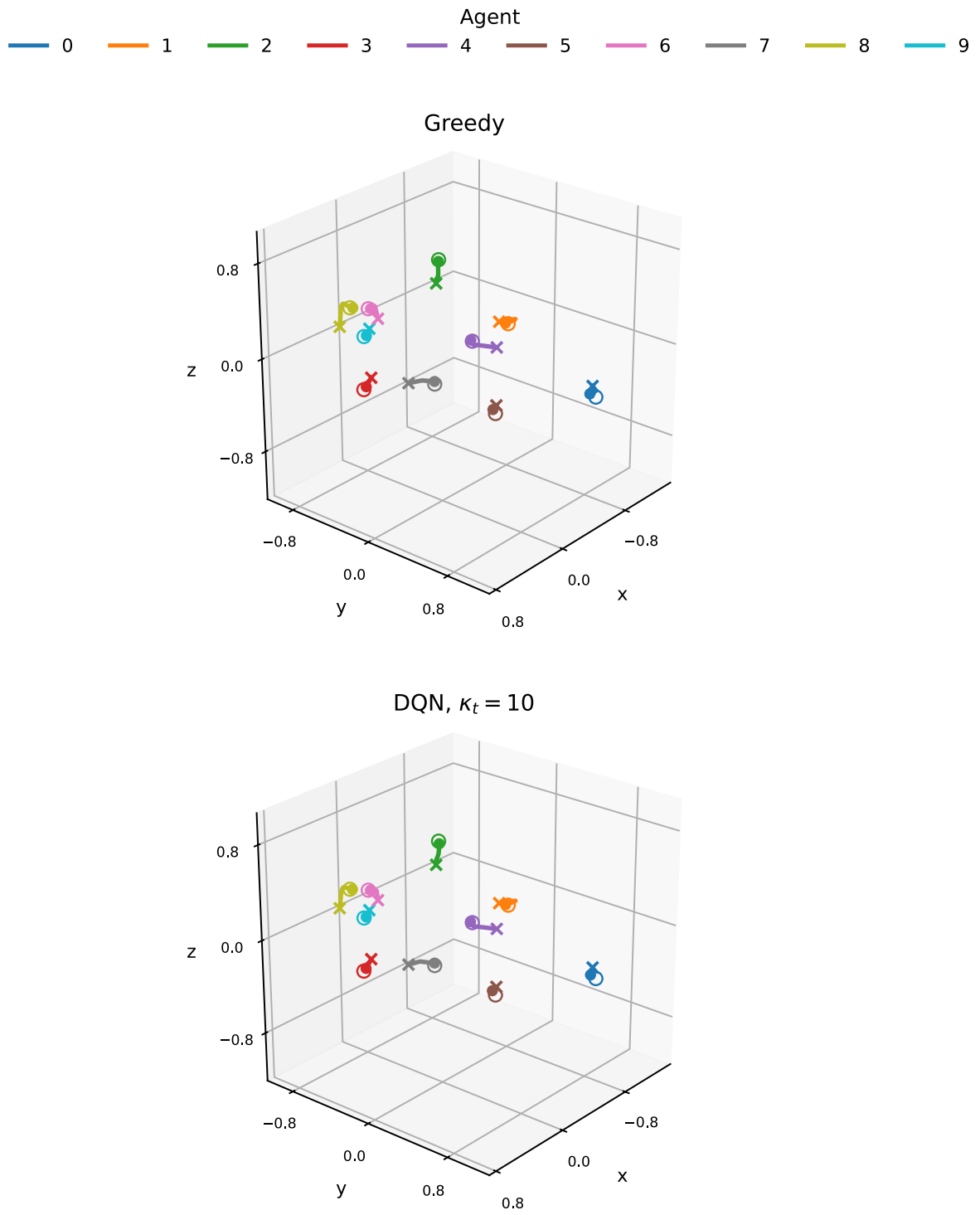


Figure 6.19: Recorded positions in the first case for greedy and the weight-10 policy. Colors identify agents, hollow markers are goals, crosses are starts, and filled markers are final positions. Arrows indicate the direction of motion.

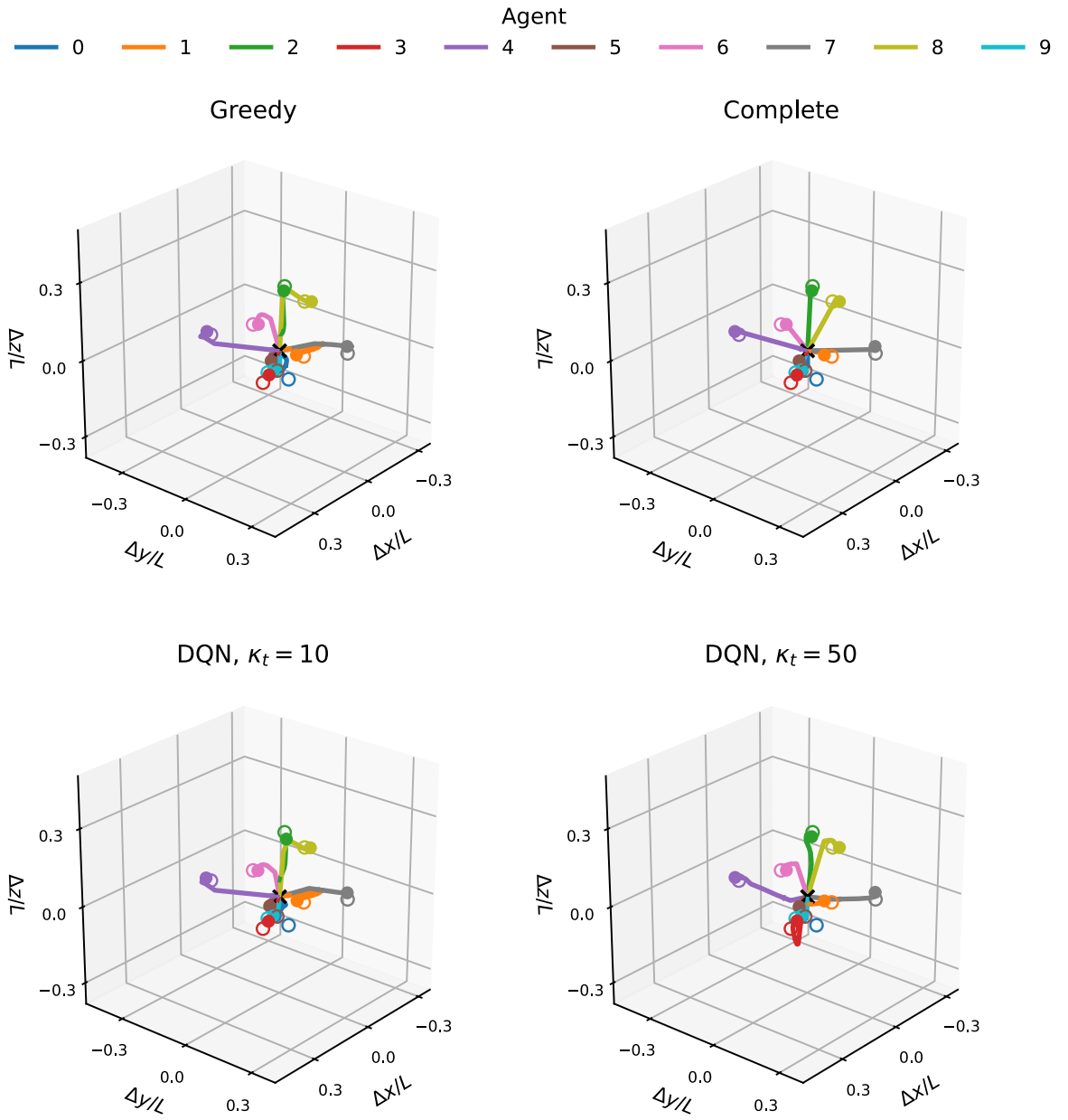


Figure 6.20: Recorded displacements $(p_i(t) - p_i(0))/L$ in the first case, with L the goal formation radius. Each path begins at the origin. Colors identify agents, filled points are final displacements, and hollow points give $(p_i^{\text{goal}} - p_i(0))/L$.

7 Conclusions and future work

7.1 Conclusions

This thesis studied the choice of the directed bearing measurements of a formation of heterogeneous agents, with the goal of constructing rigid frameworks and a balance between the number of measurements made and estimation quality of the formation shape from noisy bearings. The problem was analyzed with bearing rigidity theory, posed as a Markov decision process over edge sets, and approached with graph neural network policies trained by deep reinforcement learning that add and remove edges sequentially.

Bearing rigidity matrix of heterogeneous formations

The extended rigidity matrix of heterogeneous formations was established with the DOF restrictions applied per agent, it was shown that the rank test for infinitesimal bearing rigidity holds with the new construction for the matrix, and the matrix agrees with that of [3] on homogeneous formations.

Quantities derived from theory and the edge count algorithm

Quantities such as the rank gain of the rigidity matrix when an edge is added were derived from rigidity theory, in order to inform the optimizer policy via the observation and to define an objective function to be optimized. These oracle quantities and metrics such as the estimated shape error of the formation were normalized in order to be comparable across network sizes and formation scales. A lower bound on the minimum number of edges required for a framework to be infinitesimally bearing rigid was provided with an algorithm, extending the closed form count of [4] to directed graphs and heterogeneous compositions, and matching its results on homogeneous $\mathbb{R}^d$ frameworks. The lower bound was not shown to be attained for other domain compositions.

Structure of the objective

It was shown that two objectives, namely constructing a minimally rigid framework, and maximizing the rigidity eigenvalue, have different structures from a combinatorial optimization per-

spective. The rank of the rigidity matrix, known to be submodular for formations with planar weighted bearing-compass dynamics [34], was shown to be submodular for directed heterogeneous formations as well, meaning choosing the fewest edges for rigidity is a minimum submodular cover problem. As a result, greedy construction, which repeatedly adds the edge with the largest rank gain, reaches rigidity with the fewest possible edges when every agent is planar, and with at most 1.5 times the optimal cover cardinality otherwise. The rigidity eigenvalue for the purposes of this work was shown not to be submodular (considering it equals zero if the framework is not rigid), therefore showing that greedy does not have a guarantee on this objective. These findings located the part of the problem where a learned sequential method could improve on greedy selection, as graph reinforcement learning does not typically yield considerable improvements when shallow decision horizons are sufficient to construct optimal solutions [2].

Methodology

Topology optimization was posed as a Markov decision process in which a policy modifies the edge set at each step. An objective function combining rigidity, sparsity, and robustness to bearing noise was defined, with each term normalized to account for the number of agents or the scale of the formation. The reward used was the change in this objective between consecutive states as a potential-based shaping that gives a dense reward at every step. The observation provided the bearings for every ordered pair of agents, together with the quantities derived from the rigidity matrix. The policies were graph neural networks (GINE and EGNN) that pass messages over all pairs of agents and score every possible edit, with invalid edits masked, and they were trained with DQN and PPO on episodes with randomly drawn poses, domains, and initial edge sets with the initial edge count centered on the lower bound given by the developed algorithm.

Learned policies

The learned policies were analyzed in terms of behavior, outcome, costs, and compared with classical search methods, mainly the greedy search which repeatedly applies the single edge change that improves the objective most. On formations drawn from the training distribution, with 10 agents of randomly drawn types, and with a moderate weight on estimation quality, the policies always returned rigid graphs with about as many measurements as greedy search, and lower or equal estimation error. When estimation quality was weighted more heavily, greedy search was better overall on both counts, however the results should be further verified with repeated training. An analysis of the individual decisions of the policies showed that they behave as fast local searches, taking the same decision as greedy search about half of the time. The policies also relied on inputs computed from the rigidity matrix of the whole formation, showing that the derived quantities played their role in the decision-making process, however implying a significant limitation in terms of decentralized execution of the developed methods. The policies were also evaluated on formations outside the training distribution without retraining, remaining

comparable or better against the classical methods on homogeneous formations, with better performance in some cases. Across network sizes, they remained rigid on every formation from smaller sizes up to about three times the training size, with a lower estimation error than greedy search, but on larger formations they increasingly failed to produce rigid graphs. Neither a different weight on estimation quality nor training at larger sizes removed this loss of rigidity, implying that the generalization capabilities of the trained policies (within the scope of this work) on different network sizes are limited, whereas generalization on domain compositions did not provide evidence of a limitation.

Limitations

The policies are centralized. They are provided every pose and the whole edge set, and the quantities that most affect their outputs are computed from the rigidity matrix of the whole formation. The number of agents is fixed during training. Rigidity is rewarded but not enforced, and the policies failed to produce rigid frameworks on zero-shot transfer experiments on the largest formations tested. The formations are static and the bearings are exact during optimization, with noise added only for evaluation.

7.2 Future work

The centralized formulation studied in this thesis, which follows naturally from bearing rigidity being a global property of the framework, is meant as a first step towards a distributed method for maintaining rigid formations, hence the immediate future direction is the development of distributed methods for the formulation provided in this work. Possible approaches include decomposition into *subframeworks* as in [36], and learning using centralized training with decentralized execution (CTDE) and other multi-agent reinforcement learning techniques.

A Further analysis

This appendix carries the derivations behind two results of Chapter 4 that the main text states and uses without reproducing, and a validation of the error coefficient.

A.1 Trivial motions

Equation (4.14) counts the variations of a formation that no bearing of the complete graph reports. The count follows from first describing those variations without reference to the domains, and then intersecting them with what each agent can actually do.

Lemma A.1 (bearing-preserving variations of the complete graph)**.** *Let $n \geq 3$, let the positions be distinct, and let the agents not all be collinear. A variation (u, ω) of (p, R), with $u_i = \dot{p}_i$ and ω_i the angular velocity of agent i, preserves every bearing of $\mathcal{K}$ if and only if there are $t, w \in \mathbb{R}^3$ and $\alpha \in \mathbb{R}$ such that*

$$u_i = t + w \times (p_i - q) + \alpha\,(p_i - q), \qquad \omega_i = w, \qquad \text{for every } i, \tag{A.1}$$

where $q \in \mathbb{R}^3$ is any fixed reference point. In particular that set is seven-dimensional.

Proof. Sufficiency is a calculation. Under (A.1) the relative velocity is $u_j - u_i = w \times p_{ij} + \alpha p_{ij}$. The scaling term is parallel to the bearing and so is annihilated by the projector, $P(\hat{p}_{ij})p_{ij} = 0$, and $w \times \hat{p}_{ij}$ is orthogonal to $\hat{p}_{ij}$ and so survives it untouched, which leaves

$$\mathbf{D}_p^{(k)}(u_j - u_i) = s_{ij} R_i^\top P(\hat{p}_{ij})(w \times p_{ij}) = R_i^\top \big(w \times \hat{p}_{ij}\big)$$

for the position term of (4.4), using $s_{ij} p_{ij} = \hat{p}_{ij}$. The attitude term of (4.3) contributes $R_i^\top [\hat{p}_{ij}]_\times \omega_i = R_i^\top (\hat{p}_{ij} \times w)$, which is the negative of the first. The two cancel, so $\dot{b}_{ij} = 0$ for every ordered pair.

For necessity, fix the measuring agent i and write $q_j^{(i)} = R_i^\top (p_j - p_i)$ for the position of j in i's frame. Preserving b_{ij} for every j says that every direction $q_j^{(i)} / \|q_j^{(i)}\|$ is constant, that is $\frac{d}{dt} q_j^{(i)} = \mu_{ij} q_j^{(i)}$ for scalars μ_{ij}. Expanding the derivative and multiplying by R_i,

$$u_j - u_i - \omega_i \times p_{ij} = \mu_{ij}\, p_{ij} \qquad \text{for all } j \neq i. \tag{A.2}$$

Adding (A.2) for the ordered pair (i, j) to the same equation for (j, i), and using $p_{ji} = -p_{ij}$, gives

$$(\omega_j - \omega_i) \times p_{ij} = (\mu_{ij} - \mu_{ji})\, p_{ij}.$$

The left side is orthogonal to p_{ij} and the right side is parallel to it, so both vanish. Hence $\mu_{ij} = \mu_{ji}$ and

$$\omega_j - \omega_i \in \operatorname{span}\{p_{ij}\} \qquad \text{for every pair.} \tag{A.3}$$

Now take agent 1 as reference and set $r_i = p_i - p_1$. Equation (A.2) at $i = 1$ gives

$$u_j = u_1 + \omega_1 \times r_j + \mu_{1j}\, r_j \qquad (j \neq 1), \tag{A.4}$$

and by (A.3) we may write $\omega_1 - \omega_i = a_i\, r_i$ for scalars a_i, since $p_{i1} = -r_i$. Substituting (A.4) for u_i and u_j into (A.2) for a pair $i, j \neq 1$, and collecting terms,

$$a_i\,(r_i \times r_j) + \big(\mu_{1j} - \mu_{ij}\big) r_j + \big(\mu_{ij} - \mu_{1i}\big) r_i = 0. \tag{A.5}$$

Because the agents are not all collinear and the positions are distinct, for every $i \neq 1$ there is a j with p_1, p_i, p_j not collinear, because the line through p_1 and p_i cannot contain every agent, so some j lies off it. For such a j the vectors r_i and r_j are linearly independent, $r_i \times r_j \neq 0$ is orthogonal to both, and $\{r_i, r_j, r_i \times r_j\}$ is a basis of $\mathbb{R}^3$. Reading (A.5) in that basis gives $a_i = 0$ and $\mu_{1i} = \mu_{ij} = \mu_{1j}$. So every ω_i equals ω_1. The scalars μ then share one value α. Fix two agents k and l with p_1, p_k, p_l not collinear, so that the display above gives $\mu_{1k} = \mu_{1l}$. Any other agent i lies off at least one of the lines through p_1 and p_k and through p_1 and p_l, since these meet only at p_1, and the same display then gives $\mu_{1i} = \mu_{1k}$. Every μ_{ij} equals this common value by (A.5) with $a_i = 0$. Substituting into (A.4) and writing $w = \omega_1$ and $t = u_1 - w \times (p_1 - q) - \alpha(p_1 - q)$ gives (A.1). The parameters (t, w, α) range over $\mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}$ and (A.1) is injective in them, so the set is seven-dimensional, and changing q only reparameterizes t. □

The three parameters are the three familiar motions. Taking $w = 0$ and $\alpha = 0$ leaves a common translation, taking $t = 0$ and $w = 0$ leaves a uniform scaling about q, which every bearing misses because the relative velocity is then parallel to the bearing, and taking $\alpha = 0$ with $w \neq 0$ leaves a coordinated rotation, under which every frame and every inter-agent direction turn together so that $b_{ij} = R_i^{\top} \hat{p}_{ij}$ is unchanged. Only the third distinguishes the domains, since it moves each agent's own frame, and a frameless agent measures in the global frame, so one such agent anywhere in the formation removes the whole rotational part.

Intersecting (A.1) with admissibility gives the count. A variation is admissible when $u_i \in$ range S_i and $w \in$ range P_i for every i. The attitude condition is $w \in \bigcap_i$ range P_i and involves neither t nor α. Choosing q in the common plane of the planar agents and taking that plane as $z = 0$ makes both $\alpha(p_i - q)$ and $w \times (p_i - q)$ horizontal for a planar i, the latter because $w \in$ range $P_i \subseteq \operatorname{span}\{e_3\}$ for a planar agent, so its position condition reduces to $t \in$ range S_i, while a spatial agent imposes none. The admissible set is therefore $\big(\bigcap_i \text{range } S_i\big) \times \big(\bigcap_i \text{range } P_i\big) \times \mathbb{R}$

in the coordinates (t, w, α), giving

$$q_t = \underbrace{\dim\big(\textstyle\bigcap_i \text{range } S_i\big)}_{\text{translations}} + \underbrace{1}_{\text{scaling}} + \underbrace{\dim\big(\textstyle\bigcap_i \text{range } P_i\big)}_{\text{rotations}}. \tag{A.6}$$

Evaluating the two intersections per domain, as §4.3 does, gives (4.14).

A.2 Attitude observability

Table 4.4 gives the out-degree each domain forces on an agent. It is read off the following condition.

Proposition A.2 (attitude observability)**.** *Let $O_i = \{j : (v_i, v_j) \in \mathcal{E}\}$ be the agents that i measures. If $(\mathcal{G}, \chi)$ is infinitesimally bearing rigid, then*

$$\text{range}(P_i) \cap \bigcap_{j \in O_i} \text{span}\{\hat{p}_{ij}\} = \{0\} \qquad \textit{for every } i \in \mathcal{V}, \tag{A.7}$$

an empty intersection being read as $\mathbb{R}^3$. That is, no rotation agent i is able to perform may leave every bearing it takes unchanged.

Proof. Suppose some $w \neq 0$ lies in the intersection, and let δ be the variation that is zero everywhere except $\delta_{a,i} = w$. By (4.10) the only row blocks touching agent i's attitude columns are those of the edges $i \to j$ with $j \in O_i$, each contributing $R_i^\top [\hat{p}_{ij}]_\times w$, which vanishes exactly when $w \in \text{span}\{\hat{p}_{ij}\}$. That holds for every such j by assumption, so $\mathbf{B}_\mathcal{G}\delta = 0$. On $\mathcal{K}$ agent i measures every other agent, so unless they are all collinear with i the same intersection is $\{0\}$ and $\mathbf{B}_\mathcal{K}\delta \neq 0$. Hence $\ker \mathbf{B}_\mathcal{G} \supsetneq \ker \mathbf{B}_\mathcal{K}$. □

Reading (A.7) per domain gives Table 4.4. The condition is known in the homogeneous cases, as $d_{\text{out}}(v) \geq 1$ in $\mathbb{R}^2 \times S^1$ [21] and as $\dim \mathcal{B}_i \geq d - 1$ in $\text{SE}(d)$ [31], with $\mathcal{B}_i$ the span of the bearings agent i takes.

A.3 Error validation

The shape error (4.30) is read off the rigidity matrix, so it is a first-order prediction. It is checked against the error left after actually recovering a formation from perturbed bearings. The noise model is

$$z_{ij} = \frac{b_{ij} + \sigma\,(I - b_{ij}b_{ij}^\top)\eta_{ij}}{\big\|b_{ij} + \sigma(I - b_{ij}b_{ij}^\top)\eta_{ij}\big\|}, \qquad \eta_{ij} \sim \mathcal{N}(0, I_3), \tag{A.8}$$

which is (2.47) so σ is an angle in radians. The formation is recovered from the perturbed bearings by Gauss-Newton (§2.8.3), with $\mathbf{B}$ as the Jacobian and the steps restricted through the node projectors so that no agent leaves its domain, and the error is measured against the truth with the similarity gauge of Lemma A.1 projected out. Each domain is measured on 30 random

formations, eight agents each and ten in the mixture, with one rigid graph per formation built by the constructive method of §6.1.1 and 100 noise draws per formation and noise level. Table A.1 gives the geometric mean over formations of the ratio of the measured to the predicted position RMS error, and Table A.2 pools all formations by the product $\sigma\varepsilon_p$ of the noise level and the position error coefficient of §6.1.

Table A.1: Geometric mean over 30 formations of the ratio of measured to predicted RMS position error, against the per-axis bearing noise σ, with 100 noise draws per formation and noise level. A ratio of one is exact agreement with the prediction.

domain	0.006°	0.06°	0.6°	1.7°	5.7°
$\mathbb{R}^2$	0.988	0.952	0.769	0.595	0.330
$\mathbb{R}^3$	0.995	0.996	0.985	0.980	0.849
$\mathbb{R}^2 \times S^1$	0.930	0.823	0.507	0.313	0.153
$\mathbb{R}^3 \times S^1$	0.900	0.770	0.612	0.476	0.291
$\mathrm{SE}(3)$	0.990	1.002	0.863	0.743	0.525
mixed	1.008	0.974	0.834	0.695	0.461

Table A.2: The ratios of Table A.1 for all domains, formations, and noise levels, grouped by $\sigma\varepsilon_p$. Cells counts the formation and noise level pairs in each group, and within 10% the share with a ratio between 0.9 and 1.1.

$\sigma\varepsilon_p$	cells	geometric mean ratio	within 10%
below 0.01	269	0.997	94%
0.01 to 0.03	69	0.998	84%
0.03 to 0.1	129	0.995	88%
0.1 to 0.2	96	0.970	59%
0.2 to 0.5	128	0.864	41%
0.5 to 1	84	0.663	20%
above 1	125	0.103	0%

The measured error agrees with the prediction at the smallest noise level in every domain, and falls below it as the noise grows. The prediction therefore holds while $\sigma\varepsilon_p$ is small, with a geometric mean ratio within 1% of one below $\sigma\varepsilon_p = 0.1$, and it overstates the error beyond that. The departure starts at a lower noise level in the domains whose formations have larger error coefficients, since $\sigma\varepsilon_p$ reaches 0.1 sooner there, and within a domain the error coefficients of the 30 formations differ by factors between 45 and 590, which is why the per-domain means of Table A.1 fall below one before the pooled ones of Table A.2.

B Further results

B.1 Training behavior

Figure B.1 shows the training progress of the three reference seeds. Table B.1 lists the configurations for the trained policies whose results are presented in this thesis. Figure B.2 shows the exploration schedules used to train the DQN policies.

B.2 Training size transfer

Figure B.3 compares three policies trained at $n = 10$, $n = 16$, and $n = 20$, evaluated on the same instances.

B.3 Configuration comparisons

B.3.1 Algorithm and backbone

Table B.2 compares DQN and PPO with GINE and EGNN backbones, all trained with $\kappa_t = 50$. All configurations obtain rigid graphs on both sets. Under the same κ_e, the PPO policies have lower error than the DQN policies but use more edges on average, and greedy uses fewer edges. At weight $\kappa_e = 10$, the learned policies use more edges than greedy and have lower error, and at $\kappa_e = 50$ the DQN policies have higher error than greedy even though they used more edges on average.

B.3.2 Action space

Table B.3 compares policies with the two action spaces on the same formations. The sequential policy fails to find a rigid graph on some formations, attains the lower bound on edge count less often, and keeps more edges. Both policies are given the same number of decisions, so the sequential policy can make at most half as many edits on the graph.

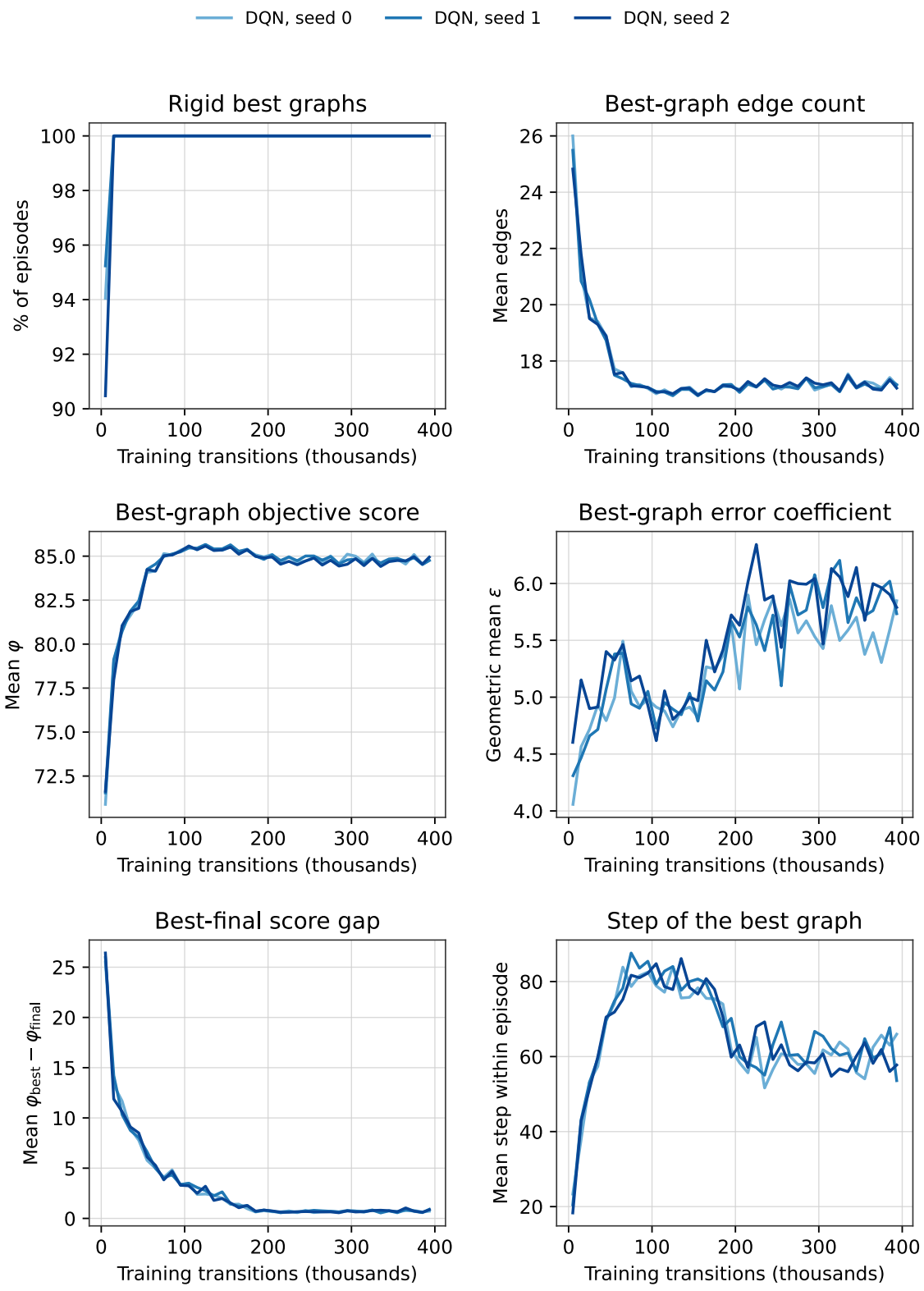


Figure B.1: Training curves of the three reference seeds with $\kappa_t = 10$. Error coefficient is shown only for rigid frameworks. Points on the plots average the actual logged episode data in disjoint bins of 10000 environment steps.

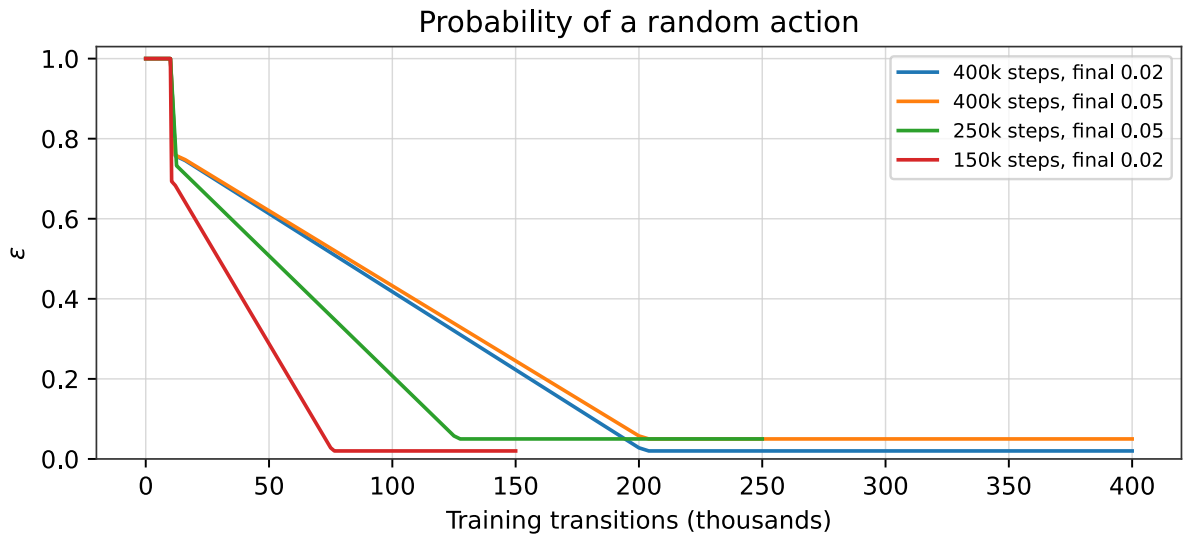


Figure B.2: Exploration schedules of the DQN policies. Actions are uniformly random for the first 10000 steps. The 400k schedule with final value 0.02 is used by the policies with $\kappa_t \in \{0, 10, 200\}$ and by the sequential, reduced, and $\kappa_t = 10$ stopping policies. The 400k schedule with final value 0.05 is used by the $\kappa_t = 50$ policies, the 250k schedule by the $n = 20$ policy, the other stopping policies, and the earlier-objective DQN policies, and the 150k schedule by the $n = 16$ policy.

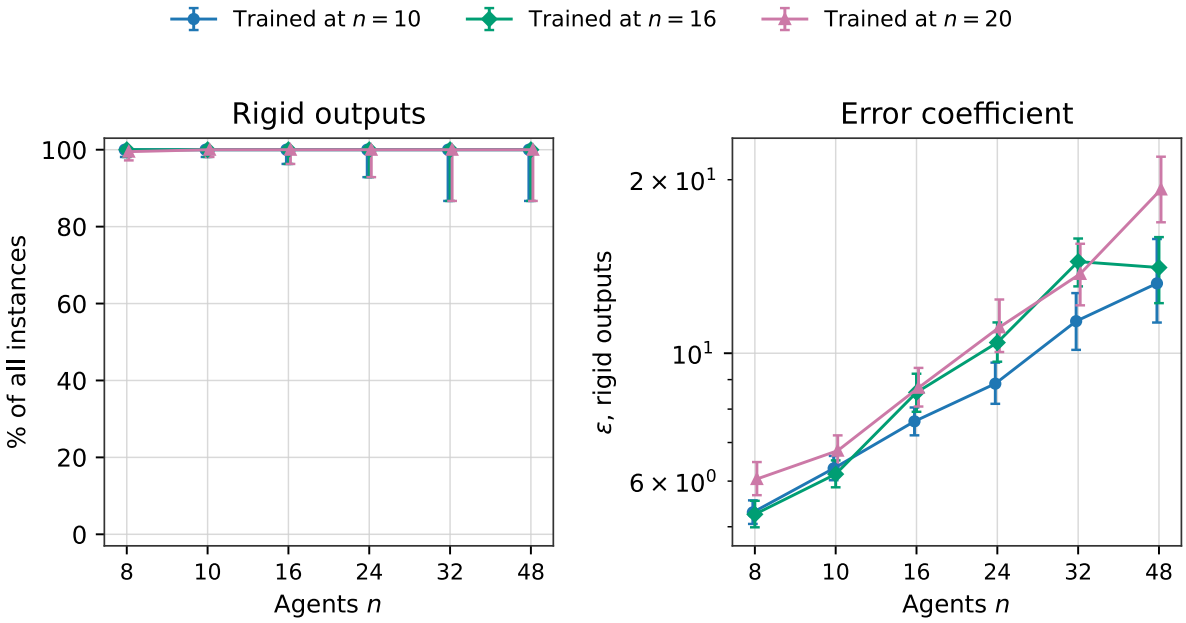


Figure B.3: Transfer of policies trained at different network sizes, evaluated with $\kappa_e = 10$. All three policies are DQN/GINE policies trained on random domains with $\kappa_t = 10$ for 150000 steps, and differ only in the training size.

Table B.1: Configurations of the trained policies. Random denotes independently sampled domains and fixed denotes a fixed hand-crafted domain mixture. The criteria λ, trace, and ε denote the forms of the robustness score (Table 5.3). E denotes the edge action space, S sequential node selection, and stop an available termination action. Reduced denotes the policy is trained without the rigidity observations.

Configuration	Formation	κ_t	Criterion	Actions	Steps (k)
DQN/EGNN	10 fixed	2	λ	E	250
DQN/GINE	10 fixed	2	λ	E	250
DQN/GINE	10 fixed	10	λ	E	250
DQN/GINE	10 fixed	200	λ	E	250
DQN/GINE	10 fixed	50	λ	E	125
DQN/GINE	20 random	2	ε	E	250
DQN/EGNN	10 random	50	ε	E	400
DQN/GINE	10 random	50	ε	E	400
PPO/EGNN	10 random	50	ε	E	400
PPO/GINE	10 random	50	ε	E	400
DQN/EGNN	10 fixed	2	λ	S	14
DQN/GINE	10 fixed	2	λ	S	250
DQN/GINE	10 fixed	200	λ	S	200
DQN/GINE	10 fixed	200	λ	S	250
DQN/GINE	10 fixed	50	λ	S	250
DQN/GINE	10 fixed	2	λ	S+stop	250
DQN/GINE	10 fixed	200	λ	S+stop	250
DQN/GINE	10 fixed	50	λ	S+stop	250
DQN/GINE	10 fixed	2	λ	E+stop	250
DQN/GINE	10 fixed	200	λ	E+stop	250
DQN/GINE	10 fixed	50	λ	E+stop	250
DQN/GINE	10 fixed	2	Trace	E	250
DQN/GINE	10 random	0	ε	E	400
DQN/GINE (seed 0)	10 random	10	ε	E	400
DQN/GINE (seed 1)	10 random	10	ε	E	400
DQN/GINE (seed 2)	10 random	10	ε	E	400
DQN/GINE	10 random	200	ε	E	400
DQN/GINE	10 random	10	ε	E	150
DQN/GINE	16 random	10	ε	E	150
DQN/GINE	20 random	10	ε	E	150
DQN/GINE (reduced)	10 random	10	ε	E	400
DQN/GINE	10 random	10	ε	S	400
DQN/GINE	10 random	10	ε	E+stop	400
DQN/EGNN	10 fixed	2	λ	E	250
DQN/GINE	10 fixed	2	λ	E	250
DQN/GINE	10 fixed	2	λ	S	250
PPO/GINE	10 fixed	2	λ	E	225
PPO/GINE	10 fixed	2	λ	E	200

Table B.2: Outcomes of the DQN and PPO policies with GINE and EGNN backbones trained with $\kappa_t = 50$, evaluated on instances with $n = 10$ and $\kappa_e = 50, 10$. The two κ_e comparisons use different random formation sets.

κ_e	Configuration	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$
50	Greedy	20/20	0/20	19.40	3.53
50	DQN/GINE	20/20	0/20	19.80	4.13
50	DQN/EGNN	20/20	0/20	19.60	4.03
50	PPO/GINE	20/20	0/20	19.95	3.14
50	PPO/EGNN	20/20	0/20	20.95	3.26
10	Greedy	200/200	84/200	17.62	7.04
10	DQN/GINE	200/200	19/200	18.55	4.48
10	DQN/EGNN	200/200	24/200	18.58	4.75
10	PPO/GINE	200/200	8/200	18.61	3.66
10	PPO/EGNN	200/200	7/200	19.18	4.19

Table B.3: Outcomes of the policies with direct-edge and sequential-node action spaces on formations at $n = 10$ with $\kappa_t = \kappa_e = 10$. Both have a horizon of 118 decisions.

Method	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$
Greedy	20/20	10/20	18.50	7.11
Direct edges, seed 0	20/20	12/20	18.20	6.37
Sequential nodes	18/20	3/20	20.44	9.58

B.4 Search budget

Figure B.4 shows the comparison of policies with different κ_t on how they perform with different budget/horizon. All four policies obtain a rigid graph on every formation within 14 edits and reach their final best score within 36 edits, well below the horizon of 118. The policies trained with $\kappa_t = 50$ and $\kappa_t = 200$ end on denser graphs with lower error than those trained with $\kappa_t = 0$ and $\kappa_t = 10$.

B.5 Selection weight

Table B.4 compares the returned graphs under two evaluation weights on the same formations. The DQN outputs are the same and only the returned best graph changes depending on κ_e, while greedy and spectral search can follow a different sequence of edits. With the larger weight every method returns denser graphs with smaller error, which shows that a single search by a trained policy visits graphs with different balances between edge count and conditioning.

B.6 Stopping policies

Policies with a stop action are evaluated on the graph they stop at instead of the best one visited during the episode. Table B.5 reports how often the episode ends before the horizon, the final outputs, and whether the final graph scores lower than a graph visited earlier.

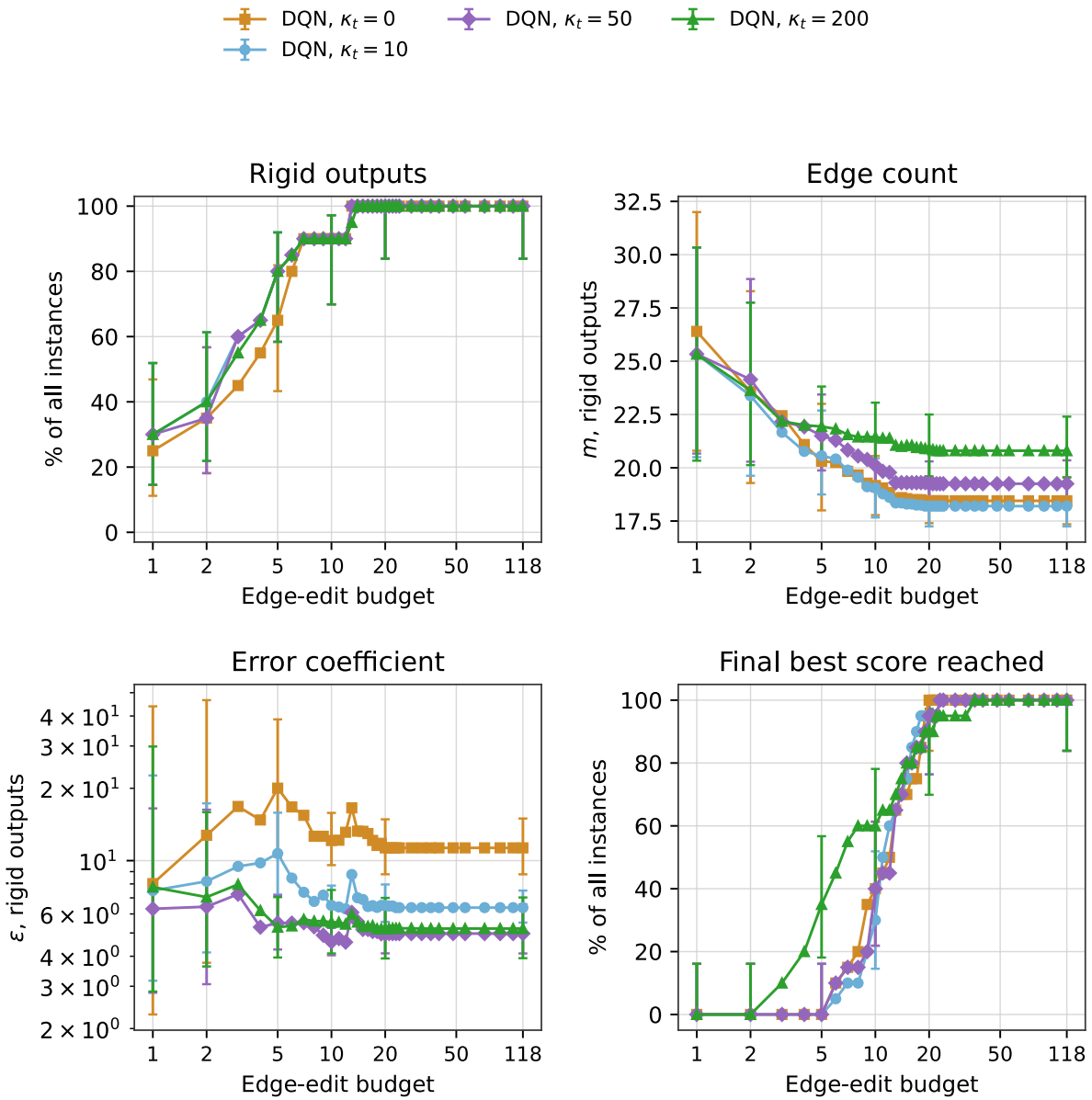


Figure B.4: Best graph found within each edit budget by policies trained with different weights. All policies are evaluated on the same 20 diagnostic formations with $\kappa_e = 10$. The $\kappa_t = 10$ policy is seed 0. Final best score reached is the fraction of results whose best score within the budget equals the best score over the full horizon.

Note. The $\kappa_t = 10$ policy is trained on randomized domains with the shape error objective, while the other policies are trained on a fixed heterogeneous mixture with the spectral objective. This means that their training weights are not comparable with the evaluation weight.

B.7 Decision quality

Figure B.5 ranks each edit of the reference policies among all legal single-edge edits, before and after the best graph of the episode is first reached. The best graph is reached within 24 edits in every episode, and after that, the policies oscillate most of the time by adding/removing until the episode terminates.

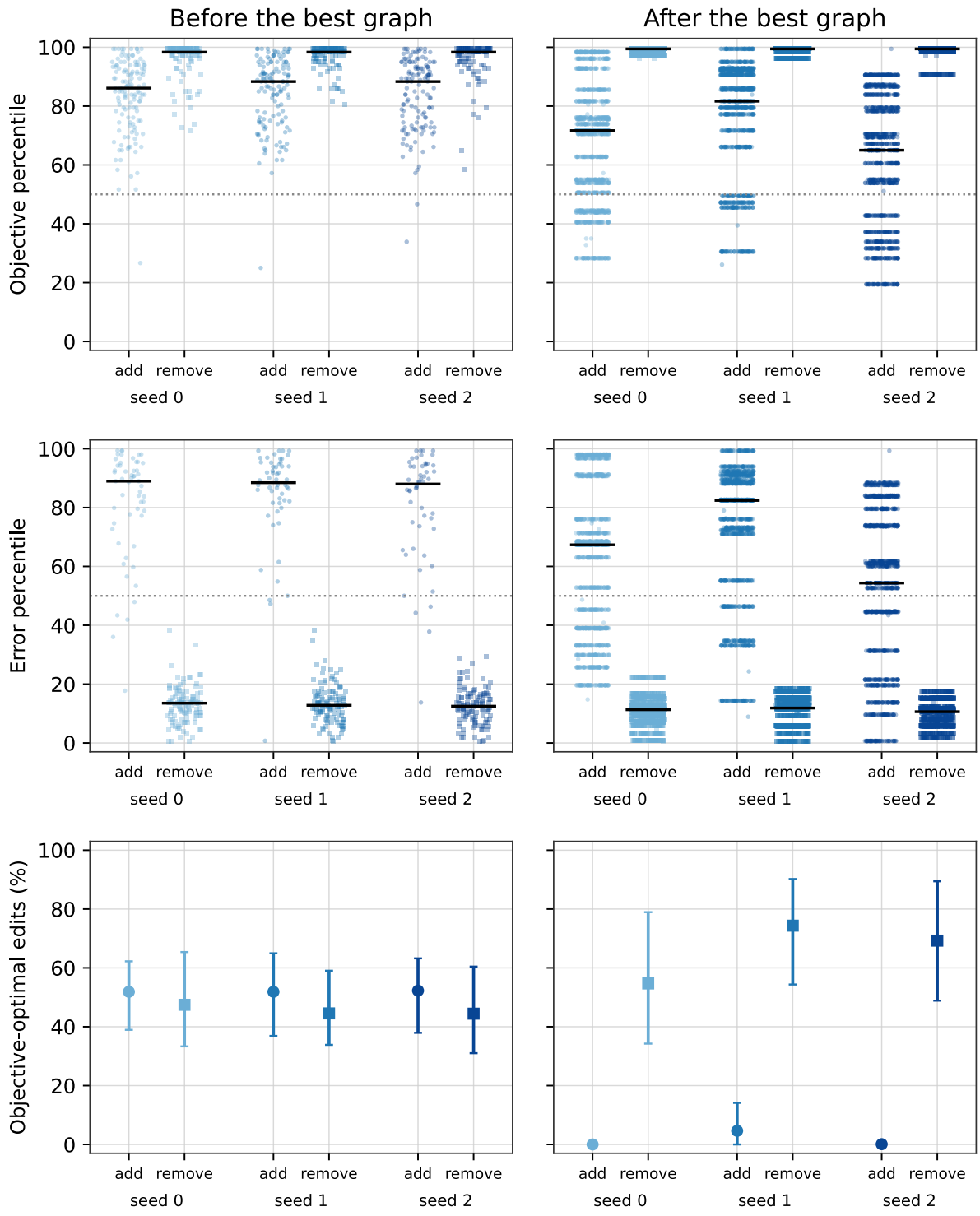


Figure B.5: Ranks of the reference policies' edits among all legal edits, before and after the best graph of the episode is first reached. Each point is one edit from 20 diagnostic episodes per seed with $\kappa_t = \kappa_e = 10$, and black lines mark the medians. Error ranking uses only edits from a rigid graph to a rigid graph. The bottom row gives the fraction of edits that are the objective-optimal edit (same as greedy), with bootstrap intervals over episodes.

Table B.4: Outputs under two evaluation weights on the same 50 formations at $n = 24$.

Method	κ_e	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$
Greedy	10	50/50	7/50	44.46	13.04
Greedy	50	50/50	0/50	45.36	5.49
Spectral	10	50/50	0/50	51.14	12.99
Spectral	50	50/50	0/50	53.96	4.64
DQN, $\kappa_t = 0$	10	50/50	14/50	43.64	25.93
DQN, $\kappa_t = 0$	50	50/50	0/50	45.26	15.80
DQN, $\kappa_t = 10$ seed 1	10	50/50	6/50	46.58	8.77
DQN, $\kappa_t = 10$ seed 1	50	50/50	0/50	47.42	7.24
DQN, $\kappa_t = 50$	10	50/50	0/50	49.12	5.77
DQN, $\kappa_t = 50$	50	50/50	0/50	49.58	4.80
DQN, $\kappa_t = 200$	10	50/50	0/50	54.44	7.46
DQN, $\kappa_t = 200$	50	50/50	0/50	56.22	4.07

B.8 Observation channels

Table B.6 compares a policy trained without the tier 3 rigidity observation channels of Table 5.7 with the informed policies and greedy. The policy trained without these channels still obtains rigid graphs, but with more edges on average, while the difference in error does not give a conclusive result. This comparison uses 20 formations at the training size only.

Figures B.6 and B.7 show how the policies trained and evaluated with $\kappa_t = \kappa_e = 0, 200$ respond when a single observation channel is perturbed and the trained policy is kept fixed. The perturbation is applied to the observation at every step in one of three ways as follows. Shuffling permutes the values of the channel among the nodes for a node channel or among the ordered pairs for a pair channel, zeroing sets every value of the channel to zero, and noise adds zero mean Gaussian noise whose standard deviation equals that of the channel's values in the same observation. The changed actions are the fraction of decisions that are different than those the policy makes without the perturbation, and the score loss is the decrease in the objective of the returned graph when the *whole episode is rerun* with the perturbed channel.

B.9 Noise validation

Figure B.8 shows the ratio of the measured to the predicted position RMS error at each bearing noise level for the graphs returned by each method. The measured error is obtained by adding tangent-plane noise to the bearings of the returned formation and recovering the agent poses by Gauss-Newton (§2.8.3), with unobservable directions projected out. The predicted error is the position part of the error coefficient (6.2) of the same graph.

Figure B.9 compares the predicted and measured position RMS of the individual formations at 1° noise.

Table B.5: Final outputs of the stopping policies under the shape error objective with $\kappa_e = 10$. Early counts episodes that end before the horizon. Edge count and error are reported only on rigid outputs. Lost best counts final graphs scoring below the best visited graph. Final best shows the count where the policy ended on a graph that has the same score as its best visited one (the tolerance is 10^{-8}, and the graphs are not necessarily the same). Later best shows the count (subset of Final best) where the policy made further edits after first reaching its best score, but ended up on the same score in the end (therefore if the policy made redundant edits). Redundant edits is the mean number of edits made after the best score was first reached, over all episodes.

Final outcomes

n	Policy	Early	Rigid	$\bar{m}$	ε	Edits
10	$\kappa_t = 10$	197/200	200/200	21.16	8.24	6.8
10	$\kappa_t = 2$	169/200	198/200	17.72	13.67	25.8
10	$\kappa_t = 50$	152/200	200/200	23.89	2.73	35.8
10	$\kappa_t = 200$	0/200	200/200	26.36	2.30	118.0
16	$\kappa_t = 10$	96/100	96/100	33.58	12.15	16.9
16	$\kappa_t = 2$	87/100	100/100	28.85	22.50	37.5
16	$\kappa_t = 50$	76/100	100/100	37.83	3.08	57.4
16	$\kappa_t = 200$	0/100	100/100	40.02	2.78	190.0
24	$\kappa_t = 10$	31/50	31/50	53.87	16.10	118.5
24	$\kappa_t = 2$	34/50	44/50	44.20	22.88	108.4
24	$\kappa_t = 50$	36/50	47/50	61.34	3.07	95.4
24	$\kappa_t = 200$	0/50	50/50	62.50	3.19	286.0

Best-score timing

n	Policy	Lost best	Final best	Later best	Redundant edits
10	$\kappa_t = 10$	15/200	185/200	0/200	1.7
10	$\kappa_t = 2$	30/200	170/200	12/200	16.6
10	$\kappa_t = 50$	162/200	38/200	5/200	28.5
10	$\kappa_t = 200$	188/200	12/200	12/200	111.7
16	$\kappa_t = 10$	14/100	86/100	0/100	7.7
16	$\kappa_t = 2$	13/100	87/100	7/100	23.0
16	$\kappa_t = 50$	92/100	8/100	1/100	46.6
16	$\kappa_t = 200$	98/100	2/100	2/100	179.2
24	$\kappa_t = 10$	23/50	27/50	0/50	106.0
24	$\kappa_t = 2$	16/50	34/50	6/50	82.1
24	$\kappa_t = 50$	44/50	6/50	3/50	79.1
24	$\kappa_t = 200$	46/50	4/50	4/50	270.5

Table B.6: Outputs of the policy trained without rigidity features, the reference policies, and greedy on 20 random formations at $n = 10$ with $\kappa_t = \kappa_e = 10$.

Method	Rigid	At bound	$\bar{m}$	$\varepsilon_{\text{geom}}$
Greedy	20/20	10/20	18.50	7.11
DQN, seed 0	20/20	12/20	18.20	6.37
DQN, seed 1	20/20	14/20	18.15	6.27
DQN, seed 2	20/20	13/20	18.20	7.66
Without rigidity features	20/20	1/20	19.95	7.22

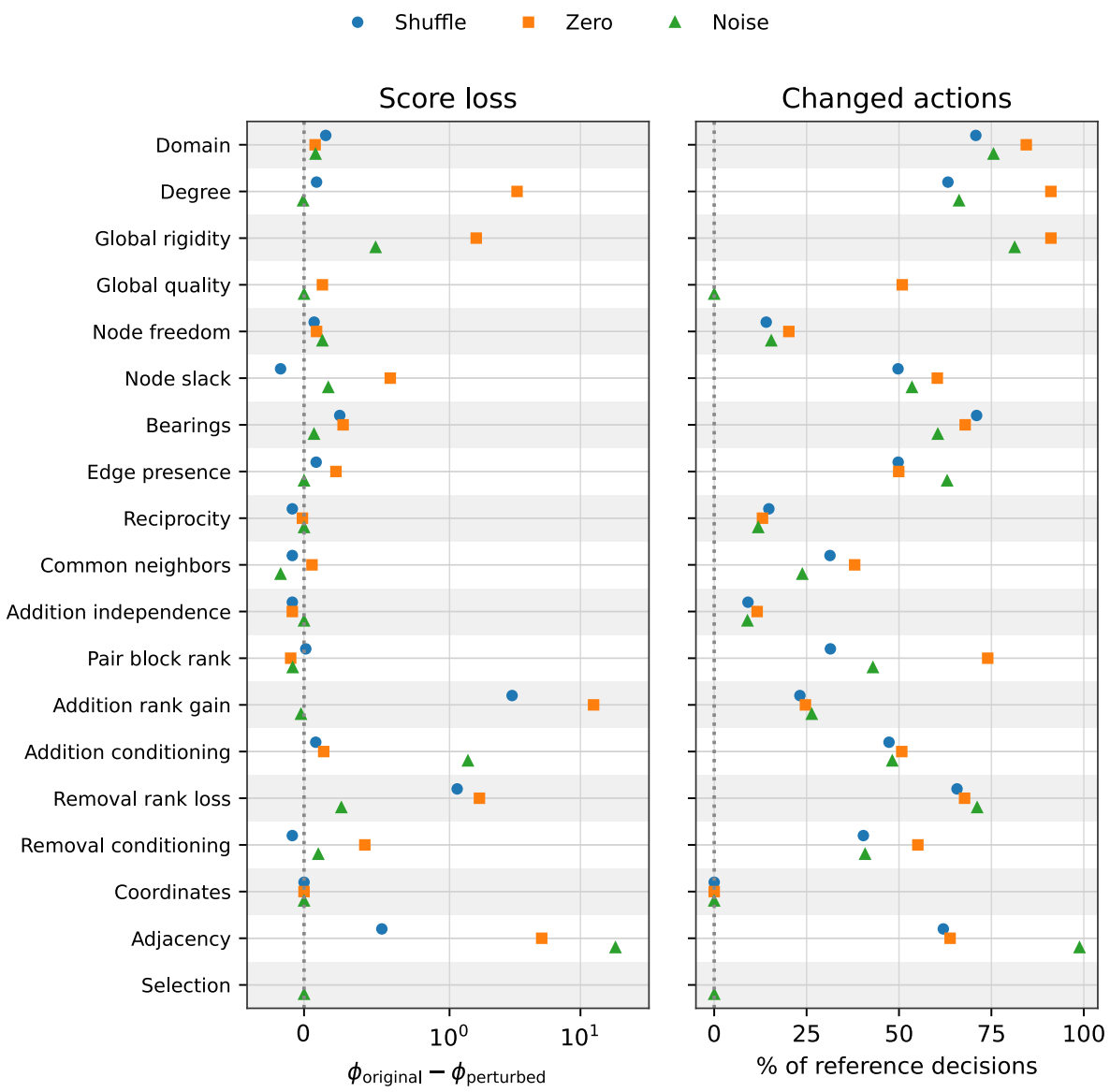


Figure B.6: Effect of perturbing observation channels on the policy with $\kappa_t = \kappa_e = 0$. Each channel of Table 5.5 is perturbed separately over 20 episodes. Global rigidity denotes the three rigidity-state channels, and addition and removal conditioning denote the add and remove stiffness channels. Perturbations that leave the channel unchanged have no marker. The score loss axis is linear within $[-1, 1]$ and logarithmic outside it.

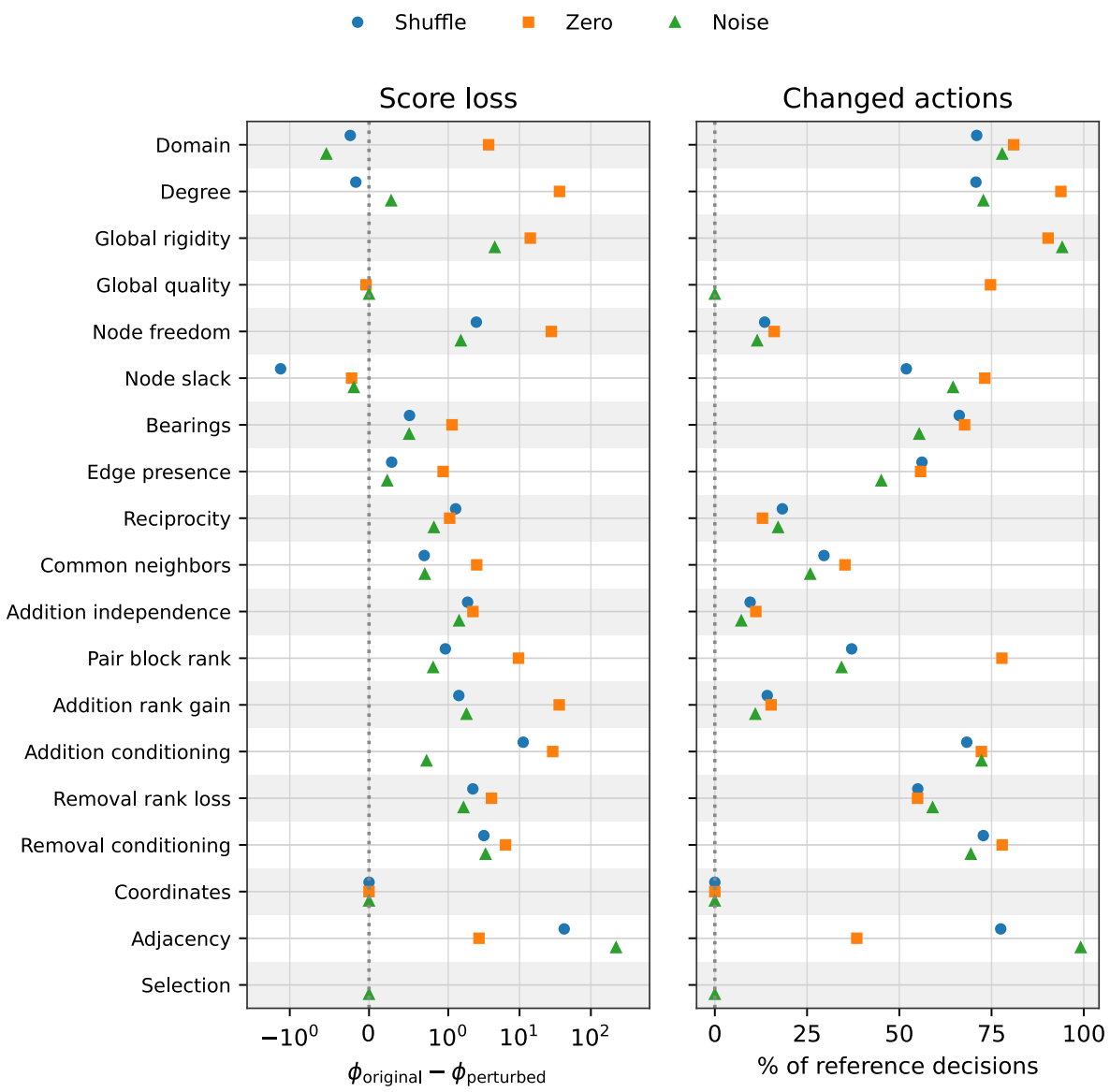


Figure B.7: Effect of perturbing observation channels on the policy with $\kappa_t = \kappa_e = 200$, as in Figure B.6. Score losses are on the scale of this policy's objective, which differs from that of Figure B.6. The score-loss axis is linear within $[-1, 1]$ and logarithmic outside it.

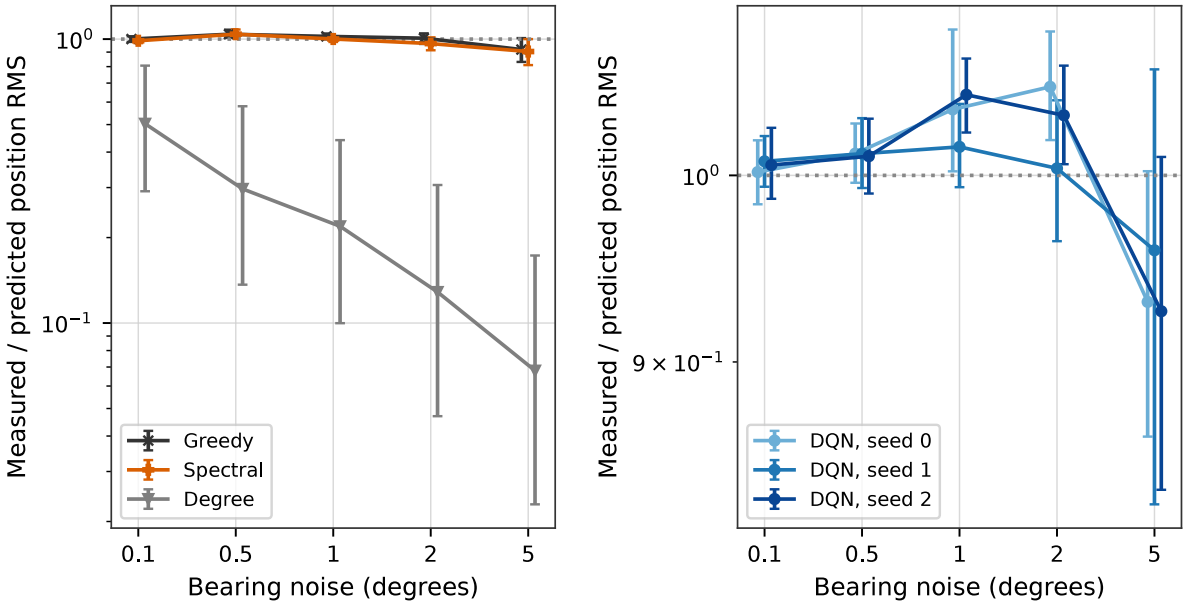


Figure B.8: Ratio of the measured to the predicted position RMS at each bearing noise level. Values are geometric means over 20 formations per method, with 100 noise trials per formation and noise level, and intervals are bootstrap intervals over formations. The reference policies have $\kappa_t = \kappa_e = 10$.

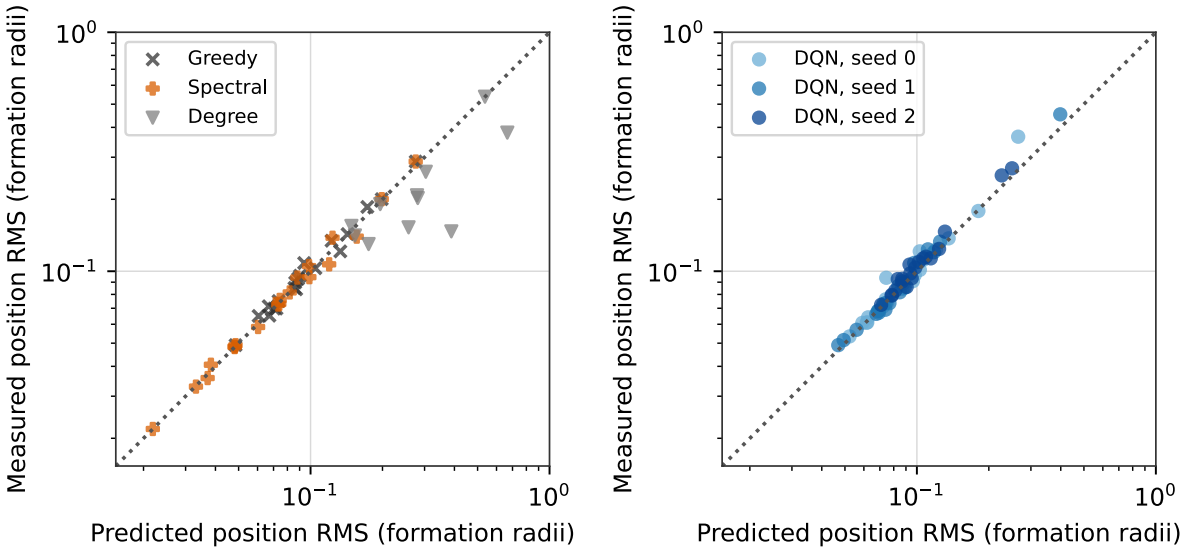


Figure B.9: Predicted and measured position RMS of the formations at 1° bearing noise, in formation radii (divided by the formation's RMS radius L).

B.10 Control trajectories

Figure B.10 shows the bearing error trajectories of all five control cases of §6.8, where every method starts from the same formation and targets the same goal.

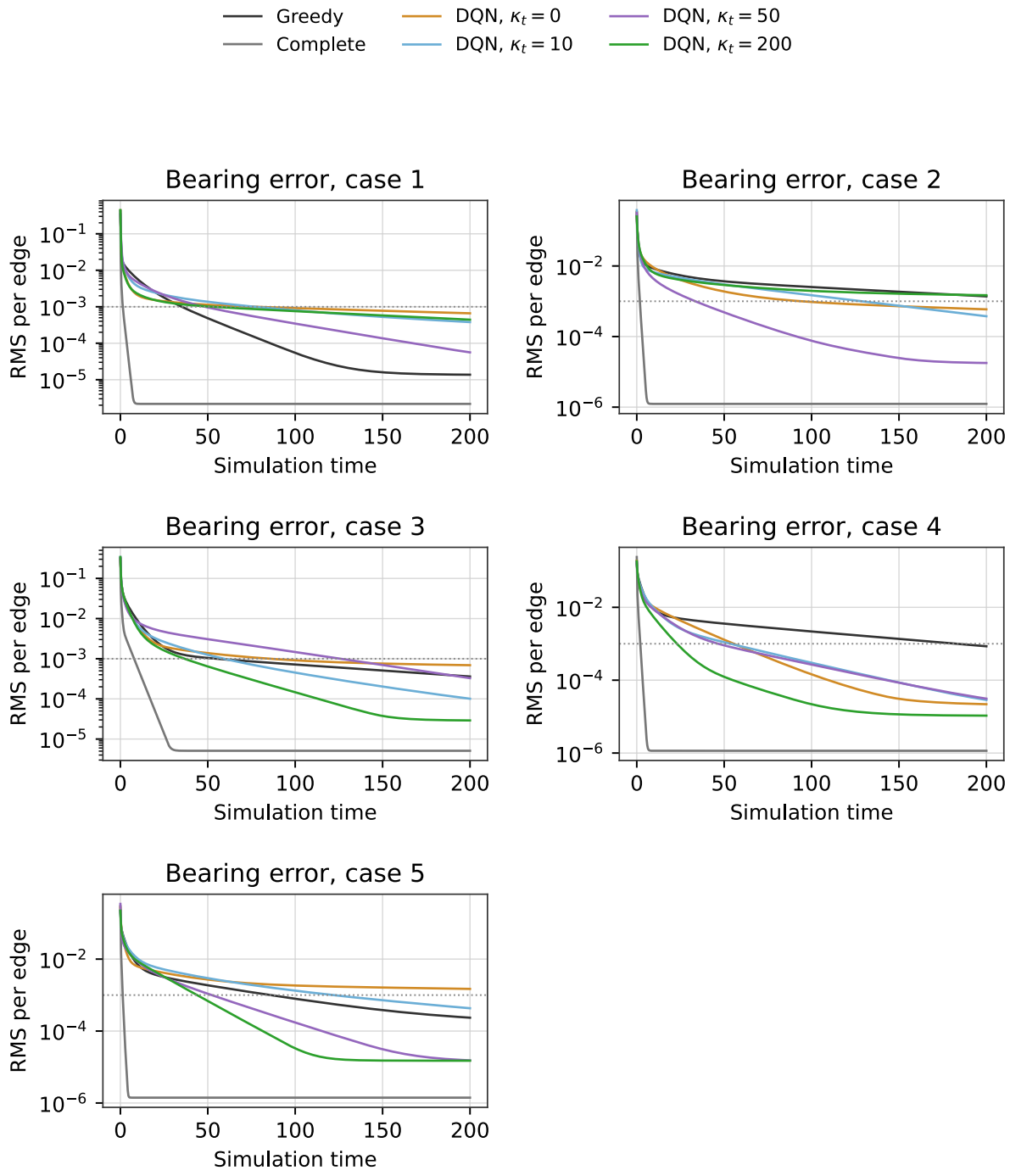


Figure B.10: Bearing error during formation control in the five control cases. The dotted line marks the convergence tolerance.

B.11 Geometric details

Figures B.11 and B.12 show additional details for the spectral and degree graphs on the formation of §6.7. The covariance display scale is shared by both panels but differs from that of Figure 6.16.

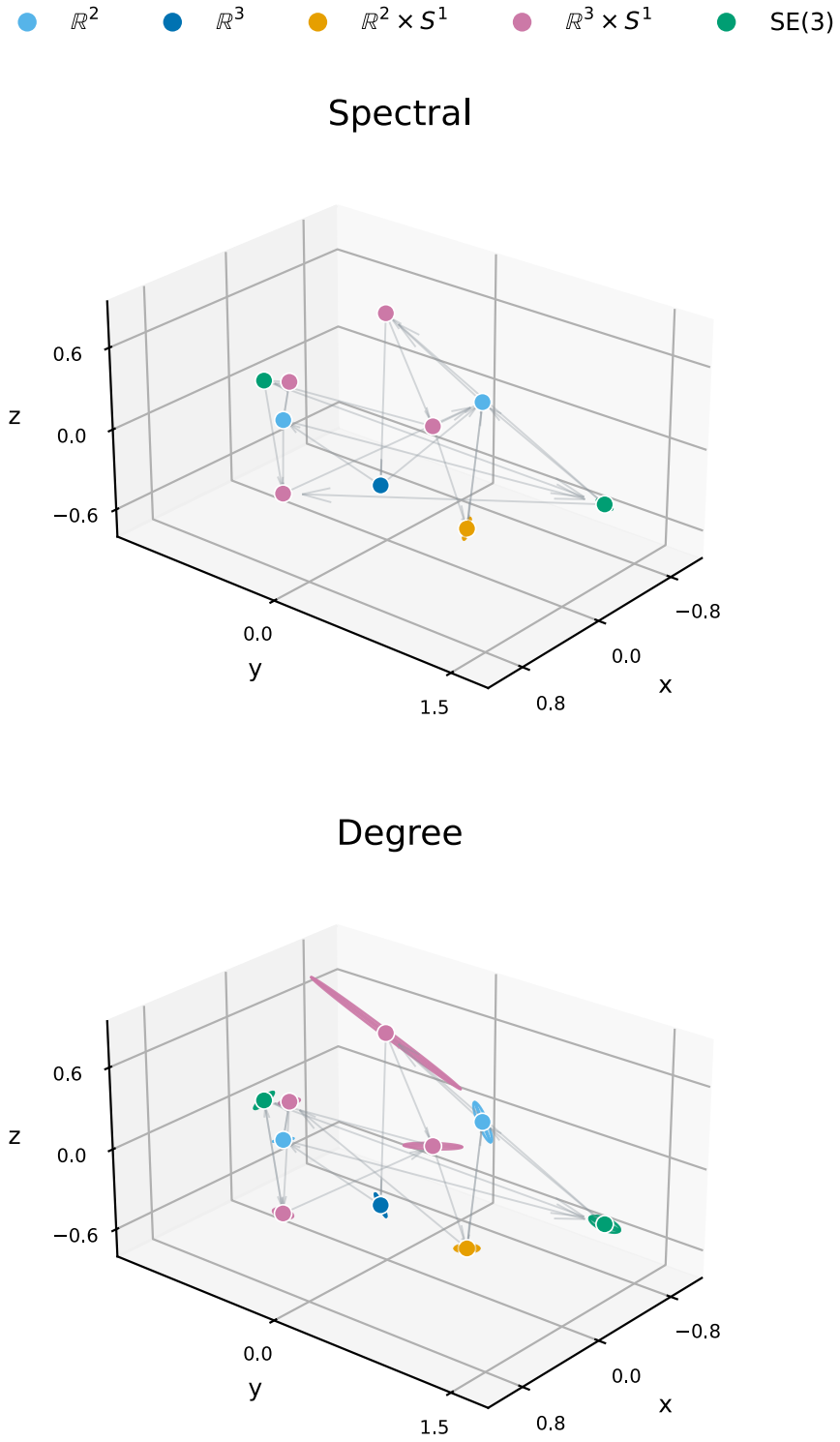


Figure B.11: Position covariance ellipsoids of the spectral and degree graphs.

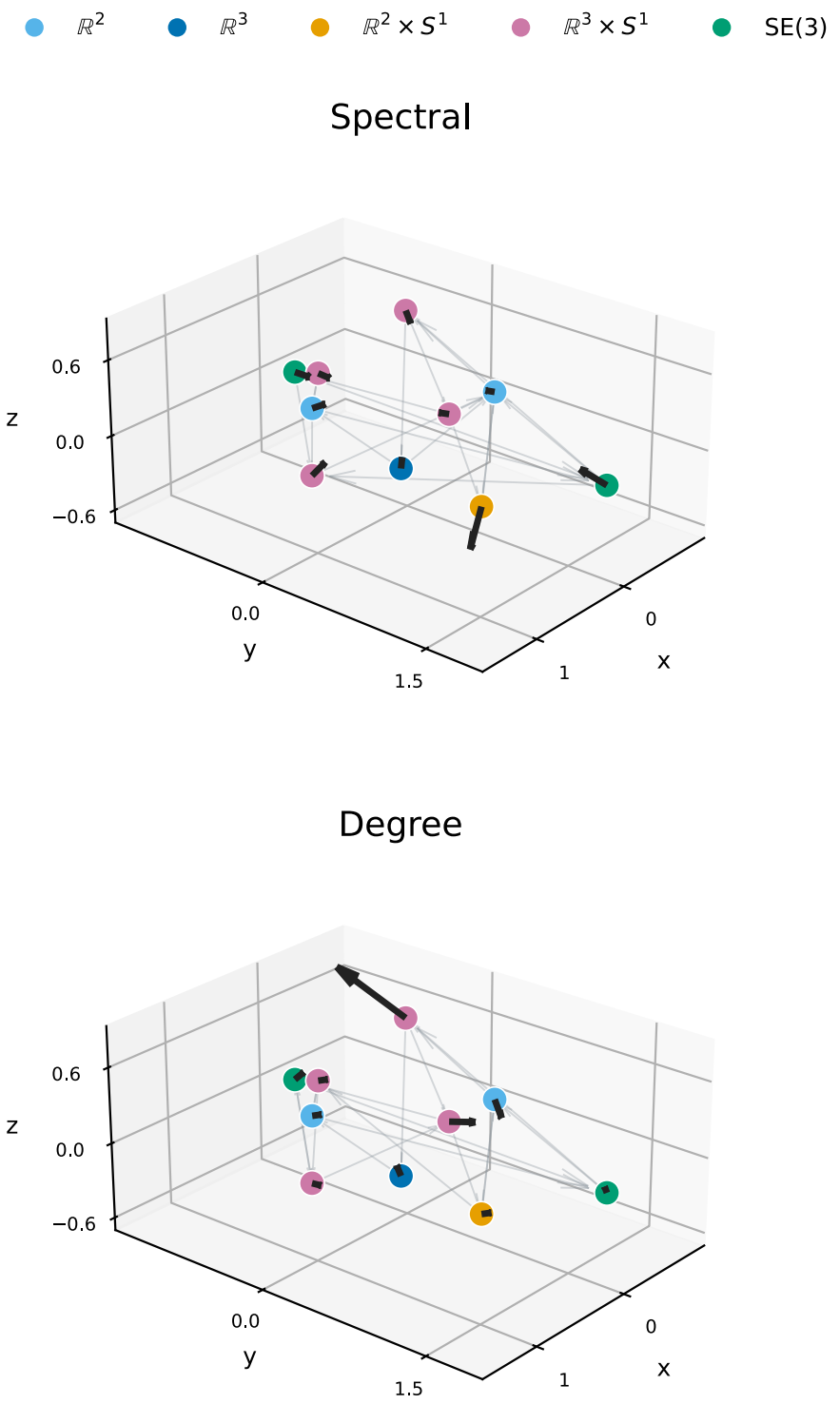


Figure B.12: Position components of the weakest observable mode of the spectral and degree graphs.

B.12 Position trajectories

Figure B.13 shows the position of each agent over time relative to its initial position in the first control case, and Figure B.14 shows the attitude error for all five cases.

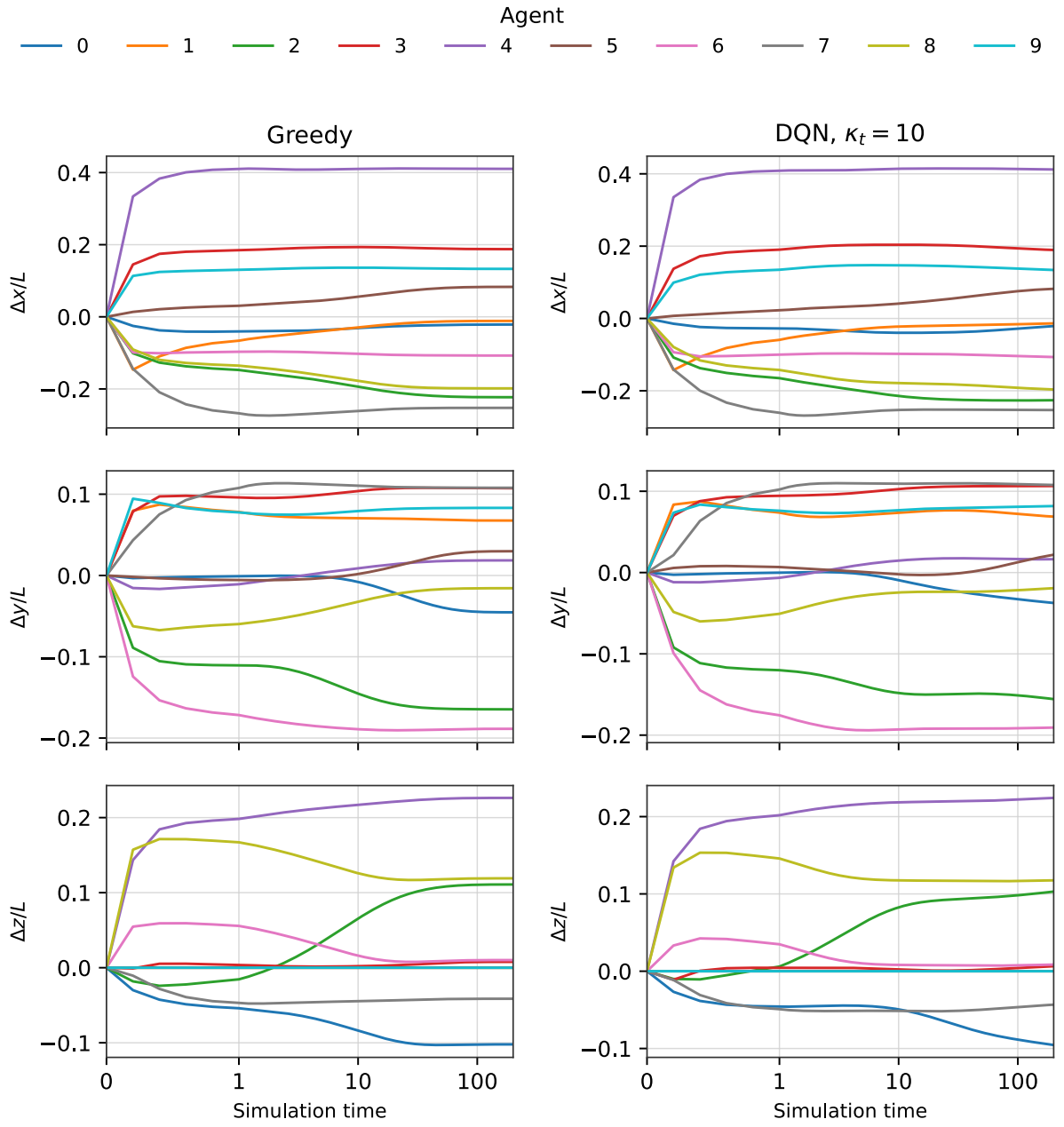


Figure B.13: Displacement of each agent from its initial position in the first control case, with L the goal formation radius. The time axis is linear up to $t = 1$ and logarithmic thereafter.

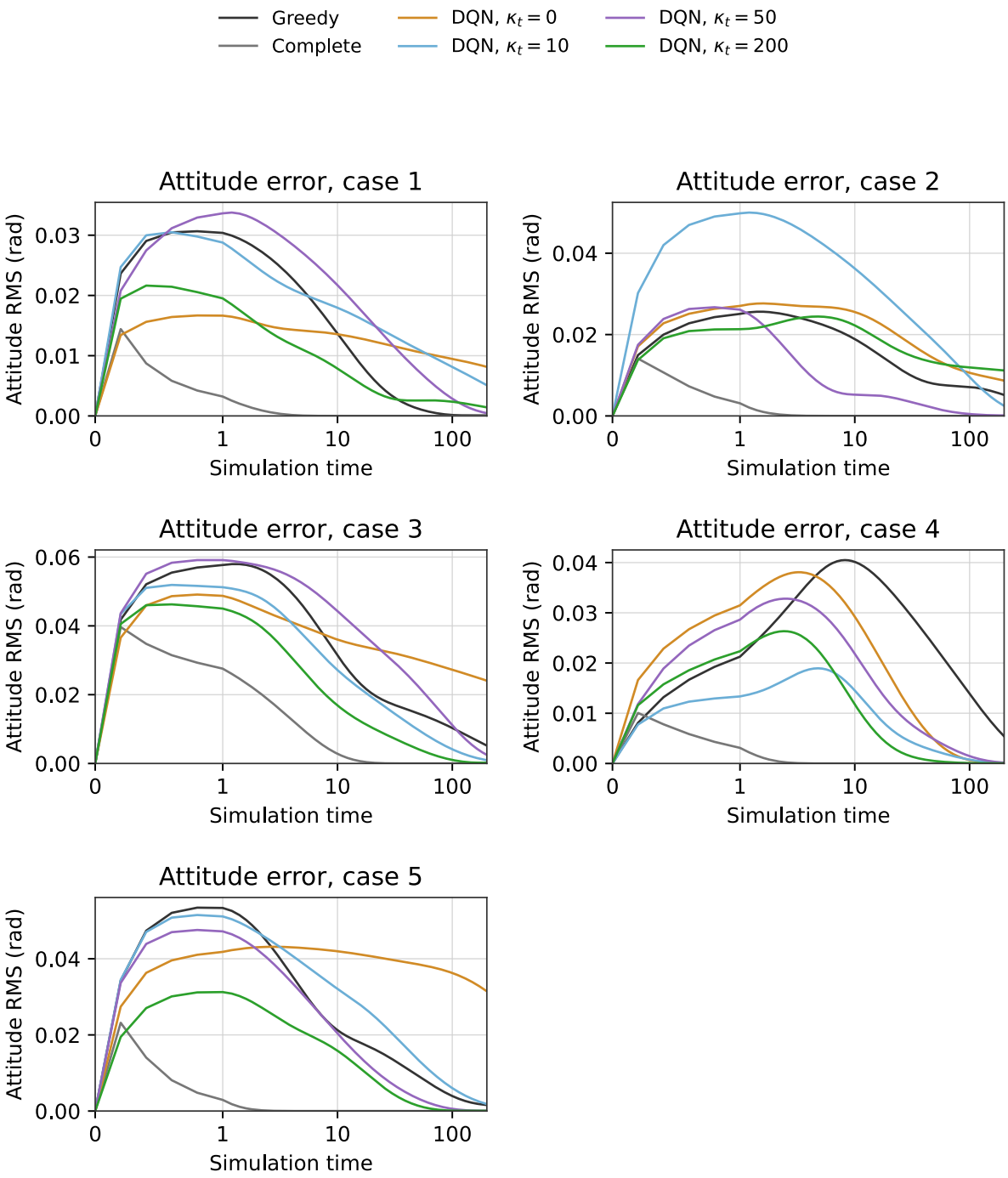


Figure B.14: Gauge-projected attitude RMS during formation control in the five control cases. The time axis is linear up to $t = 1$ and logarithmic thereafter. The initial attitudes equal the goal attitudes.

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